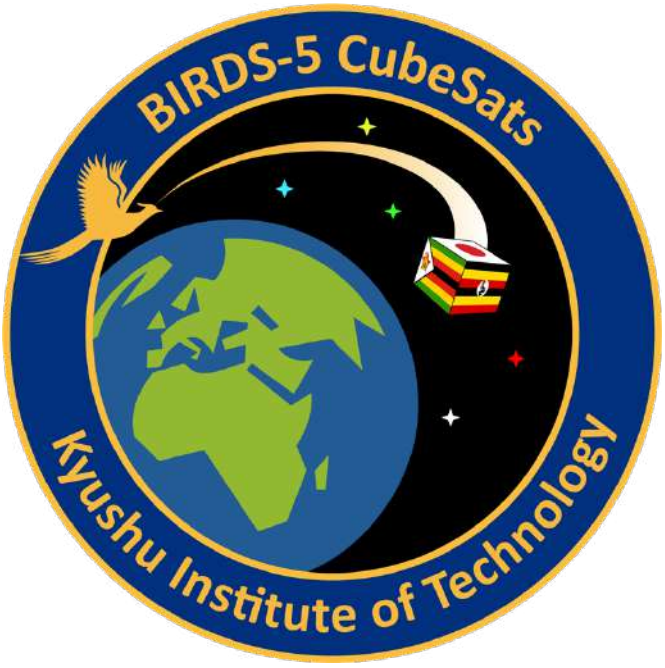



# [BIRDS-5]

# Flight Safety Assessment Report

# for Phase III

**Notice** : This technical data is furnished on the condition that it will be used by and disclosed to the receiving cooperating agency and its contractors and sub-contractors only for the purposes of fulfilling the cooperating agency's responsibilities under the Space Station Intergovernmental Agreement (IGA) and Memorandum of Understanding (MOU). It shall not be used for any other purpose, nor disclosed or retransferred to any other entity or government without prior written permission of the Japan Aerospace Exploration Agency (JAXA).

Rev. 2 : 2026/9/15

Kyushu Institute of Technology

## REVISION HISTORY

| REV. | DATE | Writer | DESCRIPTION |
|---|---|---|---|
| 1 | 2023/10/6 | Yamauchi | Initial Release |
| 2 | 2026/9/15 | Yamauchi | Inhibit diagram revised |
| | | | |

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# 1. Introduction

## 1.1 Purpose

The purpose of this document is to confirm compliance with International Space Station (ISS) safety requirements and verification results of the BIRDS-5 satellites, which will be deployed from Japanese Experiment Module (JEM) Small Satellite Deployer (hereafter called “J-SSOD”), at Phase III Payload Safety Review Panel (PSRP).

## 1.2 Scope

This document is prepared for the Phase III safety review by JAXA PSRP.

The scope of this document is to show safety design and verification results of CubeSat from its launch to deployment from ISS.

Satellite names are 001U(PearlAfricaSat-1), 001J(TAKA) and 001Z(ZIMSAT-1).

## 1.3 Applicable Documents

Current version of the following document is applicable as referred from this document.

(1) JX-ESPC-101132E JEM Payload Accommodation Handbook Vol.8 Small Satellite Deployment Interface Control Document

# 2. Safety Analysis Methodology

## 2.1 Methodology

Fault Tree Analysis (FTA) is performed to identify the possible hazard causes while reviewing detailed design of BIRDS-5 and its surrounding environment or conditions.

Hazard Report addresses all the hazards including hazard definition, hazard causes, hazard controls, verification method for the controls, and results of verification.

Compliance with ISS Jettison Policy are evaluated by J-SSOD system integrator in “Safety Assessment Report for series product.”

## 2.2 Safety Requirements

Following safety requirements are applied.

(1) SSP51721 ISS Safety Requirement Documents

(2) PPD1101 ISS Jettison Policy

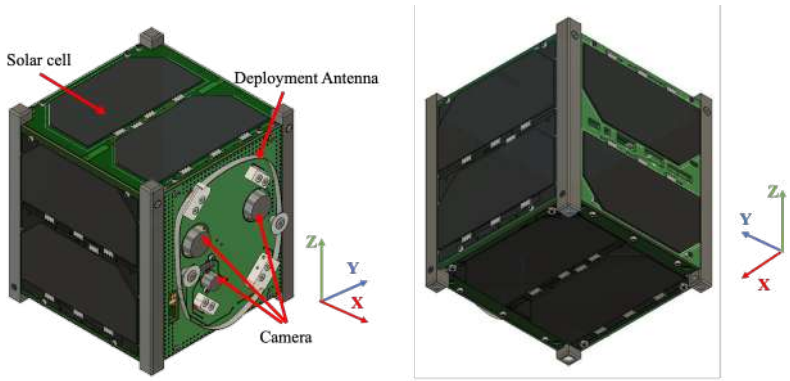
(3) SSP30599 Safety Review Process

# 3. System Description

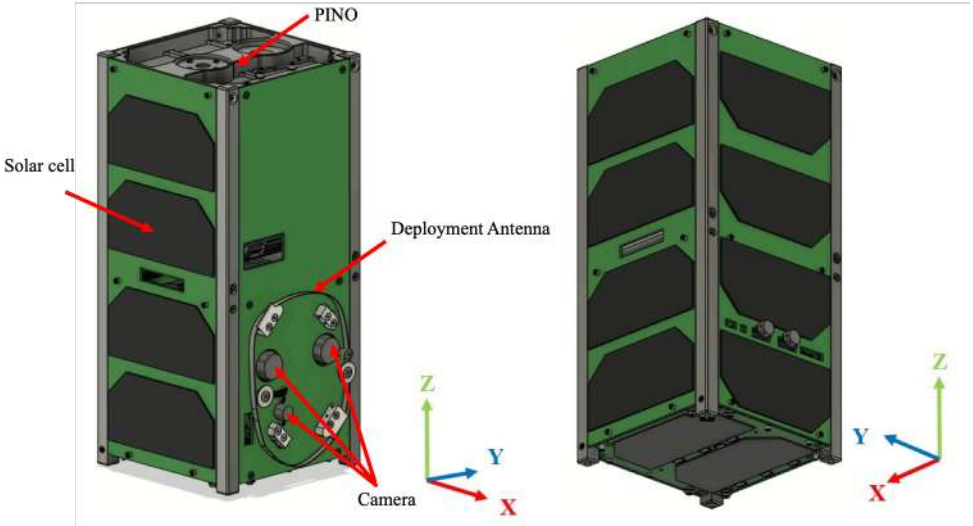
## 3.1 Overview

The BIRDS-5 project is a constellation of two 1U satellites (PearlAfricaSat-1 and ZIMSAT-1) and one 2U satellite (TAKA-1). 1U CubeSats whose dimension is 100 mm x 100 mm x 113.5 mm and weight are less than 1.33 kg, 2U CubeSat whose dimension is 100mm ×100mm × 227 mm and weight is less than 2.66 kg. External views of BIRDS-5 satellites in its stowed configuration and after deployment configuration are shown in Figure 3.1-1 and Figure 3.1-2, respectively.

[(PearlAfricaSat-1, TAKA-1, ZIMSAT-1)] shall perform several missions (Automatic Packet Reporting System (APRS) digipeater, Store-and-Forward (S&F), BIRDS-NEST, Particle Instrument for Nano-satellites (PINO), Attitude Visualization, On-board Image Classification, Land Use and Cover, Water Quality Assessment, Soil Fertility and shall be achieved with onboard mission payloads. PearlAfricaSat-1 and ZIMSAT-1 have different names, but the hardware is exactly the same.

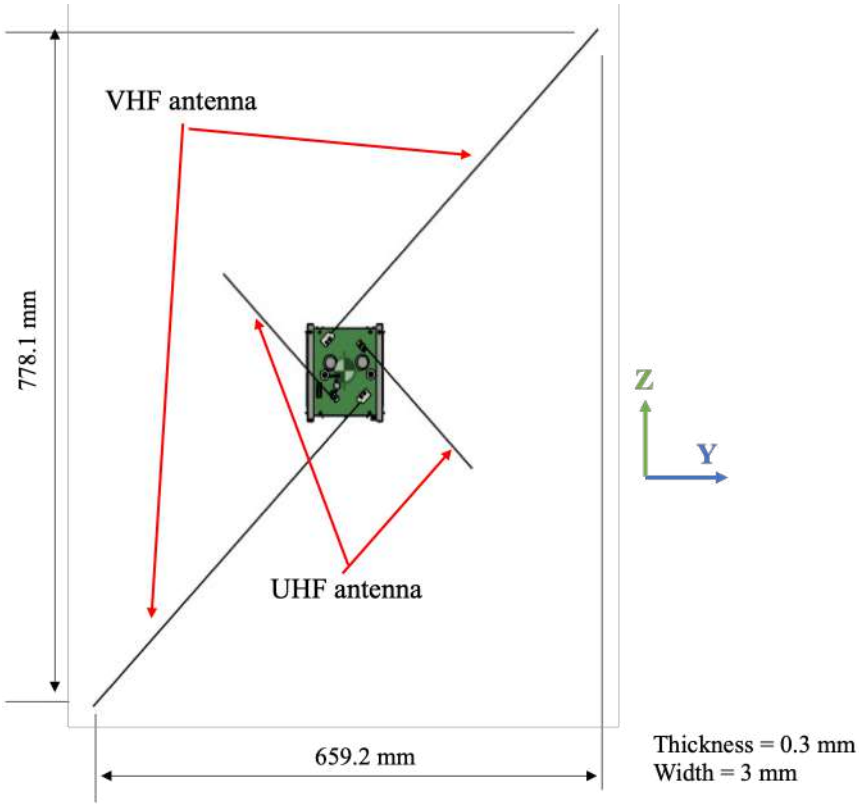



**(a) 1U CubeSat**

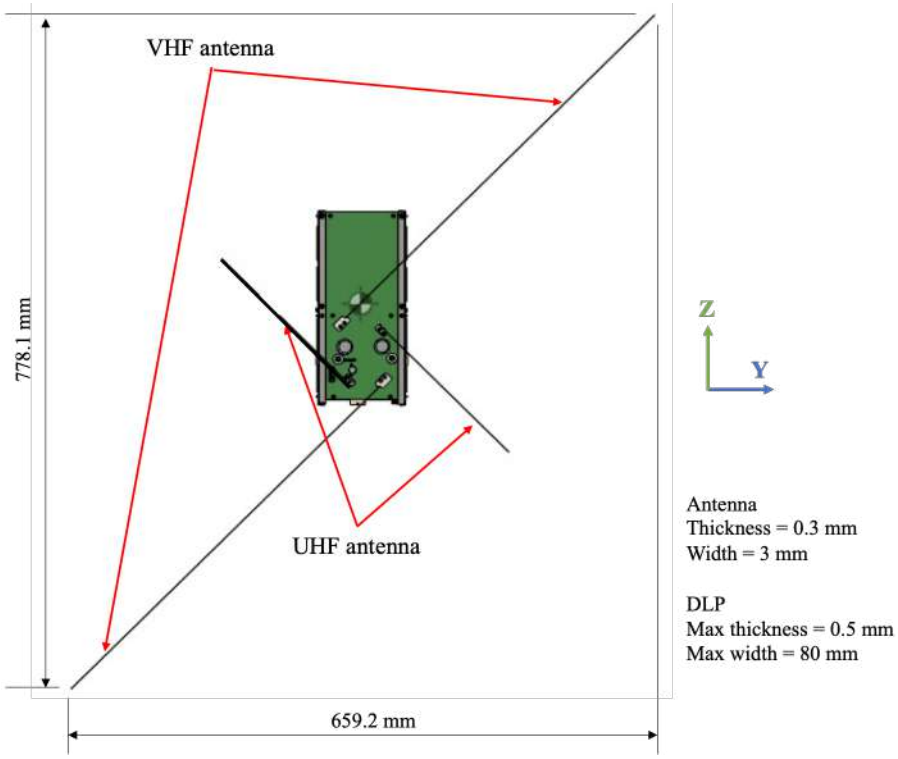



**(b) 2U CubeSat**

**Figure 3.1-1 External View of BIRDS-5 (Stowed configuration)**




**(a) 1U CubeSat**




**(b) 2U CubeSat**

**Figure 3.1-2 External View of BIRDS-5 (After deployment configuration)**

BIRDS5 Project:

BIRDS-5 mission objectives are as follows.

- COTS APRS-Digipeater demonstration on CubeSat (APRS-DP)
- Demonstration of ground data acquisition using Store and Forward (S&F)
- Demonstration of a mobile phone application displaying BIRDS satellite data (BIRDS-NEST)
- Measurement of precipitation of high-energy electrons in the radiation belt (PINO) (only TAKA-1)
- Demonstration of attitude visualization of the satellite
- Demonstration of image classification program by machine learning algorithms
- Demonstration of land use and cover by COTS multi-spectrum camera
- Demonstration of water quality evaluation of dams and lakes by COTS multi-spectrum camera
- Demonstration of analysis of soil nitrogen / fertility levels by COTS multi-spectrum camera

Figure 3.1-3 shows each satellite system. Three satellites are mostly identical with only one difference in missions, that is the PINO are mounted only on the “TAKA-1” satellite.

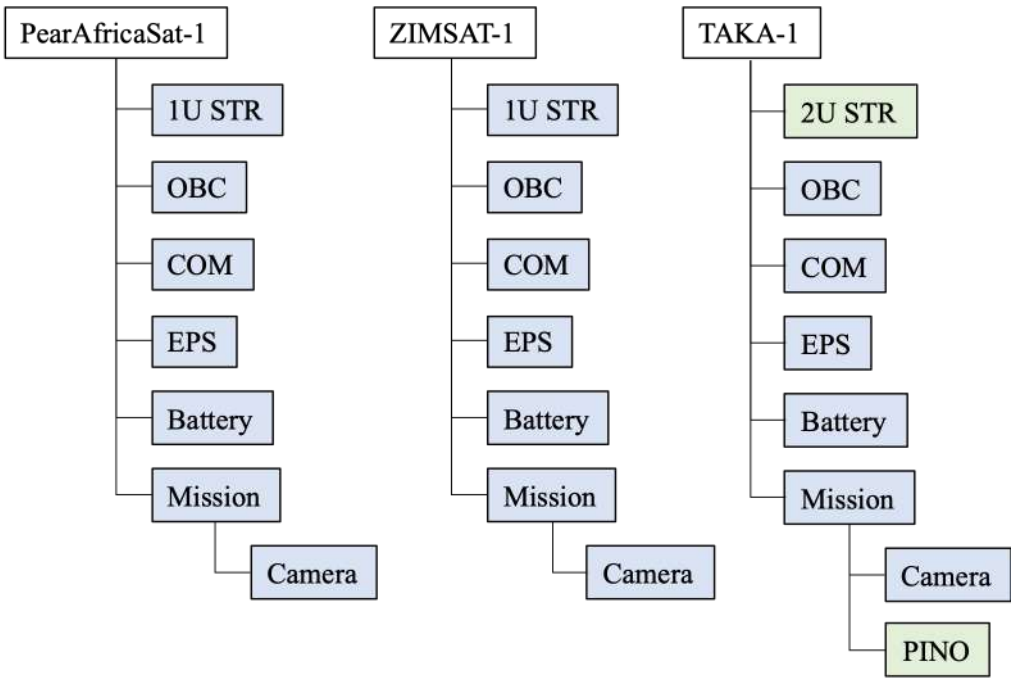



**Figure 3.1-3 System for each satellite**

## 3.2 Subsystems

### 3.2.1 Structure

Figure 3.2.1-1 and Figure 3.2.1-2 shows the primary structure and internal configuration. The primary structure is made of Aluminum alloy A6061-T6 and all four rails are hard anodized. These parts are not welded, forged, cast or quenched.

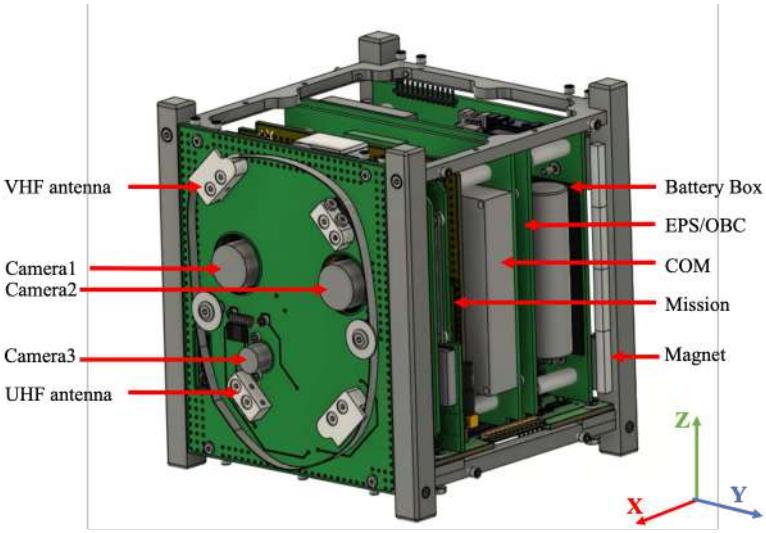



**Figure 3.2.1-1 Primary Structure and internal Configuration of 1U CubeSat**

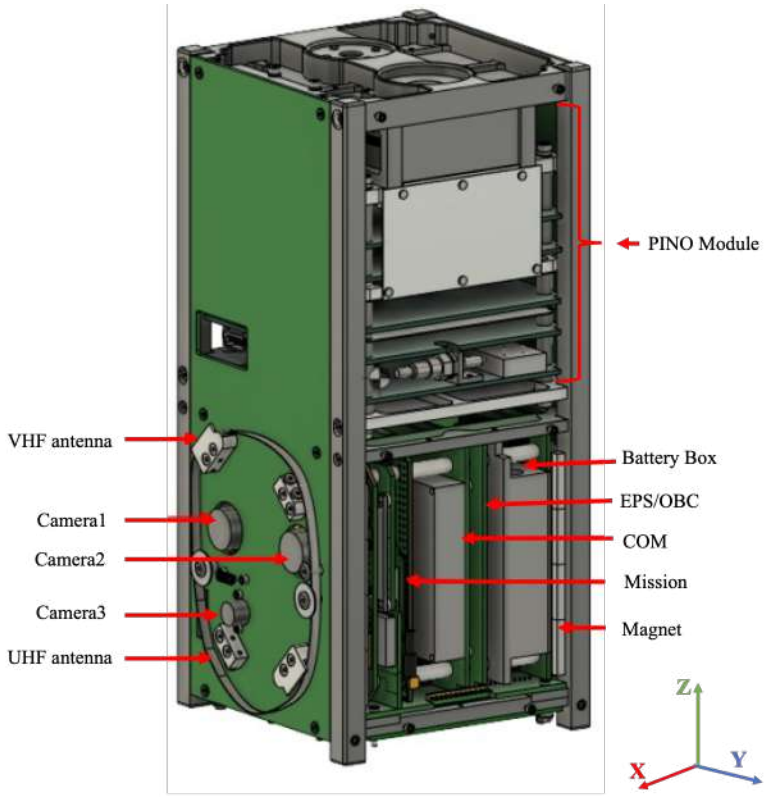
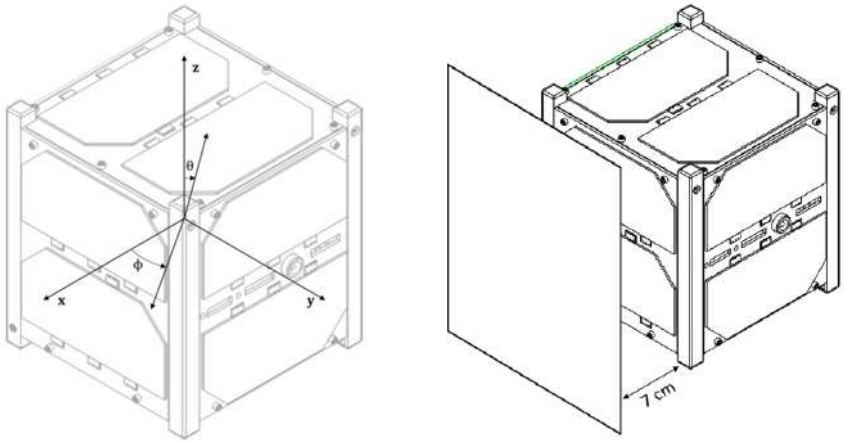



**Figure 3.2.1-2 Primary Structure and internal Configuration of 2U CubeSat**

### 3.2.2 Magnetic Field Analysis

Figure 3.2.1-1 shows the main structure of the BIRDS-5 satellite. Main structure consists of 4 rails, 4 ribs and 1 plate. Rails, ribs and the plate are composed of aluminum alloy (A6061-T6). All rails (4) are hard anodized (Anodized per MIL-A-8625 Type3). The magnet (four 3.2x3.2x19mm magnets are attached to each rail), made from an alloy of Aluminum, Nickel and Cobalt (AlNiCo) (Manufacture: MEDER electronic, Part No. 4003004018).




(a) (b)

**Figure 3.2.2-1 (a)Coordinate system and (b)calculated plane**

Each of the 4 rails will have 4 bar magnets (block), connected end to end, attached to their inner side. Thus, there will be a total of 16 bar magnets. (4 blocks). The resultant magnetic field is the superposition of the individual field of the magnetic blocks. Each block is modelled as a magnetic dipole. The magnetic moment is calculated on parallel planes at 7 cm from the satellite surface as shown in Figure 3.2.2-1.

**Theoretical consideration for four bar magnets as a Single Dipole**

Calculating the Volume of the Alnico 5 permanent magnets

Volume =Length*Breadth*Height

=1.90 cm * 0.32 cm * 0.32 cm

$= 0.19\ cm^3\ each\ bar$

For Four magnets

$= 4 * 0.2 = 0.78\ cm^3$

**Calculating the Magnetic Moment**

M: Magnetic moment [$Am^2$]

B: Remanence [T]

V: Volume [$m^3$]

μ: Permeability

$$\mu M = \mathrm{BV}$$
$$= 1.25 * 0.78 * 10^{-6}$$
$$= 0.98 \times 10^{-6}\ [Wb \cdot m]$$

The magnetic flux density field of magnets oriented along the z axis is given by

$$B = \frac{|\mu M|}{4 * \pi * r^3}\left(2cos\theta\hat{r} + sin\theta\hat{\theta}\right)$$

Where

r: radial distance from the center of the magnetic moment

λ: Inclination angle (deg)

Since each rail is equipped with magnets, the total magnetic field is calculated as the sum of the magnetic fields from each magnet. The total magnetic field is calculated on the planes at 7 cm from the surface .at this distance, the maximum magnetic field is 2.27 Gauss. The analysis results of the magnetic field distribution are shown in Figure 3.2.2-2 and Figure 3.2.2-3. The maximum magnetic field is formed at a position close to the rail on a plane 7 cm from the satellite. The magnetic field is less than the required 3.16 Gauss so there is no hazard.

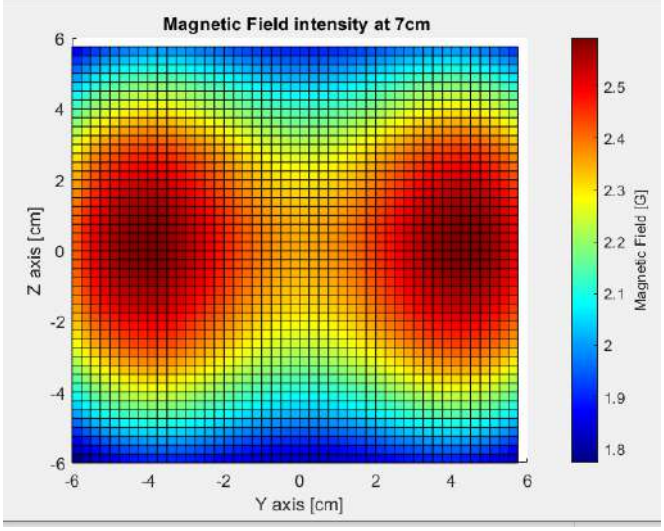



**Figure 3.2.2-2 2D Map of magnetic field intensity of 16 bar magnets a parallel plane 7cm apart**

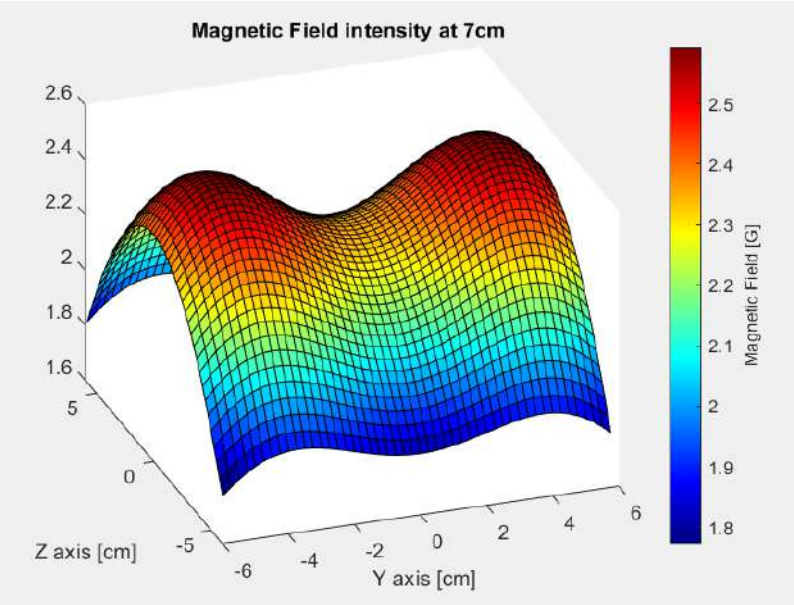



**Figure 3.2.2-3 3D Map of magnetic field intensity of 16 bar magnets at a parallel plane 7cm apart**

### 3.2.3 Mechanism

**Antennae**

BIRDS-5 has four deployable antennas. UHF and VHF antennas are dipole type and secured on the +X panel. Figure 3.2.3-1 shows the antenna deployment mechanism. UHF antennas (inside) are restrained by VHF antennas (outside). VHF antennas are restrained by double strings for redundancy. Each VHF antenna is 514 mm length, 4mm width and 0.3 mm thickness while each UHF antenna is 171 mm length, 4mm width and 0.3 mm thickness. The antenna folding is guided by guide rails with washers to prevent it from untangling.

**Safety of the Antenna Deployment System**

**Antennae**

The power supplied from the battery is required to activate the antenna deployment system. When the antennas are inadvertently deployed inside the J-SSOD, CubeSat may not be deployed from J-SSOD appropriately due to contact of the antenna with J-SSOD inner surface. To prevent this mis-operation, this CubeSat used two strings for VHF antennas (Figure 3.2.3-1). The UHF antenna is covered by a long outer VHF antenna, so the UHF antenna cannot be deployed unless the VHF antenna is deployed.

The heat cutter circuit is connected to the electrical power system. In order to release the satellite successfully the deployment switch (Dep-SW) has to be closed. Then, electric power is supplied to the burner circuit, the heat cutter is powered, and the wire is cut.

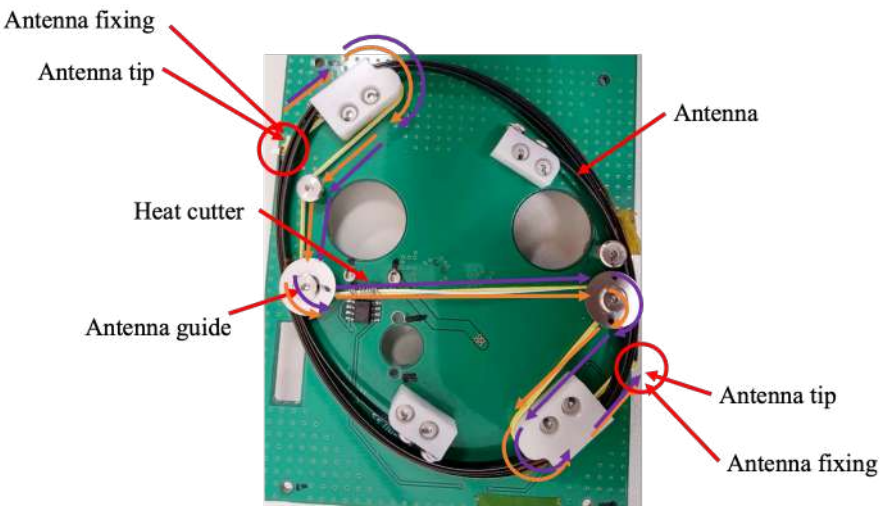



**Figure 3.2.3-1 Stowed antenna configuration**

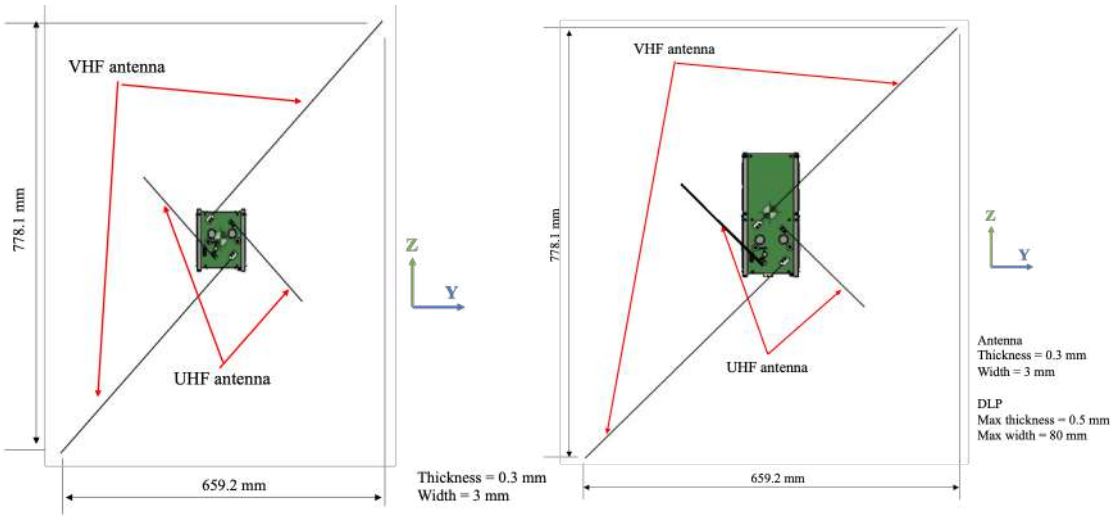
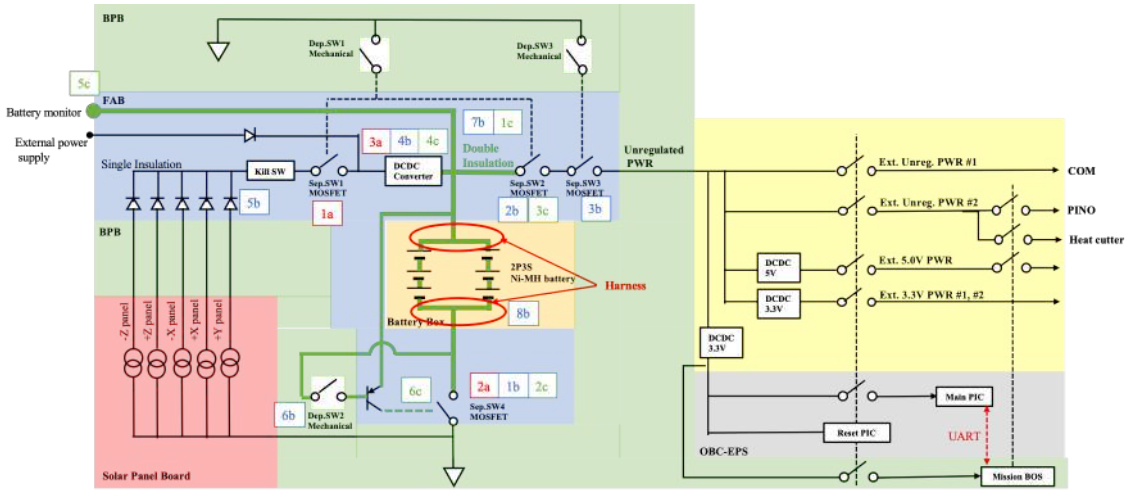



**Figure 3.2.3-2 1U and 2U Deployed antenna configuration**

### 3.2.4 Electrical Power System

Figure 3.2.4-1 shows inhibit schematic. There are three deployment switches to cut power for all loads before deployment, and these switches controls to prevent hazards.

Six Nickel Metal Hydride - NiOOH/metal alloy/KOH batteries are connected by three-series and two-parallel. The CubeSat does not use wet Electrolytic Capacitor in EPS.




| Hazard | | Hazard Control #1 | Hazard Control #2 | Hazard Control #3 |
|---|---|---|---|---|
| Over-charge | | SepSW1[1a] | SepSW4[2a] | DCDC converter[3a] |
| Over-discharge | Load side | SepSW4[1b] | SepSW2[2b] | SepSW3[3b] |
| | Solar cell side | | DCDC converter[4b] | Diode[5b] |
| | DepSW2 side | DepSW2 [6b] | Proper Insulation[7b] | Proper Charging [8b] |
| External short | Load side | Proper Insulation[1c] | SepSW4[2c] | SepSW2[3c] |
| | Solar cell side | | | DCDC converter[4c] |
| | External power supply side | | | DCDC converter[4c] |
| | Battery monitor side | | | Proper Insulation[5c] |
| | DepSW2 side | | | Proper Insulation[6c] |

Note: Proper insulation (double isolation is shown by green line in figure above, single isolation is black line)
All wires and components between the battery and the first power functions are assembled as double insulation. .
The DCDC converter (LTC3119) used for inhibit is a buck-boost converter and its internal FET configuration prevents reverse current.

**Figure 3.2.4-1 Inhibit Schematic**

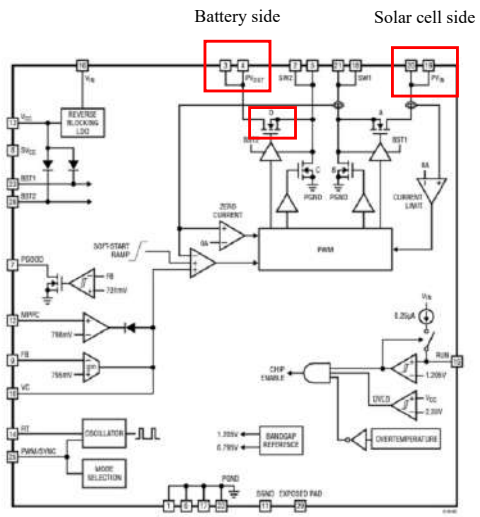



**Figure 3.2.4-2 Inhibit Schematic of DCDC converter**

### 3.2.4.1 Deployment Switch

Table 3.2.4.1-1 shows the design characteristics of the deployment switch. Figure 3.2.4.1-1 shows locations of the deployment switches.

**Table 3.2.4.1-1 Deployment Switch Properties**



| | |
|---|---|
| Manufacture | C & K |
| Part Number | SDS002 |
| Rated current (A) | 100 mA (DC) |
| Rated voltage-DC | 12 V |
| Actuator type | Overtravel plunger |
| Actuating Force | 0.74 N |
| Releasing Force | 0.25 N |
| Overtravel | 2.15 mm |
| Operating temperature | -40 ° C to 85 ° C |

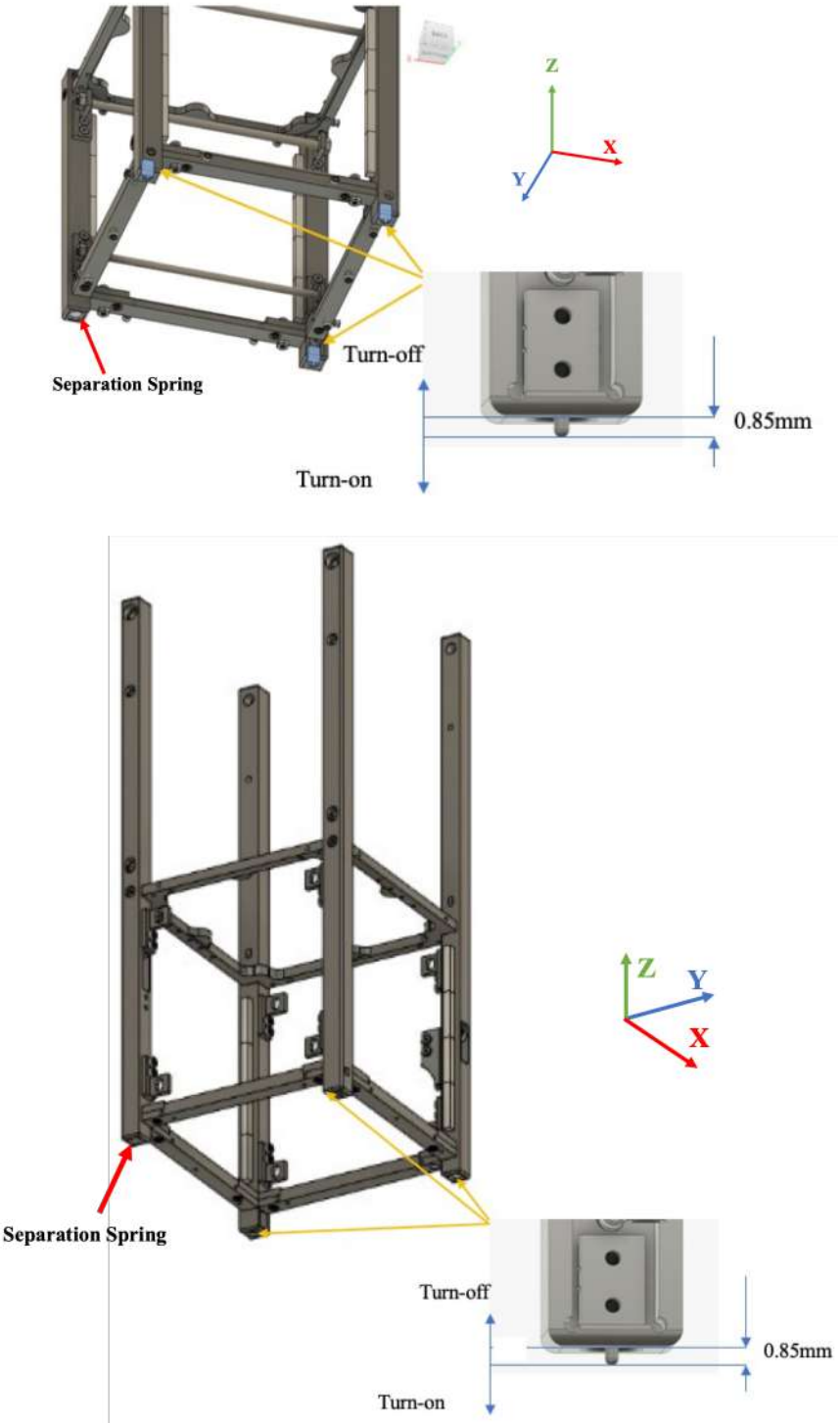



**Figure 3.2.4.1-1 Deployment Switch**

As separation springs, a spring plunger (P/N 251D939002-1) and deployment switches were used at the positions shown in Fig. 3.2.4.1-1. The spring force of the spring plunger is 0.6±0.06 [N]. The return force of the deployment switch is 0.25 [N]. The total spring force is 1.35 [N], which meets the requirements.

### 3.2.5 Communication

The Communication Subsystem as shown in Figure 3.2.5-1, receives command uplink from the ground station, sends telemetry/housekeeping/mission data to the ground station, and transmits Morse-coded CW beacon. Its command uplink receiver operates in the UHF amateur band (within 435-438MHz) and GMSK demodulation, 4,800 bps in AX.25 protocol. Similarly, the telemetry/housekeeping/mission data downlink transmitter operates in the UHF amateur band ([redacted] MHz) and supports GMSK modulation, 9,600 bps in AX.25 protocol, with transmission power of 0.8W which conforms RF radiation criteria. The uplink command to the CubeSats is sent from the own ground station in Japan.

The APRS-DP/S&F Payload, which supports both the Automatic Packet Reporting System Digipeater Mission and Store-and-Forward Mission, operates in the VHF amateur band ([redacted] MHz). The payload communicates with APRS-DP and S&F users at [redacted] MHz in both uplink and downlink. It supports AX.25 protocol, AFSK on FM demodulation/modulation, 1,200 bps data rate. The payload consists of a half-duplex VHF FM transceiver, an APRS-DP/AX.25 TNC with modem module, APSF MCU and flash memory for data storage. It utilizes the VHF antenna for operation. Activation and deactivation of the payload is controlled by the main OBC which also controls the power supply to the payload.

Below is the sequence of operation of the CubeSat after release from the ISS.

1) Deployment switches are closed.
2) OBC starts operation.
3) OBC waits for 2,000 seconds and confirms that batteries generate enough output voltage.
4) OBC activates the heater circuit connecting to nichrome wire for antenna deployment mechanism.
5) The heater circuit stops after antenna deployment.
6) The transceiver turns ON after antenna deployment and commands can be received. CW beacon is transmitted.

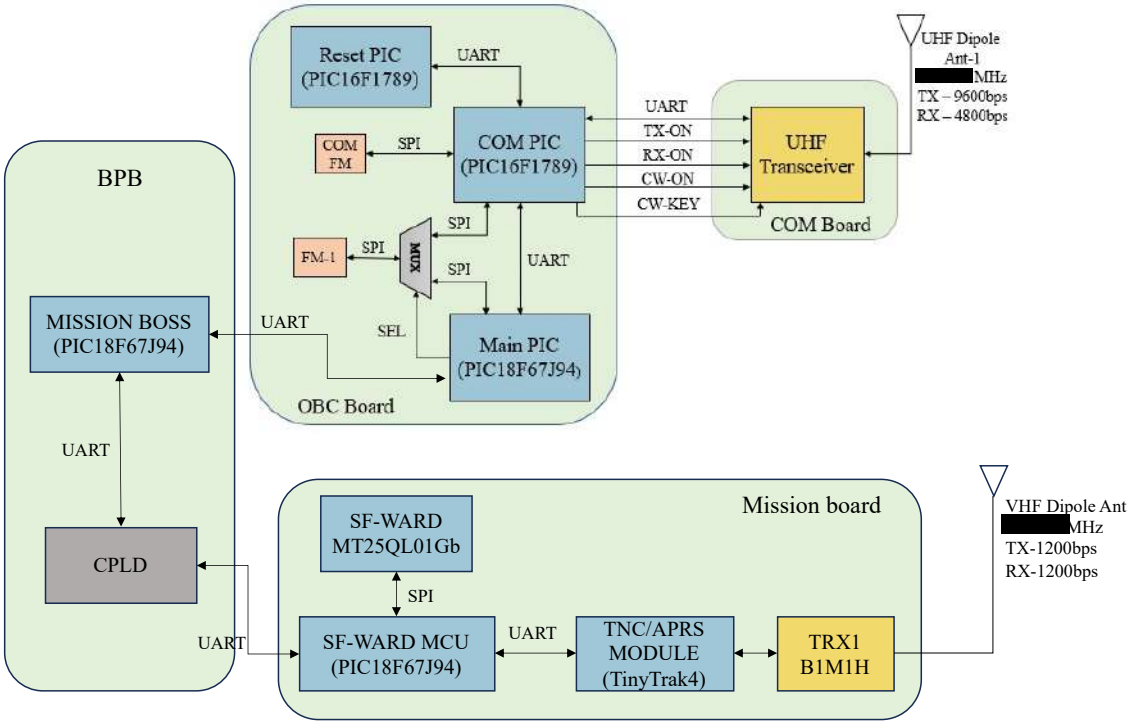



**Figure 3.2.5-1 Communication Subsystem Block Diagram**

### 3.2.6 Mission

#### 3.2.6.1 PINO

PINO will observe high-energy electrons coming from the Van-Allen radiation belts. PINO carries solid-state detectors, which are designed to detect high-energy electrons in space. PINO can detect high-energy electrons precipitating into the Earth's upper atmosphere. PINO's field-of-view directs along the geomagnetic field since the high-energy electrons precipitate along the magnetic field lines. PINO will provide monitoring of the space environment at an altitude of approx. 400km, which could be useful for forecasting space weather.

PINO consists of stacked semiconductor detectors and five circuit boards. PINO is supplied with an unregulated voltage from the EPS subsystem. Figure 3.2.6.1-1 shows a high voltage part of the PINO module. From the unregulated voltage, PINO generates all the required regulated voltages including high voltage (max. 2kV) which is needed to bias the semiconductors. The PINO module is connected to the LOAD side of EPS system, and EPS system has 3 inhibit before the satellite deployment (Figure 3.2.4-1). Because the EPS system inhibits to supply the unregulated voltage before the satellite deployment, PINO is not turned on before the deployment.



**Figure 3.2.6.1-1 High voltage parts in the sectional view of PINO module**

Figure 3.2.6.1-2 shows the +Z-plate. PINO has two holes, from which electrons come into the semiconductor detectors. The diameters of the holes are less than 10 mm, and the ISS crews cannot touch the inside of PINO.

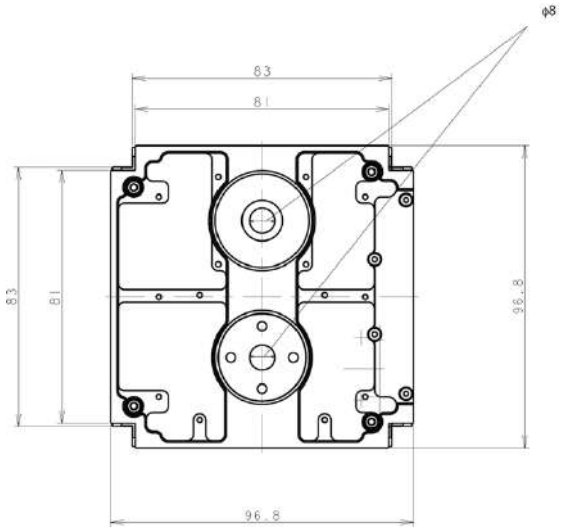



**Figure 3.2.6.1-2 +Z plate of the 2U CubeSat**

## 4. Launch Configuration

On ground, CubeSat is installed into Satellite Install Case of J-SSOD or J-SSOD-R Launch Case, and there is packed in a Cargo Transfer Bag (CTB) with cushion foam as shown in Figure 4-1.

The CTB is launched by HTV-X, NG Cygnus or Space-X Dragon and transferred to ISS as a pressurized cargo.

When CubeSat is inside J-SSOD or J-SSOD-R Launch Case, deployable antennas are restrained by wire so that the antenna is within the allowable satellite envelope defined in JX-ESPC-101132E “JEM Payload Accommodation Handbook Vol.8”.

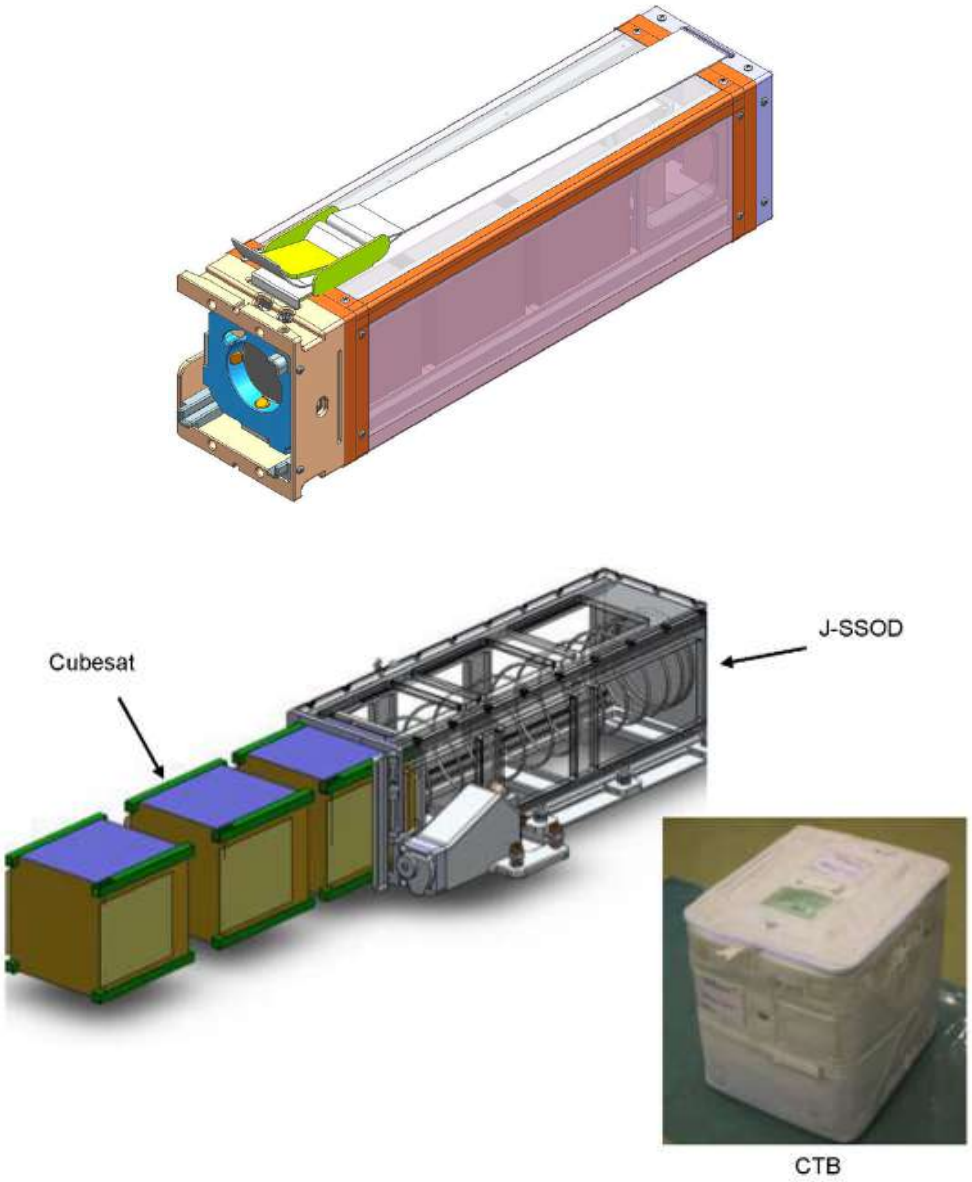



**Figure 4-1 J-SSOD, J-SSOD-R Launch Case and CTB**

## 5. Operation

**[In case launched with J-SSOD]**

CubeSat is cold-launched. After arrival of the CTB to ISS, the CTB is stowed inside ISS pressurized module. Before the CubeSat deployment mission, J-SSOD is picked up from the CTB and installed onto Japanese Experiment Module Remote Manipulator System (JEMRMS) Multi-Purpose Experiment Platform (MPEP), which is attached on JEM Airlock Slide Table. Then the J-SSOD with MPEP is transferred outside JEM Pressurized Module (PM) through JEM Airlock and manipulated by JEMRMS for the CubeSat deployment position and orientation. Finally, CubeSat is deployed from J-SSOD by command from IVA crew or ground operator to J-SSOD.

No on-orbit maintenance for CubeSat is planned.

**[In case launched with J-SSOD-R]**

CubeSat is cold-launched installed in Satellite Launch Case, and stowed in CTB. After arrival of the CTB to ISS, the CTB is stowed inside ISS pressurized module. Before the satellite deployment mission, Satellite Launch Case is picked up from the CTB and CubeSat is transferred from Satellite Launch Case to Satellite Deploy case. And Satellite Deploy case is installed onto JEMRMS Multi-purpose Experiment Platform (MPEP), which is attached on to the JEM Airlock Slide Table. Then the J-SSOD-R with MPEP is transferred outside JEM Pressurized Module (PM) through JEM Airlock and manipulated by JEMRMS for the CubeSat deployment position and orientation. Finally, CubeSat is deployed from J-SSOD by command from IVA crew or ground operator to J-SSOD.

No on-orbit maintenance for CubeSat is planned.

## 6. Hazard Analysis Results

### 6.1 Safety Feature

Fault Tree Analysis was performed for hazard identification as shown in Figure 6.2-1. Six kinds of standard hazard are identified for CubeSat as shown in Table 6.2-1 and summarized in Appendix B-1 with using JSC Form 1298. Also, five kinds of unique hazard are identified as shown in Table 6.1-2 and summarized in Appendix B-2 with using JSC Form 851.

## [Standard Hazards]

(1) Flammable Material

There is no propagation path of fire since the CubeSat is kept inside the CTB within the Satellite Install Case of the J- SSOD during on-orbit stowage, this feature was reviewed through the standard MIUL and MUA process. (See Appendix B-1 JSC Form 1298, 1. Ignition of Flammable Material)

(2) Material Offgassing

There are many COTS parts and materials used for the CubeSat, however, the CubeSat design contains less than 20 pounds mass of polymeric material and does not contain any of the exclusions. (See Appendix B-1 JSC Form 1298, 2. Material Offgassing in Habitable Areas Exceed Allowable Concentrations)

(3) Inadvertent Release of Dust, Toxic, or Biological Hazardous Material into Habitable Volume

Battery has electrolyte that is identified toxic material. The material was reviewed through the standard HMST process.

(4) Inadvertent Release of Sharp Particles into Habitable Volumes

The CubeSat has camera lenses, filters, and cover glass for solar cells as shatterable materials.

**<Camera lenses and filters>**

Camera lenses are non-stressed (no delta pressure) and recessed, which reduce the risk of breakage on-orbit. (See Appendix B-1 of BIRDS5-UNQ-01) The filter is contained inside the camera barrel.

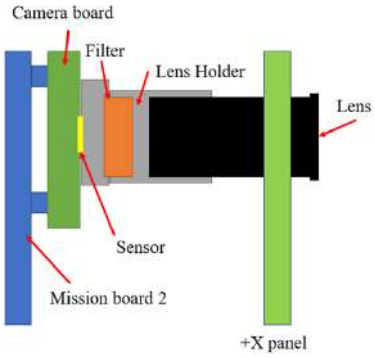



**Figure 6.1-1 Configuration of Multi Spectral Camera**

**<Cover glass of solar cells>**

The CubeSat is kept inside the Satellite Install Case of the J-SSOD. Cover glass of solar cells are non-stressed (no delta pressure), and not exposed IVA crew. J-SSOD HR (No. J-SSOD-03) and associated NCR-JAXA-JSSOD-03 are already approved with EVA No Touch Area (NTA) on the J-SSOD satellites.

In addition, the CubeSats were tested under the predicted launch vibration environment. (See Appendix B-2 BIRDS5-UNQ-01 Structural Failure)

(5) IVA Crew Exposure to Mechanical Hazards and Translation Path Obstructions

There are no potential sharp edges on the surfaces of the CubeSat. The CubeSat is kept inside the Satellite Install Case of the J-SSOD. The touch test was conducted to verify that there are no sharp

edges on the surfaces of deploying direction.
In addition, there is no planned and contingency scenario for EVA crew to access the CubeSat. J-SSOD Hazard Report (No. J-SSOD-03) and associated NCR-JAXA-JSSOD-03 are already approved with no touch area on the J-SSOD satellites. (See Appendix B-1 JSC Form 1298, 5. IVA Crew Exposure to Mechanical Hazards and Translation Path Obstructions)

(6) IVA Crew Exposure to Touch Temperature Exceedances
N/A – There are no sources of heating and/or cooling in the CubeSat design. The CubeSat remains deactivated until the deployment from ISS by three deployment switches. It is not credible to exceed the temperature over 45degC at the outer panel with the maximum expected power consumption (4.8W).

[Note for EVA] J-SSOD Hazard Report (No. J-SSOD-03) and associated NCR-JAXA-JSSOD-03 are already approved with no touch area on the J-SSOD satellites.

(7) IVA Crew Exposure to Light Amplification by Stimulated Emission of Radiation (LASER) and/or Incoherent Electromagnetic Radiation Emissions.
N/A – The CubeSat has no LASERs and/or sources of incoherent electromagnetic radiation.

(8) IVA Crew Exposure to Noise Limit Exceedances
N/A – The CubeSat has no noise sources.

(9) Injury/Damage as a Result of Improperly Bonded and Grounded Equipment
N/A – No electrical power or data interface is provided to the CubeSat while on ISS and all electrical power is wholly contained within the CubeSat and J-SSOD.

(10) Injury/Damage as a Result of Improper Power Distribution Circuitry and Circuit Protection Devices
The CubeSat does not interface with ISS power and is unpowered until deployment. There are SepSW2, DCDC converters on the HOT side of the battery, and SepSW4 on the RTN side to serve as inhibits. Appropriate isolation was provided between the battery and the first inhibits. (See BIRDS5-STD-Attachment-02) The CubeSat provides safety inhibits to prevent inadvertent powering during IVA phases (including stowage and installation). (See Appendix B-1 JSC Form 1298, 10. Injury/Damage as a Result of Improper Power Distribution Circuitry and Circuit Protection Devices.) The mission payload contains a high voltage part of 32V or higher. (See Appendix B-2 BIRDS5-UNQ-05 Electric Shock)

(11) Mating and Demating of Energized Connector
N/A – No planned mating or demating of low powered CubeSat connectors.

(12) Non-Ionizing Radiation Interference

The EMI compatibility Test was not performed since the CubeSat is de-activated with three inhibits in place. This is approved by TIA process. (See Appendix B-1 JSC Form 1298, 12. Non-Ionizing Radiation Interference). Regarding the RF radiation hazard to both ISS hardware and ISS crew, the RF frequency, power level, and field strength for UHF ([redacted] MHz) are OE-14-002 and SSP50005 (evaluated by SRAG sheet) compliant. VHF ([redacted] MHz) is satisfied SSP50005 compliant but not satisfied OE-14-002, so BIRDS5-UNQ-03 is drafted.

**<FET activation under radiated electric field>**

On the other hand, there is likelihood to activate FET and transistor under radiated electric field in ISS. But, since all FETs and transistor involved in the inhibit activate at more than 0.1 Volts, they are acceptable. Refer to STD-12, BIRDS5-STD-Attachment-01 and JDX-2020277.

**<Magnetic Field induced by magnet, magnetic torquers>**

BIRDS-5 has 16 magnets, but the maximum magnitude of the magnetic field is 2.27 Gauss which meets the requirement (less than 3.16 gauss at 7 cm distance from the surface of equipment defined on SSP30237), so there is no hazard (See 3.2.2 Magnetic Field Analysis).

(13) Injury/Damage as a result of Rotating Equipment Failure

N/A – No rotating equipment is used in the CubeSat design.

(14) Injury/Damage as a result of Sealed Container Failure

N/A – No sealed containers (inherently pressurized) are used in the CubeSat design.

## [Unique Hazards]

(1) Structural Failure

Since the CubeSat is launched in the CTB with cushion material, the launch environment is structurally moderate. However, based on the FTA, if a structural failure occurs in the Satellite Install Case of the J-SSOD, the direction of the deployment may be affected due to inadvertent contact between the CubeSat and the case, which may result in a collision of the deployed CubeSat against the ISS structure. Thus, the unique hazard is provided to control this potential hazard.

(2) Battery Leakage / Rupture

COTS rechargeable NiMH Batteries are used. The Battery Description Form is submitted and reviewed by JAXA battery expert.

(3) Exposure of the ISS to Excessive Levels of EMI radiation and RF radiation

Regarding the RF radiation hazard to both ISS hardware and ISS crew, the RF frequency, power level, and field strength for UHF ([redacted] MHz) are OE-14-002 and SSP50005 (evaluated by SRAG sheet) compliant. VHF ([redacted] MHz) is satisfied SSP50005 compliant but not satisfied OE-14-002, so BIRDS5-UNQ-03 is drafted and controlled by 3 inhibits.

(4) Impact / Collision to ISS due to inappropriate CubeSat deployment from J-SSOD by inadvertently-deployment

If the antennas are inadvertently deployed inside Satellite Install Case, the 1U CubeSat antennas may invade the envelope of the ±Z side, 2U CubeSat antenna may invade the envelope of the -Z side. Therefore, we use two strings to fix the antenna, respectively. By controlling the inadvertent deployment, the antenna will not invade the envelope of the ±Z side. The deployment mechanism of the antenna is controlled according to BIRDS5-UNQ-04.

Even if the heat cutter is powered immediately after the release from J-SSOD (immediately after the inhibit is released) and the antennas are inadvertent deployed, the distance between the CubeSats and, between the CubeSat and J-SSOD will be sufficiently far apart in the time it takes for the wire to be cut.

Assuming CubeSat mass: 2.66 kg (2U size maximum) and total separation force: 1.08 N (the minimum value defined in JX-ESPC-101132E), and the time of force application is 0.2 s, the relative velocity v for adjacent CubeSat is

$$v=F/m*t=1.08/2.66*0.2=0.08m/s$$

This value is less than 0.77 m/s which is the lower limit of J-SSOD release speed (JX-ESPC-101132-D1 Table 4.3.1.2.2-1). The time until the CubeSat's heat cutter is turned on and the antenna is deployed is about 5 s.

Therefore, the relative distance x is

$$x = 0.40 \text{ m}$$

Since the Z directional antenna length of the CubeSat is about 0.34 m, there will be no interference with the JSSOD or the adjacent CubeSat even if the antenna is mis-deployed immediately after ejection.

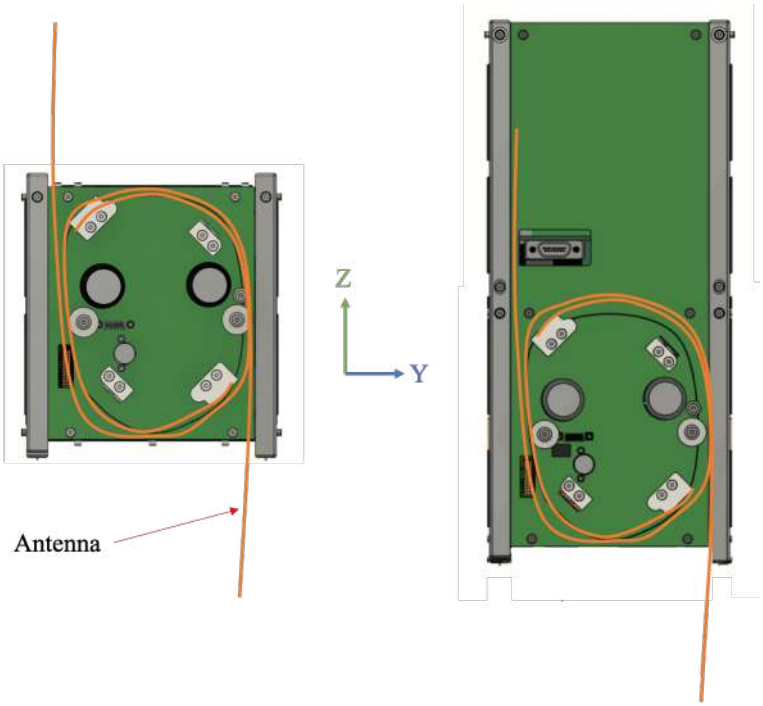



**Figure 6.1-3 Location of inadvertent deployed antenna**

(It shows the state when inadvertent deployment occurs)

(5) Electric Shock

If the ISS crew touches high-voltage parts caused by inadvertent power-on, there is a risk of electric shock or injury. BIRS5-UNQ-05 is drafted and controlled by 3 inhibits.

NOTE: The high voltages are NOT applied in the +Z panel of PINO module for the following two reasons.

-The high voltages component board is covered by a Kapton sheet. (Figure 6.1-4)

-There is enough space for insulation between the high voltages' component boards and +Z panel of PINO module. (Figure 6.1-5)

The high-voltage module is covered with Kapton tape and connected to the semiconductor by a coaxial

high voltage cable (Max 18kV DC. The high voltage is applied to the lower part of the semiconductor. The distance between the semiconductor detector and the satellite +Z plate is 1 mm or more and is enough space for insulation.

+Z panel of PINO is connected to the ground. PINO monitors the high-voltage output as house-keeping

data. If the high-voltage output is short-circuited to the +Z plane (GND), this high-voltage output monitor

will drop. However, no such status was detected when 2kV was applied.



**Figure 6.1-4 Insulation for High voltage generation parts**



**Figure 6.1-5 High voltage parts in the sectional view of PINO module**

## 6.2 Safety Issue

All identified hazard causes are well controlled as shown in the standard hazard report and the unique hazard reports. There is no non-conformance identified.

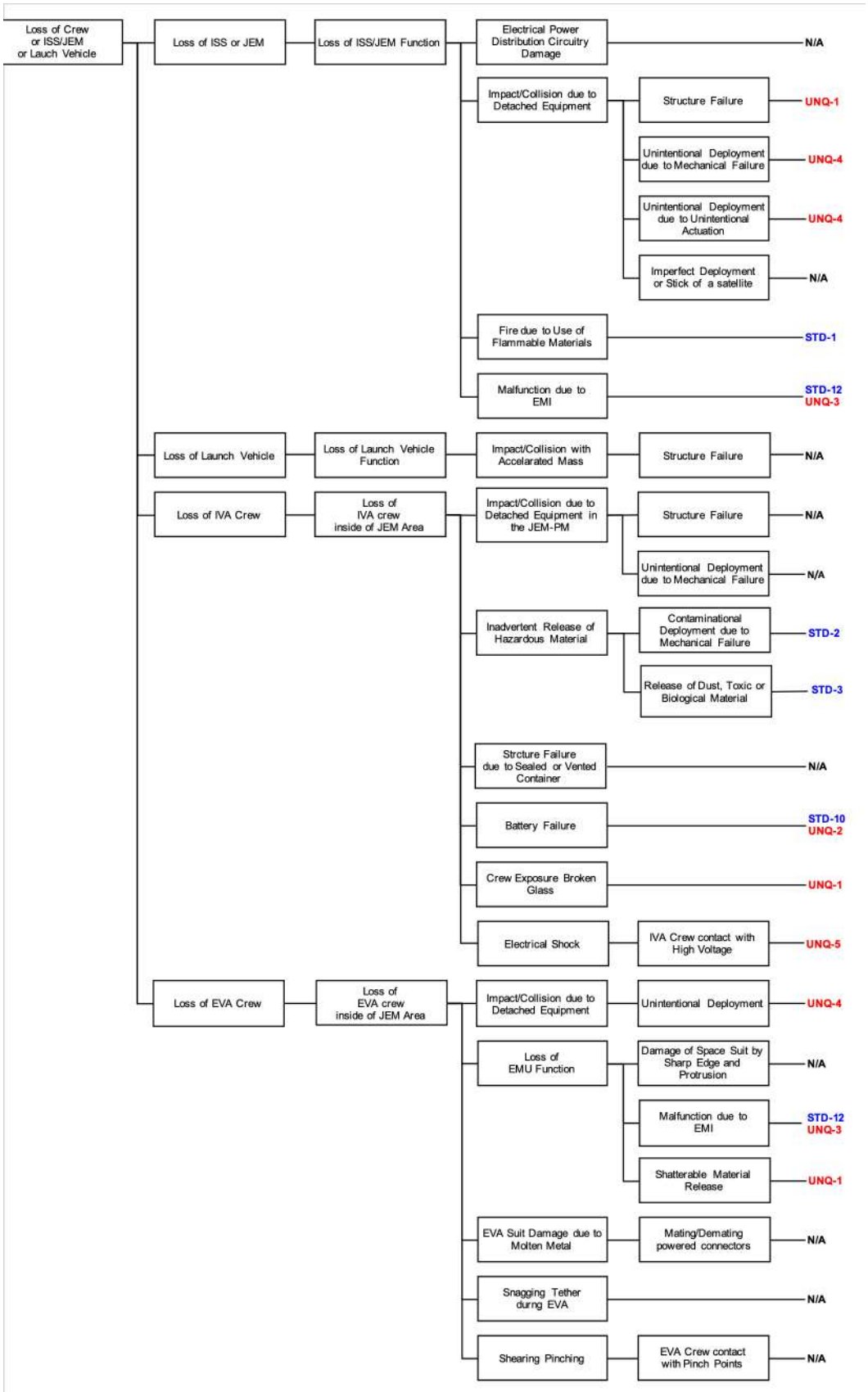



**Figure 6.2-1 Result of FTA (1/2)**

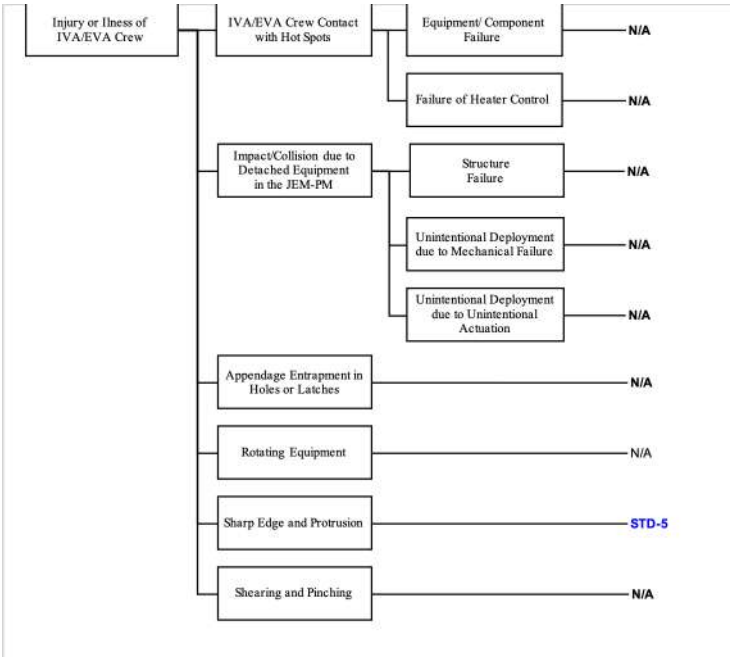



**Figure 6.2-1 Result of FTA (2/2)**

## Table 6.2-1 Standard Hazard Report List

| Hazard Number | Title | | Identified Hazard (subsystem and phase) | | | | | | Hazard Report Applicability | | Safety Feature Description [N/A] Rationale for Not Aplicable or [UNQ HR] Reason for transferring to UNQ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Identified subsystems (parts in the item list) | Launch | Stow | Operation | Return | Disposal | STD HR | UNQ HR | |
| 1 | Ignition of Flammable Material | | all Items | √ | √ | √ | | | App. | N/A | |
| 2 | Material Offgassing | | all Items | √ | √ | √ | | | App. | N/A | |
| 3 | Inadvertent Release of Dust, Toxic, or Biological Hazardous Material | | Experiment regents | √ | √ | √ | | | App. | N/A | |
| 4 | Inadvertent Release of Sharp Particles | | Camera lens and Cover glasses | √ | √ | √ | | | N/A | App. | The CubeSat has camera lenses and cover glass for solar cells as shatterable materials. See UNQ-1 |
| 5 | IVA Crew Exposure to Mechanical Hazards, etc | Sharp Edges, Corners, Holes, Etc | all Items | | | √ | | | App. | N/A | |
| | | Translation Path Loads | | | | | | | N/A | N/A | The CubeSat is not a device exposed inside the pressurization module. (This is evaluated on the J-SSOD side) |
| | | Translation Path Protrusions and Entanglements | | | | | | | N/A | N/A | The CubeSat is not a device exposed inside the pressurization module. (This is evaluated on the J-SSOD side) |
| 6 | IVA Crew Exposure to Touch Temperature Exceedances | | | | | | | | N/A | N/A | There are no sources of heating and/or cooling in the CubeSat design. The CubeSat remains deactivated until the deployment from ISS by three deployment switches. |
| 7 | IVA Crew Exposure to LASER / Incoherent Emissions | LASER Emissions | | | | | | | N/A | N/A | The CubeSat has no LASERs. |
| | | Incoherent Emissions | | | | | | | N/A | N/A | The CubeSat has no sources of incoherent electromagnetic radiation. |
| 8 | IVA Crew Exposure to Noise Limit Exceedances | | | | | | | | N/A | N/A | The CubeSat has no noise sources. |
| 9 | Improperly Bonded and Grounded Equipment | | | | | | | | N/A | N/A | No electrical power or data interface is provided to the CubeSat while on ISS and all electrical power is wholly contained within the CubeSat and J-SSOD. |
| 10 | Improper Power Distribution Circuitry and Circuit Protection Devices | | Inhibit circuit | √ | √ | √ | | | App. | App. | UNQ-05 is drafted and controlled by 3 inhibits. See UNQ-5 |
| 11 | Mating and Demating of Energized Connectors | | | | | | | | N/A | N/A | No planned mating or demating of low powered CubeSat connectors. |
| 12 | Non-Ionizing Radiation Interference | Electromagnetic Emissions | all powered Items | √ | √ | √ | | | App. | N/A | |
| | | Radio Frequency Transmitter | Communication system | √ | √ | √ | | | App. | App. | UHF satisfies SSP5005 and OE-14-002. VHF satisfies SSP5005, but not OE-14-002. |
| 13 | Injury/Damage as a Result of Rotating Equipment Failure | | | | | | | | N/A | N/A | No rotating equipment is used in the CubeSat design. |
| 14 | Injury/Damage as a Result of Sealed Container Failure | | | | | | | | N/A | N/A | No sealed containers (inherently pressurized) are used in the CubeSat design. |

1 ページ

## Table 6.2-2 Unique Hazard Report List

| Hazard Report No. | Hazard Report Title |
|---|---|
| BIRDS5-UNQ-01 | Structure Failure |
| BIRDS5-UNQ-02 | Battery Leakage / Rupture |
| BIRDS5-UNQ-03 | Exposure of the ISS to Excessive Levels of EMI radiation and RF radiation |
| BIRDS5-UNQ-04 | Impact / Collision to ISS due to inappropriate CubeSat deployment from J-SSOD by inadvertently-deployment |
| BIRDS5-UNQ-05 | Electric Shock |

**Table 6.2-3 RF radiation**

| Freq. [MHz] | Transmit Power [W] (*2) | Antenna Gain [dBi] | Max. Radiation Power [W] | Electrical Field Strength [V/m] (*1) | Criteria (*3) | |
|---|---|---|---|---|---|---|
| | | | | | Max. Output Power [W] | Electrical Field Strength [V/m] (*1) |
| [redacted] | 0.5 | 2 | 0.79 | 4.88 | 0.075 | 1.55 |
| [redacted] | 0.8 | 0.5 | 0.90 | 5.19 | 7 | 19 |

(*1) at 1[m] away from the source.

(*2) The transmitter capable max output.

(*3) Evaluations for the human body are evaluated using the SRAG evaluation sheet (Confirmation of conformity to SSP50005).

**Note:**

- **If Electrical Field Strength has to be calculated, the following equation can be used.**

$$\text{Electrical Field Strength[V/m]} = \frac{\sqrt{30 \times 10^{(\text{Antenna Gain[dBi]}/10)} \times \text{Transmit Power[W]}}}{1[m]}$$

- **Max. Radiation Power is so called Effective Isotropic Radiated Power (EIRP). It can be calculated by the following equation.**

$$\text{EIRP[W]} = 10^{[\{10\log(\text{Transmit Power[W]})+\text{Antenna Gain[dBi]}\}/10]}$$

# Appendix A

# Abbreviation and Acronyms

# List of Abbreviations and Acronyms

| | |
|---|---|
| BPB | Back Plane Board |
| COTS | Commercial Off-The-shelf |
| COM | Communication |
| CTB | Cargo Transfer Bag |
| DC/DC | Direct Current to Direct Current |
| EPS | Electrical Power System |
| EVA | Extra-Vehicular Activity |
| FAB | Front Access Board |
| ISS | International Space Station |
| IVA | Inter-Vehicular Activity |
| J-SSOD | JEM Small Satellite Orbital Deployer |
| J-SSOD-R | JEM Small Satellite Orbital Deployer Resuppliable |
| JEM | Japanese Experiment Module |
| JSC | Johnson Space Center |
| MIUL | Materials Identification Usage List |
| MPEP | Multi-Purpose Experiment Platform |
| MUA | Material Usage Agreement |
| NASA | National Aeronautics and Space Administration |
| OBC | On-Board Computer |
| PINO | Particle Instrument for Nanosatellite |
| PSRP | Payload Safety Review Panel |
| RF | Radio Frequency |
| SAR | Safety Assessment Report |

# Appendix B-1

# Standard Hazard Report

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |
| **Flight Applicability: HTV-X, Cygnus or Dragon** | **Report POC: Yamauchi Takashi** |
| **Interfaces: No direct interface with ISS, only with Deployer(J-SSOD) which interfaces with JEM Airlock and JEMRMS** | |

**Hardware *Name (include part number(s))***

**Satellite Name: BIRDS-5-FM-01**
**P/N: BIRDS-5-FM-01**

| APPROVAL | HARDWARE ORGANIZATION Printed Name, Signature, Date | ISS SAFETY REVIEW PANEL (ISRP) Printed Name, Signature, Date |
|---|---|---|
| Phase I | - | N/A |
| Phase II | - | - |
| Phase III | Takshi Yamauchi 2026.9.15 | |

*Signatures about are effective for all the following pages.*

| | | | |
|---|---|---|---|
| **HR #: BIRDS5-STD-01** | | | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | | | **Status:** |
| **Phase: Phase III** | | | **Revision Date: September 15 2026** |
| | **CONTROL(S):** | √ | **VERIFICATION(S):** |

## 1. Ignition of Flammable Material

App. (STD only)

**Hazard Description**: Fire may occur due to improper material selection in the presence of an ignition source. Use of material that does not meet International Space Station (ISS)/Visiting Vehicle (VV) flammability requirements can lead to fatal injury to the crew and/or damage to the ISS/VV through fire, smoke, and heat.

**[UNQ HRが必要な条件]Transfer to UNQ HR : In cases that require any additional unique Controls/Verifications for fire propagation.**

**[選択必須] Ctl-1.1：Materials Do Not Propagate Fire in Use Configuration** — √

Materials are selected such that a fire propagation path is not created by their in use configuration.

Select of Verifications:
(In all cases) Flammability Assessment = (a)
(Optional) Flammability Test = (b)

If need, Ctl-1.2 or 1.3 be applied.

| **V-1.1(a)** | **Flammability Assessment** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to identify all non-metallic materials, their worst-case use conditions, and their flammability characteristics.<br><br>The restriction of propagation paths by covering flammable material with a non-flammable material, or by separation of flammable material or other acceptable means (please describe other means) has been identified for all flammable material.<br><br>MIUL has been reviewed and approval by JAXA M&P. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| Closed | **August 3,2021** | BIRDS5-MIUL-01 |

**and (in needed)**

| **V-1.1(b)** | **Flammability Test** | |
|---|---|---|
| Test<br>Verify Once | Flammability testing (worst-case exposure conditions and representative use thicknesses and product form) was performed on the non-metallic material and it has been reviewed and approval by JAXA M&P. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**Option (Select with Ctl-1.1) [必要に応じて選択] Ctl-1.2：Ops Control (NASA OCAD) Stowage of Flammable Material**

When not in use, flammable materials will be stowed in non-flammable containers or compartments. Non flammable stowage containers include CTBs, JSBs, other bags or containers made of non flammable materials, ZSR/RSR compartments, and designated stowage areas in payload or other racks/compartments without power connectors. When deployed in the open cabin for use, flammable items will be kept away from rack power outlets, UOPs/SUPs, and power strips (120V or 28V) regardless of configuration (capped, mated, open, powered, or unpowered). A guideline of 6 inches will be provided in crew training. Measurement in real-time is not required. This operational control applies to the list of items below. This list is not inclusive.
1) Plastic or Trash Bags (Ziplocs, waste bags, food packaging, etc)
2) Fabric and Foam (clothing, towels, Velcro, foam packaging material, etc)
3) Off-the-shelf plastics (camcorders, mp3 players, inflatable globe, etc) (not laptops)
4) Paper (procedure books, wipes, reading materials, pictures, post-its, etc)
5) Bungees

| **V-1.2** | **[OPS CONTROL] Stowage of Flammable Material [Crew Preference Stowage Location, not Procedure Driven** |
|---|---|
| NASA OCAD<br>Verify Once | Verification is completed once formal acceptance is provided by FOD through NASA OCAD #122607. |

**Option (Select with Ctl-1.1) [必要に応じて選択] Ctl-1.3：Ops Control (JAXA OCAD) Stowage of Flammable Material**

<Extract operational control from MIUL/MUA# and insert here.>

[Note] the examples of operational control are as follows;
- When in use, flammable material shall be kept away from ignition source.
- When in use, flammable material shall be superviced by crew.
- Usage time shall be limted less than 1 hour/day.

[Note]

| **V-1.3** | **[OPS CONTROL] Stowage of Flammable Material** | |
|---|---|---|
| JAXA OCAD<br>Verify Once | Verification is completed once formal acceptance is provided by JAXA Operation community through JAXA OCAD. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | Addendix-XX, OCM# |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 2. Material Offgassing in Habitable Areas Exceed Allowable Concentrations

App. (STD only)

**Hazard Description**: Use of offgassing products within the habitable areas may result in concentrations of gaseous contaminants reaching levels which may cause immediate illness or long term adverse effects on crew health.

**[UNQ HRが必要な条件]Transfer to UNQ HR : N/A**

**[選択必須]** **Ctl-2：Polymeric Material Selection**

Polymeric materials have been selected to minimize the evolution of toxic gaseous products from the hardware.

Select of verifications:
(Optional) Apply 20-lb Offgassing Exemption Rule = (a)
(Optional) Offgassing Assessment = (b)
(Optional) Offgassing Test = (c)

[Note] If apply some verifications in a control, refer to attachment for summary of applicated items to each verification.

√

| **V-2(a)** | **20-lb Offgassing Exemption Rule** | |
|---|---|---|
| Analysis<br>Verify Once | The design contains less than 20 pounds mass of polymeric material and does not contain any of the exclusions. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| Closed | **August 3,2021** | BIRDS5-MIUL-01 |

**and/or**

| **V-2(b)** | **Offgassing Assessment** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to identify all polymeric materials, their worst-case use conditions, and their offgassing characteristics by consulting the Materials and Process Technical Information System (MAPTIS) database, SSP 30233 or other applicable IP material process/segment specification.<br><br>MIUL has been reviewed and approval by JAXA M&P. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**and/or**

| **V-2(c)** | **Offgas Testing** | |
|---|---|---|
| Test<br>Verify Once | Offgas testing has been performed on hardware or on hardware materials lacking data in the MAPTIS database, SSP 30233 or other applicable IP material process/segment specification.<br>Offgas testing has been reviewed and approval by JAXA M&P. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 3. Inadvertent Release of Dust, Toxic, or Biological Hazardous Material into Habitable Volume

App. (STD only)

**Hazard Description**: Inadvertent release of chemicals, biological material and/or nuisance particles (dust) may result in a hazard to the crew, including irritation to the skin, eyes, and respiratory tract, and/or International Space Station (ISS) and or Visiting Vehicle (VV) hardware damage.

**[UNQ HRが必要な条件]Transfer to UNQ HR : In case that exceedance of marginal level of V-3(a).**

[選択必須]

**Ctl-3：Material Selection**

Hazardous inert, insoluble nuisance particles, such as dust generated as a result of a failure, chemical and/or biological material are contained for the appropriate hazard level.

Select of verifications:
(Optional) Meet V-3(a) criteria= (a)
(Optional for batteries) Meet Battery Common HMST = (b)

| | |
|---|---|
| **V-3(a)** | **HMST (preliminary, V-1)** |
| Analysis<br>Verify Once | The Hazardous Material Summary Table (HMST) confirms the hazardous material used in the design does not exceed any one of the applicable criteria outlined below:<br><br>(criteria)<br>(1) Toxicity Hazard Level (THL)=0;<br>(2) Biological Safety Level (BSL)=1;<br>(3) inert, insoluble particles > 10 μm with a concentration, C ≤ 10 mg/m3;<br>(4) inert, insoluble particles ≤ 10 μm with a concentration, C ≤ 3 mg/m3.<br><br>Note: This is a marginal hazard.<br><br>Note(JAXA only)： For the final V2 HMST, instead of using a VTL, the JAXA ISS Safety Office tracks closure via the JAXA internal process document JSX-2012029. Refer to the flight-specific master HMSTs posted to IHS Record #44176 |

| **Status** | **Completion Date** | **Closure Documentation** |
|---|---|---|
| | | Addendix-XX, HMST# |

**and/or**

√

| | |
|---|---|
| **V-3(b) (JAXA only)** | **HMST for Battery** |
| Analysis<br>Verify Once | Toxicity Hazard Level associated with electrolytes contained in batteries can be assessed to use Battery Common HMST. |

| **Status** | **Completion Date** | **Closure Documentation** |
|---|---|---|
| Closed | **August 3,2021** | Refer to [ Appendix G-3 ] for the Common Batteries HMST Applicable Sheet.<br><br>(Note) Leakage Hazard of Battery electrolytes is assessed in Battery Unique HR. HMST of battery electrolytes which cannot be included in Battery Common HMST will be attached to Battery Unique HR. |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 4. Inadvertent Release of Sharp Particles into Habitable Volumes

Transfer to Unique Hazard Report

**Hazard Description**:
Sharp hazards, such as glass shards or other shatterable material, greater than 50 µm may imbed in or cut/damage the eye, resulting in a disabling injury, even for a short time.

**[UNQ HRが必要な条件]Transfer to UNQ HR : In cases that require any additional unique Controls/Verifications to reduce the risk of release of sharp particles.**
**[NOTE] If require to withstand for a specificaliy load (e.g. mission load, crew load, pressure load) to the item throught all phase, the item should be assessed in UNQ HR to ensure it can withstand for a specificaliy load.**

**[制御の選択条件]Select of controls :**
**For Launch Phase, Ctl-4.1 shall be applicated.**
**For On-orbit Phase, Ctl-4.1 or 4.2 shall be applicated.**
**If need to contain when not in use, Ctl-4.3 should be applied.**
**If need to remove the containment(Ctl-4.1) during on mission, Ctl-4.4 should be applied.**

| Option [必要に応じて選択] | | | | |
|---|---|---|---|---|
| **Option [必要に応じて選択]** | **Ctl-4.1：Containment or Positive Protection** | **V-4.1(a)** | **Design of As-Built Hardware for Containment or Positive Protection** | |
| | Glass or frangible components inside of the habitable volume are contained or have positive protection to prevent the inadvertent release of sharp particles > 50µm.<br><br>Select of verifications:<br>(In all cases) Review of Design and Inspection = (a) and (b) | Analysis<br>Verify Once | An assessment of the design has been performed to ensure the glass or frangible material is contained or has positive protection. The containment or positive protection preclude crew exposure to frangible material. | |
| | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | | |
| | | **and** | | |
| | | **V-4.1(b)** | **Inspection of As-Built Hardware for Containment or Positive Protection** | |
| | | Inspection<br>Verify Each Flight | An inspection of each flight unit has been performed to ensure the hardware is in accordance with approved engineering drawings, and containment or positive protection is present. | |
| | [Note] Refer to Attachment-XX for Control Methods Summary | **Status** | **Completion Date** | **Closure Documentation** |
| | | | | |
| **Option (For On-orbit only) [必要に応じて選択]** | **Ctl-4.2：Non-Stressed Design for Optical Equipment** | **V-4.2(a)** | **Design Features** | |
| | Camera lenses or similar pieces of optical equipment are non-stressed (no delta pressure) and have design features, such as recessed lenses or otherwise protected by design, which reduce the risk of breakage on-orbit.<br><br>Select of verifications:<br>(In all cases) Review of Design and Inspection = (a) and (b) | Analysis<br>Verify Once | An assessment of the design has been performed to determine that the camera lenses or similar pieces of optical equipment are non-stressed (no delta pressure) and have design features, such as recessed lenses or otherwise protected by design, which reduce the risk of breakage on-orbit.<br>The design features are accomplished via the design features of the design camera lenses or similar pieces of optical equipment which reduce the risk of breakage on-orbit. | |
| | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | | |
| | | **and** | | |
| | | **V-4.2(b)** | **Inspection of As-Built Hardware for Design Features** | |
| | | Inspection<br>Verify Each Flight | An inspection of each flight unit has been performed to ensure the hardware is in accordance with the approved design, and design features has been verified. | |
| | [Note] Refer to Attachment-XX for Control Methods Summary | **Status** | **Completion Date** | **Closure Documentation** |
| | | | | |
| **Option (Select with Ctl-4.1 or 4.2) [必要に応じて選択]** | **Ctl-4.3：Ops Control (JAXA OCAD) Stowage of Optical Equipment** | **V-4.3** | **[OPS CONTROL] Stowage of Optical Equipment** | |
| | Camera lenses and similar pieces of optical equipment are stowed or the glass elements contained/covered, when not in use. | JAXA OCAD<br>Verify Once | Verification is completed once formal acceptance is provided by JAXA Operation community through JAXA OCAD. | |
| | | **Status** | **Completion Date** | **Closure Documentation** |
| | [Note] Refer to Attachment-XX for Control Methods Summary | | | Addendix-XX, OCM# |
| **Option (Select with Ctl-4.1) [必要に応じて選択]** | **Ctl-4.4：(JAXA only)**<br>**Ops Control (JAXA OCAD) Check of Optical Equipment** | **V-4.4 (JAXA only)** | **[OPS CONTROL] Check of Optical Equipment** | |
| | Crew shall check no breakage for glass or frangible components before removal of containment or positive protection. | JAXA OCAD<br>Verify Once | Verification is completed once formal acceptance is provided by JAXA Operation community through JAXA OCAD. | |
| | | **Status** | **Completion Date** | **Closure Documentation** |
| | [Note] Refer to Attachment-XX for Control Methods Summary | | | Addendix-XX, OCM# |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 5. IVA Crew Exposure to Mechanical Hazards and Translation Path Obstructions

App.(STD only) : Sharp Edges, Corners, Holes, Etc
N/A(STD and UNQ) : Translation Path Loads
N/A(STD and UNQ) : Translation Path Protrusions and Entanglements

**Hazard Description**: Sharp edges, protusions, and translation path obstructions may result in crew injury or entanglement.

**[UNQ HRが必要な条件]Transfer to UNQ HR : ・For EVA Crew Exposure to Mechanical Hazards and Translation Path Obstructions.**
**・In cases that require any additional unique Controls/Verifications to reduce the risk of release of Mechanical Hazards and Translation Path Obstructions.**

**[選択必須]** **Ctl-5.1:Sharp Edges, Corners, Holes, Etc**

All hardware within a pressurized module is designed to be free from sharp edges, corners, holes, etc., during all crew operations.

Select of verifications:
(In all cases (※)) Review of Drawing and Inspection = (a) and (b)
(※) For COTS Hardware which has no Drawing, Inspection Only = (b)

√

| **V-5.1(a)** | **Drawing Analysis for Sharp Edges, Corner Protection, Holes, etc.** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to comply the requirement to the intent of SSP51721, Secion 4.9.3, or IRD/ICD, the hardware is free from crew hazards created by sharp edges, corners, holes, etc. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| Closed | **December 9,2021** | BIRDS5-AD-01<br>Note: Solar cell edges on the +Z plane are verified by touch testing. |

**and**

√

| **V-5.1(b)** | **Inspection of As-Built for Sharp Edges, Corner Protection, Holes, etc** | |
|---|---|---|
| Inspection<br>Verify Each Flight | An inspection of each flight unit has been performed to ensure the hardware is in accordance with the approved design, and the manufacturing of the item has not created a hazard.<br><br>Note(JAXA Only): COTS Hardware Inspection of As-Built shall be include touch test. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| Closed | **February 15,2022** | BIRDS5-SEIR-01 |

**[選択必須]** **Ctl-5.2:Translation Path Loads**

Hard mounted/rigidly attached hardware exposed in the translation path, which could result in a critical/catastrophic hazard upon failure, meet the crew-imposed minimum design and ultimate loads.

[Note]This control apply for hardware which is exposed to cabin.

| **V-5.2** | **IVA Crew Loads Assessment** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure the exposed to the translation path will withstand a design load of 556 N (125 lbf) and an ultimate load of 778 N (175 lbf) applied over a 10.16 cm X 10.16 cm (4 in. X 4 in.) area. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**[選択必須]** **Ctl-5.3:Translation Path Protrusions and Entanglements**

Hardware exposed in the translation paths are designed to minimize the possibility of entanglement or injury to crewmembers and damage to adjacent equipment.

[Note]This control apply for hardware which is exposed to cabin.

| **V-5.3** | **Translation Path Protrusions and Entanglements** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure all of the flex hoses, lines, and cables are tethered or otherwise restrained to prevent injury to crew and damage to adjacent hardware.<br>A minimum emergency translation corridor of 32 X 45 inches (81 X 114 cm) is maintained within the USOS modules and 32 x 32 inches (81 x 81 cm) within the Russian Segment. A 32 X 45 in (81 X 114 cm) corridor allows for a crewmember to reverse direction at any point along the corridor during emergency situations. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

| HR #: BIRDS5-STD-01 | System/Payload: Payload |
|---|---|
| Item Name: BIRDS5 FM | Status: |
| Phase: Phase III | Revision Date: September 15 2026 |

## 6. IVA Crew Exposure to Touch Temperature Exceedances

N/A (STD and UNQ)

**Hazard Description**: Crew contact with surfaces of excessively high or low temperatures may result in skin damage.

**[UNQ HRが必要な条件]Transfer to UNQ HR : ・For EVA Crew Exposure to Touch Temperature Exceedances**
**・In cases that require any additional unique Controls/Verifications to reduce the risk of IVA Touch Temperature Exceedances**

**Option**
**[必要に応じて選択]**

**Ctl-6.1：IVA Touch Temperatures**

Exposed surfaces in the crew IVA environment do not have surface temperatures greater than 45°C (113°F), less than 0°C (32°F), or are found to be acceptable by means of Permissible Material Temperature Calculation.

Note: If active thermal management (such as fans, heaters, etc.) is present in the design, the thermal analysis/test must address the worst-case failure scenario of the single fault tolerant design.

Select of Verifications:
(Optional) Thermal Analysis = (a)
(Optional) Thermal Testing = (b)

[Note] If apply some verifications, Refer to Attachment for Applicated items Summary, which identify for items.

| V-6.1(a) | Thermal Analysis | |
|---|---|---|
| Analysis<br>Verify Once | A thermal analysis of the design has been performed to ensure the exposed surface temperatures of the item do not exceed 45°C (113°F), are not less than 0°C (32°F) or are found to be acceptable by Permissible Materials Temperature Calculation. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | [Thermal analysis summary] Attachment- |
| | | |

**and/or**

| V-6.1(b) | Thermal Testing | |
|---|---|---|
| Test<br>Verify Once | A thermal test has been performed to ensure the exposed surface temperatures of the item do not exceed 45°C (113°F), are not less than 0°C (32°F) or are found to be acceptable by Permissible Materials Temperature Calculation. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**Option**
**[必要に応じて選択]**

**Ctl-6.2：IVA Touch Temperature Exceedances**

Exposed surfaces in the crew IVA environment have surface temperatures greater than 45°C (113°F), less than 0°C (32°F), or are found to be acceptable by means of Permissible Material Temperature Calculation.

Note: If active thermal management (such as fans, heaters, etc.) is present in the design, the thermal analysis/test must address the worst-case failure scenario of the single fault tolerant design.

Select of Verifications:
(Optional) Thermal Analysis = (a) and (b)
(Optional) Thermal Testing = (c) and (d)

And, Ctl-6.3 and/or Ctl-6.4 should be applied.

| V-6.2(a) | Thermal Analysis | |
|---|---|---|
| Analysis<br>Verify Once | A thermal analysis of the design has been performed which identified the worst case thermal extremes present on accessible surfaces. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | [Thermal analysis summary] Attachment- |
| | | |

**and**

| V-6.2(b) | Thermal/Operational Analysis | |
|---|---|---|
| Analysis<br>Verify Once | The thermal analysis of the design has indicated that a minimum wait time. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**or**

| V-6.2(c) | Thermal Testing | |
|---|---|---|
| Test<br>Verify Once | A thermal test has been performed which identified the worst case thermal extremes present on accessible surfaces. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**and**

| V-6.2(d) | Thermal/Operational Testing | |
|---|---|---|
| Test<br>Verify Once | The thermal testing has indicated that a minimum wait time. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

| HR #: BIRDS5-STD-01 | | | System/Payload: Payload | | |
|---|---|---|---|---|---|
| Item Name: BIRDS5 FM | | | Status: | | |
| Phase: Phase III | | | Revision Date: September 15 2026 | | |
| Option (Select with Ctl-6.2) [必要に応じて選択] | **Ctl-6.3：Ops Control (JAXA OCAD) Personal Protective Equipment** | | **V-6.3** | **[OPS CONTROL] Personal Protective Equipment** | |
| | Crew shall wear [the PPE] during [operation] for [the item].<br><br>[Note]<br>The PPE=<br>operation=<br>the item= | | JAXA OCAD Verify Once | Verification is completed once formal acceptance is provided by JAXA Operation community through JASA OCAD. | |
| | | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | | | Addendix-XX, OCM# |
| Option (Select with Ctl-6.2) [必要に応じて選択] | **Ctl-6.4：Ops Control (JAXA OCAD) Isolation** | | **V-6.4** | **[OPS CONTROL] Isolation** | |
| | Crew must not touch the item until a period of the [indicated time] has passed to allow it to [recover] to the [acceptable temperature].<br>This period is measured from the time that [indicated time or operation task].<br>Or, Crew must not touch the item unless the observable temperature sensor at crew contact points shows 0 to 45 ° C.<br><br>[Note]<br>indicated time=<br>recover=Warm up or Cool down<br>acceptable temperature=<br>indicated time or operation task= | | JAXA OCAD Verify Once | Verification is completed once formal acceptance is provided by JAXA Operation community through JAXA OCAD. | |
| | | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | | | Addendix-XX, OCM# |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 7. IVA Crew Exposure to Light Amplification by Stimulated Emission of Radation (LASER) and/or Incoherent Electromagnetic Radiation Emissions

N/A(STD and UNQ) : LASER Emissions Exposure

N/A(STD and UNQ) : Incoherent Light Emissions Exposure

**Hazard Description**: Crew exposure to high-intensity LASER and/or incoherent electromagnetic radiation emissions may result in biological damage to the eye or skin. Sustained damage to the eye is a common effect of exposure leading to crew incapacitation or blindness. Skin tissue destruction can also occur.

**[UNQ HRが必要な条件]Transfer to UNQ HR：Class 1M, 2M, 3R, 3B and 4 Laser or Incoherent Electromagnetic Radiation Emissions for over $10^7$ nits**

**[選択必須] (For laser)**

**Ctl-7.1：LASER Emissions Exposure**

The crew will not be exposed to high-intensity LASER emissions.

Select of Verifications:
(Optional) Use Class 1 Laser = (a)
(Optional) Use Class 2 Laser = (b)

[Note] If apply some verifications, Refer to Attachment for Applicated items Summary, which identify for items, each class, activation Phase(operation task)

| | | | |
|---|---|---|---|
| **V-7.1(a)** | **Class 1 LASER** | | |
| Analysis<br>Verify Once | An assessment has been performed to ensure the LASER design is not capable of emitting in excess of the Class 1 Accessible Emission Limit (AEL), which varies by wavelength and pulse duration,<br>OR<br>For COTS hardware only, the LASER (COTS hardware labeled as Class 1, but unable to confirm LASER strength) is enclosed via means of describe containment and features of the system to preclude crew exposure to the LASER emissions, which prohibits or limits access to the LASER radiation. | | |
| **Status** | **Completion Date** | **Closure Documentation** | |
| | | | |

**and/or**

| | | | |
|---|---|---|---|
| **V-7.1(b)** | **Class 2 LASER** | | |
| Analysis<br>Verify Once | An assessment has been performed to ensure that:<br>(a) the LASER design Continuous Wave (CW) and repetitive-pulse LASERs emissions are in the visible region of the spectrum (0.4 to 0.7 μm), and<br>(b) the LASERs can emit accessible radiant energy exceeding the Class 1 AEL for the maximum duration inherent in the LASER, but not exceeding the Class 1 AEL for any pulse duration < 0.25 s (the time estimated to blink or look away), and<br>(c) not exceeding an average radiant power of 1 mW.<br>OR<br>For COTS hardware only, the LASER (COTS hardware labeled as Class 2, but unable to confirm LASER strength) is enclosed via means of describe containment and features of the system to preclude crew exposure to the LASER emissions, which prohibits or limits access to the LASER radiation. | | |
| **Status** | **Completion Date** | **Closure Documentation** | |
| | | | |

**[選択必須] (For Incoherent Electromagnetic Radiation)**

**Ctl-7.2：Incoherent Electromagnetic Radiation Emissions Exposure**

The crew will not be exposed to incoherent electromagnetic radiation emissions.

| | | | |
|---|---|---|---|
| **V-7.2** | **Low Intensity Design** | | |
| Analysis<br>Verify Once | An assessment has been performed to ensure the Incoherent Electromagnetic Radiation design is not capable of emitting in excess of $10^7$ nits (nits=cd/$m^2$) to confirm the design is a marginal hazard. | | |
| **Status** | **Completion Date** | **Closure Documentation** | |
| | | | |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 8. IVA Crew Exposure to Noise Limit Exceedances

N/A (STD and UNQ)

**Hazard Description**: Crew exposure to noise in exceedance of acceptable sound pressure levels (SPL) may result in crewmember hearing damage or crew injury/death, as a result of communication interference or inability to detect caution and warning alarms.

**[UNQ HRが必要な条件]Transfer to UNQ HR : In cases that require any additional unique Controls/Verifications to Noise Limit Exceedances.**

[選択必須] **Ctl-8：Noise Exposure**

Hardware within the habitable volume will not generate noise that may cause injury or hearing loss, interfere with voice communications, cause fatigue, or in any other way degrade overall human-machine system effectiveness.

Select of Verifications:
(Optional) Continuous acoustic testing = (a)
(Optional) Intermittent acoustics testing = (b)
(Optional) Continuous/Intermittent acoustic analysis = (c)
(Optional) Noise is above Noise Criteria for the end item = (d)

| **V-8(a)** | **Continuous Noise Limits** | |
|---|---|---|
| Test<br>Verify Once | Continuous acoustics testing has been performed to ensure the design is compliant to SSP 51721, Section 4.9.1, or the IRD/ICD process. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**or**

| **V-8(b)** | **Intermittent Noise Limits** | |
|---|---|---|
| Test<br>Verify Once | Intermittent acoustics testing has been performed to ensure the design is compliant to SSP 51721, Section 4.9.1, or the IRD/ICD process. The noise duration is the total time that the non-integrated equipment item produces intermittent noise above the NC-40 limit during a 24-hour time period. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**or**

| **V-8(c)** | **Continuous and Intermittent Noise Limits** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed of all noise sources to ensure the is compliant to SSP 51721, Section 4.9.1, or the IRD/ICD process. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**or**

| **V-8(d)** | **Continuous Noise Limits (Integrated Vehicle System Level Acoustic Limit)** | |
|---|---|---|
| Analysis<br>Verify Once | An integrated assessment of the design has been performed of all noise sources to ensure the is compliant to SSP 51721, Section 4.9.1.5. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 9. Injury/Damage as a Result of Improperly Bonded and Grounded Equipment

N/A (STD and UNQ)

**Hazard Description**: In the event of failure of electrical insulation, an overcurrent event may lead to generation of hazardous products due to pyrolysis, ignition source, touch temperature, or propagation of hazard, which may result in critical/catastrophic injury to the crew and/or critical/catastrophic damage to the International Space Stations (ISS) or Visiting Vehicle (VV).

**[UNQ HRが必要な条件]Transfer to UNQ HR：Input or internal power circuits is greater than 32V.**

**[選択必須]** **Ctl-9.1：Bonding**

All electrical equipment is properly bonded to comply with ISS Bonding requirement of SSR 30245.

Select of verifications:
(In all cases) Analysis and Inspection = (a) and (b)

| | | |
|---|---|---|
| **V-9.1(a)** | **Bond Path Analysis** | |
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure:<br>・the item provides a fault current return path back to source (or electrical/mechanical interface) and each electrical bond within the return paths is a Class H bond<br>・the interface bonding methodology is compatible with the electrical/mechanical interface to ISS. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | [Bond Path Analysis] Attachment/Figure- |
| | | |

**and**

| | | |
|---|---|---|
| **V-9.1(b)** | **Inspection of As-Built Hardware Bonding Surfaces/Paths** | |
| Inspection<br>Verify Once | An inspection of each flight unit has been performed to ensure the bond paths are in accordance with the approved design.<br>The inspection ensured:<br>1) The as-built provides a fault current return path back to the source (or electrical/mechanical interface);<br>2) Each as-built electrical bond within the return path is a class H bond. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**[選択必須]** **Ctl-9.2：Grounding**

All electrical equipment is properly grounded to comply with ISS Grounding requirement of SSP 30240.

Select of verifications:
(In all cases) Analysis and Inspection = (a) and (b)

| | | |
|---|---|---|
| **V-9.2(a)** | **Grounding Analysis** | |
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure:<br>1) Electrical power consuming input power and data/signal isolation prevents inadvertent fault current return paths which may propagate a fault;<br>2) The item, which contain electrical power sources or electrical power converters that distribute power or signals external to the item, has outputs that are properly isolated from the electrical power inputs to prevent inadvertent fault current return paths which may propgate a fault;<br>3) The electrical power sources or converters, which route electrical power external to the item or provide electrical power to other external users, have electrical power outputs that are properly referenced to chassis/structure to facilitate a fault current return path back to the source to allow the proper operation of overcurrent protection devices.<br><br>and, the interface grounding/isolation is compatible with the ISS electrical power and signal interface. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | [Grounding Analysis] Attachment/Figure- |
| | | |

**and**

| | | |
|---|---|---|
| **V-9.2(b)** | **Inspection of As-Built Hardware Grounding Scheme** | |
| Inspection<br>Verify Once | An inspection of each flight unit has been performed to ensure the grounding and isolation schemes are in accordance with the approved design.<br>This inspection has ensured:<br>1) The as-built electrical power consuming input power and return is isolated and data/signal grounding/isolation scheme prevents ground loops and inadvertent fault current return paths;<br>2) The as-built hardware, which contain electrical power sources or electrical power converters that distribute power or signals external to the item, have outputs that are properly isolated from the electrical power inputs;<br>3) The as-built electrical power sources or converters, which route electrical power externally or provide | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 10. Injury/Damage as a Result of Improper Power Distribution Circuitry and Circuit Protection Devices

App. (STD and UNQ)

**Hazard Description**: Improper fusing, circuit protection, current limits, wire sizing and derating may result in critical/catastrophic injury to the crew and/or critical/catastrophic damage to the International Space Stations (ISS) or Visiting Vehicle (VV).

**[UNQ HRが必要な条件]Transfer to UNQ HR：Input or internal power circuits is greater than 32V.**

**Option**
**[必要に応じて選択]**

**Ctl-10.1：Proper Circuit Protective Devices and Wire/Cable Derating**

All electrical equipment contain properly sized circuit protective devices, wire and cable to comply with SSP 51721 section4.3.1 or TM102179 .

Select of verifications:
(Optional) Wire Length less than Six Inches between the battery power source and first power function = (a)
(Optional) Wire/Cable Derating Analysis =(b) and (c)

√ **V-10.1(a)** **Wire Length less than Six Inches when Battery Powered**

Analysis
Verify Once

When the item only utilizes battery power for all any phases of launch, on-orbit operations, and landing, 18 AWG or smaller wire (non-silicone insulation) in length less than 6" between the battery power source and first power function, and the item does not contain safety critical function will not require additional verification provided to JAXA.

Batteries will not cause damage to this wiring, or require additional circuit protection to prevent overheating/pyrolization of the wiring, and will not cause a touch temperature hazard.

| Status | Completion Date | Closure Documentation |
|---|---|---|
| Closed | **December 9. 2021** | [Wire size and length analysis]BIRDS5-STD-Attachment-02 |
| | | |

**or**

√ **V-10.1(b)** **Protective Devices Sizing and Wire/Cable Derating Analysis**

Analysis
Verify Once

Circuit protection and wire sizing analysis of the design has been performed to ensure that the maximum available upstream and/or worst-case current conditions are assessed and appropriately controlled. All analysis was based on IVA environmental conditions, in the specific flight system configuration (single or bundled wiring).

See Protective Devices Sizing and Wire/Cable Derating Analysis <File Name.File Extension>

| Status | Completion Date | Closure Documentation |
|---|---|---|
| Closed | **September 15. 2023** | [Wires and Circuit Protection Devices analysis] BIRDS5-STD-Attachment-03 |
| | | |

**and**

√ **V-10.1(c)** **Inspection of As-Built Hardware Circuit Protective Devices and Wire/Cable**

Inspection
Verify Once

An inspection of each flight unit has been performed to ensure the hardware is in accordance with the approved design and proper installation of fusing and protective devices are verified.

| Status | Completion Date | Closure Documentation |
|---|---|---|
| Closed | **September 15, 2023** | BIRDS5-STD-Attachment-03 |

| HR #: BIRDS5-STD-01 | System/Payload: Payload |
|---|---|
| Item Name: BIRDS5 FM | Status: |
| Phase: Phase III | Revision Date: September 15 2026 |

## 11. Mating and Demating of Energized Connectors

N/A (STD and UNQ)

**Hazard Description**: Mating and demating of energized connectors may result in critical/catastrophic injury to the crew. Molten metal may be generated by mating and demating of connectors under load and lead to crew eye or skin injury.

**[UNQ HRが必要な条件]Transfer to UNQ HR：Input or internal power circuits is greater than 32V (for electrical shock) or 65A (for molten metal)**

**Option [必要に応じて選択]**

**Ctl-11.1：Low Power Design Features**

Electrical equipment design protects the crew from exposure to a maximum continuous current to 3A and less with an open circuit voltage to no greater than 32V.

| V-11.1 | Low Power Exception Analysis | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure the item has current limitation design features or upstream circuit protection that limits the maximum continuous current to 3A or less with an open circuit voltage no greater than 32V. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | [Mating / Demating Power Connector analysis] Attachment- |
| | | |

**Option [必要に応じて選択]**

**Ctl-11.2：Molten Metal Hazard**

Electrical equipment design protects the crew from exposure to a maximum continuous current greater than 3A but less than 65A and an open circuit voltage no greater than 32V.

For Molten Metal Hazard, Electrical equipment connectors employ the all following：
[Connector Design Features]
The design features that completely enclose or shroud the pins and sockets during mating/demating.

[Protection from Powered Side]
The design provides protection of the power side from debris/inadvertent shorting.

[One Verifiable Upstream Inhibit ]
The design feature provides physical interruption of power to the connector. The physical interruption must be able to be checked or verified by the crew or ground personnel.

Select of verifications:
(In all cases) Select ALL : (a) ～ (d)

And, Ctl-11.3 should be applied.

| V-11.2(a) | Molten Metal Hazard Analysis | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure the maximum continuous current is greater than 3A but less than 65A and an open circuit voltage no greater than 32V. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | [Mating / Demating Power Connector analysis] Attachment- |
| | | |

**and**

| V-11.2(b) | Molten Metal Hazard Connector Design Features／Protection from Powered Side | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure connectors employ design features：<br>・The completely enclose or shroud the pins and sockets during mating/demating. A "scoop proof" like design is one where the power between the pins and the sockets is disconnected prior to the conductors being exposed to the outside environment.<br>・The power side of the connectors are terminated in sockets. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**and**

| V-11.2(c) | Inspection of As-Built Hardware Connector Design Features／Powered Side of Connectors Terminated in Sockets | |
|---|---|---|
| Inspection<br>Verify Once | An inspection of each flight unit has been performed to ensure the hardware is in accordance with the approved connector design features, the power side of the connectors are terminated in sockets. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | [Mating / Demating Power Connector analysis] Attachment- |
| | | |

**and**

| V-11.2(d) | Molten Metal Hazard One Verified Upstream Inhibit | |
|---|---|---|
| Analysis<br>Verify Once | An assessment of the design has been performed to ensure the power circuit has at least one physical interruption (i.e. inhibit) which is capable of being verified at the time the interruption is applied. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**Option (Select with Ctl-11.2) [必要に応じて選択]**

**Ctl-11.3：Ops Control (JAXA OCAD) Molten Metal**

Prior to mating/demating of the energized connector, the power shall be removed per attachment.

[Note] Refer to attachment -XXXXX

| V-11.3 | [OPS CONTROL] Molten Metal Hazard PS28 | |
|---|---|---|
| JAXA OCAD<br>Verify Once | Verification is completed once formal acceptance is provided by JAXA Operation community through JAXA OCAD. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | Addendix-XX, OCM# |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 12. Non-Ionizing Radiation Interference

App.(STD only) : Electromagnetic Emissions
App.(STD and UNQ) : Radio Frequency Transmitter

**Hazard Description**: Non-ionizing radiation from electrical, electronic, or electromechanical devices can cause interference with International Space Station (ISS) or Visiting Vehicle (VV) critical systems necessary for daily ISS quiesent and VV proximity operations. Interference to critical systems could either be intermittent in nature or result in halting of system operations, which may resulting in a safety hazard for ISS or operational delays.

**Transfer to UNQ HR: For Non-Ionizing Radiation Interference, in cases that require any additional unique Controls/Verifications.**
**For Radio Frequency Transmitter, in cases that not meet SSP51721 requirement.**

| Option [必要に応じて選択] | | | | | |
|---|---|---|---|---|---|
| | **Ctl-12.1: Electromagnetic Emissions are Limited** | √ | **V-12.1** | **Compliance to the IRD/ICD or SSP30237/SSP30243** | |
| | Electronic, electrical, and electromagnetic equipment and subsystems are designed to meet emissions and susceptibilities. | | Analysis and/or Test Verify Once | Compliance is documented in the IRD/ICD process or SSP30237/SSP30243.<br><br>If susceptibility testing is not conducted, provide the rationale document or describe in attachment (failure of the item to function or erratic/unintended functioning of the item does not create a hazard, or The IRD/ICD process determines that susceptibility testing is not required, etc.). | |
| | | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | Closed | **December 9,2021** | [EMC assessment summary] BIRDS5-STD-Attachment-01 |
| | | | Closed | **September 15. 2023** | BIRDS5-SAR-01 3.2.2 |
| Option [必要に応じて選択] | **Ctl-12.2: Electromagnetic Emissions are Limited (with TIA)** | √ | **V-12.2** | **Tailoring/Interpretation Agreement (TIA)** | |
| | The designs (which not meet emissions and susceptibilities) for Electronic, electrical, and electromagnetic equipment and subsystems are acceptable by TIA assessment.<br><br>If needed, Ctl-12.3 be applied. | | Analysis Verify Once | A violation was noted during the IRD/ICD process, a Tailoring/Interpretation Agreement (TIA) was required and an operational control was identified.<br><br>If susceptibility testing is not conducted, provide the rationale document or describe in attachment (failure of the item to function or erratic/unintended functioning of the item does not create a hazard, or The IRD/ICD process determines that susceptibility testing is not required, etc.). | |
| | | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | Closed | **May 30. 2014** | [EMC assessment summary] Attachment-G-4 |
| | | | | | |
| Option (Select with Ctl-12.2) [必要に応じ | **Ctl-12.3: Ops Control (JAXA OCAD) Non-Ionizing Emission** | | **V-12.3** | **[OPS CONTROL] Non-Ionizing Emission** | |
| | <Extract operational control from TIA # and insert here.> | | JAXA OCAD Verify Once | Verification is completed once formal acceptance is provided by JAXA Operation community through JAXA OCAD. | |
| | | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | | | Addendix-XX, OCM# |
| [選択必須] (For Radio Frequency Transmitter) | **Ctl-12.4: Radio Frequency Transmitter are Limited** | √ | **V-12.4** | **Compliance to the IRD/ICD and SSP51721** | |
| | Radio Frequency Transmitter are designed to meet emissions. | | Analysis and/or Test Verify Once | Compliance is documented in the IRD/ICD process and SSP51721, Section 4.3.7, 4.3.8, to confirm the emission design is a marginal hazard. | |
| | | | **Status** | **Completion Date** | **Closure Documentation** |
| | | | Closed | **February 22, 2022** | BIRDS5-SAR-02, Table6.2-3 |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 13. Injury/Damage as a Result of Rotating Equipment Failure

N/A (STD and UNQ)

**Hazard Description**:
Failure and liberation of rotating elements at a high velocity may result in critical/catastrophic injury to the crew and/or critical/catastrophic damage to the International Space Stations (ISS) or Visiting Vehicle (VV).

**Transfer to UNQ HR: The rotating equipment has kinetic energy more than 14,240 ft-lbs (19,307 Joules) or not containmnet.**

**Option [必要に応じて選択]**

**Ctl-13.1: Limiting the Potential Energy of Rotating Equipment**

Rotating equipment designs limit kinetic energy for mass release and/or have containment for liberated rotating elements.

Select of verifications:
(In all cases) Containment Design and Potential Energy Analysis= (a) and (b)

| **V-13.1(a)** | **Containment Capabilities Assessment** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment has been performed to ensure the design provides containment or has positive protection from all rotating equipment. The containment or positive protection is accomplished via containment/positive protection and the features of the system which preclude crew exposure to rotating equipment. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |
| | | |

**and**

| **V-13.1(b)** | **Potential Energy (Non-Fracture Critical) Analysis** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment has been performed to ensure the rotating equipment has kinetic energy less than 14,240 ft-lbs (19,307 Joules).<br><br>Note: Rotating parts that do not exceed 200 mm in diameter and 8000 rpm in speed meet this requirement, therefore, no calculation is required. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |
| | | |

**Option (For HDD) [必要に応じて選択]**

**Ctl-13.2: Limiting the Potential Energy of Rotating Equipment (HDD)**

HDD have integrity containment for liberated rotating elements based on UL or IEC ratings.

| **V-13.2** | **Unmodified Computer Disk Drive Assessment** | |
|---|---|---|
| Analysis<br>Verify Once | An assessment has been performed to ensure that the contains a rotating hard drive within a computer chassis that meets UL or IEC ratings. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | UL or IEC ratings: <Provide the actual rating.> |
| | | |

| | |
|---|---|
| **HR #: BIRDS5-STD-01** | **System/Payload: Payload** |
| **Item Name: BIRDS5 FM** | **Status:** |
| **Phase: Phase III** | **Revision Date: September 15 2026** |

## 14. Injury/Damage as a Result of Sealed Container Failure

N/A (STD and UNQ)

**Hazard Description**: Structural failure of a sealed container and liberation of fragments at a high velocity may result in critical/catastrophic injury to the crew and/or critical/catastrophic damage to the International Space Stations (ISS) or Visiting Vehicle (VV). A sealed container is defined as any single, independent (not part of a pressurized system) container, component, or housing that is sealed to maintain an internal non-hazardous environment and that has a stored energy of less than 14,240 foot-pounds (19,310 Joules) and an internal pressure of less than 100 psia (689.5 kPa).

**Transfer to UNQ HR: ：Internal pressure value is greater than 100 psia (689.5 kPa) or stored energy is exceedance of 14,240 ft-lbs (19,307 Joules).**

**Option [必要に応じて選択]**

**Ctl-14.1: Kinetic Energy Limited (less than 1.5 atm)**

Sealed container(s) design limit kinetic energy for mass release and pressure.

Note: Sealed containers are inherently pressurized hardware (e.g., a sealed electronic box, cylinders) that are not a part of a pressure system and contain no hazardous materials (limited to THL=0, BSL=1).

Select of verifications:
(In all cases)
Stored Energy/Pressure and maximum delta pressure Analysis= (a) and (b)

| **V-14.1(a)** | **Stored Energy/Pressure Analysis of Sealed, Non-Fracture Critical Container** | |
|---|---|---|
| Analysis<br>Verify Once | For a sealed container containing non-hazardous material, an assessment has been performed to ensure the sealed container's expected internal pressure value is less than 100 psia (689.5 kPa) and contains less than 19,310 Joules (14,240 foot-pounds) of stored energy. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**and**

| **V-14.1(b)** | **Pressure Analysis of Sealed Container** | |
|---|---|---|
| Analysis<br>Verify Once | An analysis of the design has been performed to ensure that the sealed container's maximum delta pressure less than 1.5 atm (22 psia, 1.5 bars). | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**Option [必要に応じて選択]**

**Ctl-14.2: Kinetic Energy Limited (over 1.5 atm )**

Sealed container(s) design limit kinetic energy for mass release and pressure and design which MDP is over 1.5 atm provide appropriate strength.

Select of verifications:
(Optional) Pressure Analysis (2.5 x MDP) = (a)
(Optional) Proof Test (1.5 x MDP) = (b)

| **V-14.2(a)** | **Stored Energy/Pressure Analysis of Sealed, Non-Fracture Critical Container and Pressure Analysis of Sealed Containers (2.5 x MDP)** | |
|---|---|---|
| Analysis<br>Verify Once | An analysis of the design has been performed to ensure followings<br>・the sealed container's expected internal pressure value is less than 100 psia (689.5 kPa) and contains less than 19,310 Joules (14,240 foot-pounds) of stored energy.<br>・the sealed container's maximum delta pressure is over 1.5 atm (22 psia, 1.5 bars).<br>・the sealed container has an MDP greater than 1.5 atm (22 psia, 1.5 bars), but less than 6.81 atm (100 psia, 6.9 bars) and shows a minimum safety factor for the design is 2.5 X MDP. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

**or**

| **V-14.2(b)** | **Stored Energy/Pressure Analysis of Sealed, Non-Fracture Critical Container and Proof Test of Sealed Container (1.5 x MDP)** | |
|---|---|---|
| Analysis and Test<br>Verify Once | An analysis/test of the design has been performed to ensure followings<br>・the sealed container's expected internal pressure value is less than 100 psia (689.5 kPa) and contains less than 19,310 Joules (14,240 foot-pounds) of stored energy.<br>・the sealed container's maximum delta pressure is over 1.5 atm (22 psia, 1.5 bars).<br>・the sealed container has an MDP greater than 1.5 atm (22 psia, 1.5 bars), but less than 6.81 atm (100 psia, 6.9 bars) and each flight unit has passed a proof test to 1.5 X MDP. | |
| **Status** | **Completion Date** | **Closure Documentation** |
| | | |

There is likelihood to activate FET and transistor under radiated electric field in ISS. But, since all FETs and transistor involved in the inhibit activate at more than 0.1 Volts, so there is no chance of malfunction.

Figure 1 shows the inhibit circuit diagram. The red boxes indicate the FETs and transistors involved in the inhibit.

The specifications for these are shown.

The DCDC converter used as an inhibit uses four FETs inside.

These FETs are controlled by a logic circuit inside the DCDC converter, and the lines connecting them to the gates are extremely short. Therefore, no evaluation is required.

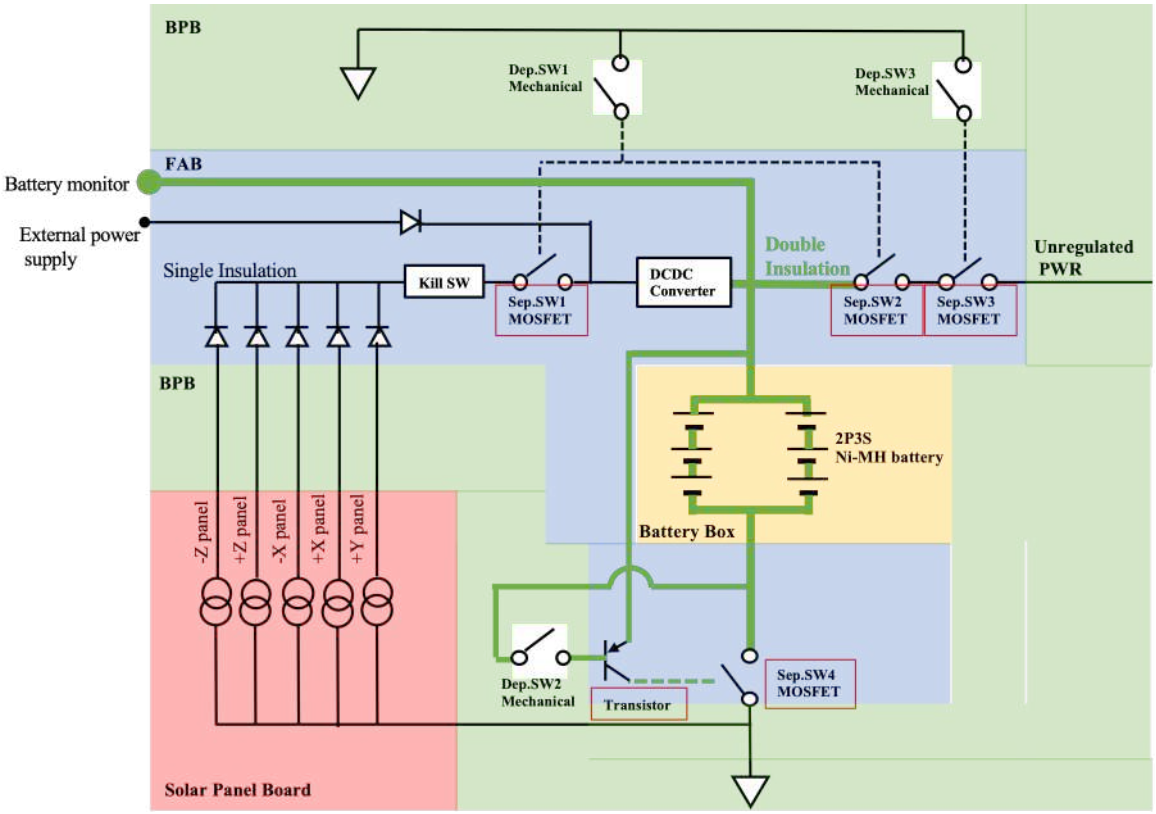



**Figure 1 EPS diagram**

Transistor: PDTA144E

SepSW1, 3, 4:SiA447DJ

SepSW2:Si7232DN

## 7. Characteristics

Table 8. Characteristics

$T_{amb}$ = 25 °C unless otherwise specified.

| Symbol | Parameter | Conditions | | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|---|
| $I_{CBO}$ | collector-base cut-off current | $V_{CB}$ = −50 V; $I_E$ = 0 A | | - | - | −100 | nA |
| $I_{CEO}$ | collector-emitter cut-off current | $V_{CE}$ = −30 V; $I_B$ = 0 A | | - | - | −1 | μA |
| | | $V_{CE}$ = −30 V; $I_B$ = 0 A; $T_j$ = 150 °C | | - | - | −5 | μA |
| $I_{EBO}$ | emitter-base cut-off current | $V_{EB}$ = −5 V; $I_C$ = 0 A | | - | - | −90 | μA |
| $h_{FE}$ | DC current gain | $V_{CE}$ = −5 V; $I_C$ = −5 mA | | 80 | - | - | |
| $V_{CEsat}$ | collector-emitter saturation voltage | $I_C$ = −10 mA; $I_B$ = −0.5 mA | | - | - | −150 | mV |
| $V_{I(off)}$ | off-state input voltage | $V_{CE}$ = −5 V; $I_C$ = −100 μA | | - | −1.2 | −0.8 | V |
| $V_{I(on)}$ | on-state input voltage | $V_{CE}$ = −0.3 V; $I_C$ = −2 mA | | −3 | −1.6 | - | V |
| R1 | bias resistor 1 (input) | | | 33 | 47 | 61 | kΩ |
| R2/R1 | bias resistor ratio | | | 0.8 | 1 | 1.2 | |
| $C_c$ | collector capacitance | $V_{CB}$ = −10 V; $I_E$ = $i_e$ = 0 A; f = 1 MHz | | - | - | 3 | pF |
| $f_T$ | transition frequency | $V_{CE}$ = −5 V; $I_C$ = −10 mA; f = 100 MHz | [1] | - | 180 | - | MHz |

[1] Characteristics of built-in transistor



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| SPECIFICATIONS ($T_J$ = 25 °C, unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **TEST CONDITIONS** | **MIN.** | **TYP.** | **MAX.** | **UNIT** |
| **Static** | | | | | | |
| Drain-source breakdown voltage | $V_{DS}$ | $V_{GS}$ = 0, $I_D$ = -250 µA | -12 | - | - | V |
| $V_{DS}$ temperature coefficient | $\Delta V_{DS}/T_J$ | $I_D$ = -250 µA | - | -7 | - | mV/°C |
| $V_{GS(th)}$ temperature coefficient | $\Delta V_{GS(th)}/T_J$ | | - | 3 | - | |
| Gate-source threshold voltage | $V_{GS(th)}$ | $V_{DS}$ = $V_{GS}$, $I_D$ = -250 µA | -0.4 | - | -0.85 | V |
| Gate-source leakage | $I_{GSS}$ | $V_{DS}$ = 0 V, $V_{GS}$ = ± 8 V | - | - | ± 100 | nA |
| Zero gate voltage drain current | $I_{DSS}$ | $V_{DS}$ = -12 V, $V_{GS}$ = 0 V | - | - | -1 | µA |
| | | $V_{DS}$ = -12 V, $V_{GS}$ = 0 V, $T_J$ = 55 °C | - | - | -10 | |
| On-state drain current [a] | $I_{D(on)}$ | $V_{DS}$ ≤ -5 V, $V_{GS}$ = -4.5 V | -10 | - | - | A |
| Drain-source on-state resistance [a] | $R_{DS(on)}$ | $V_{GS}$ = -4.5 V, $I_D$ = -7 A | - | 0.0110 | 0.0135 | Ω |
| | | $V_{GS}$ = -2.5 V, $I_D$ = -5 A | - | 0.0150 | 0.0194 | |
| | | $V_{GS}$ = -1.8 V, $I_D$ = -3 A | - | 0.0230 | 0.0344 | |
| | | $V_{GS}$ = -1.5 V, $I_D$ = -1 A | - | 0.0400 | 0.0710 | |
| Forward transconductance [a] | $g_{fs}$ | $V_{DS}$ = -6 V, $I_D$ = -7 A | - | 35 | - | S |
| **Dynamic [b]** | | | | | | |
| Input capacitance | $C_{iss}$ | $V_{DS}$ = -6 V, $V_{GS}$ = 0 V, f = 1 MHz | - | 2880 | - | pF |
| Output capacitance | $C_{oss}$ | | - | 590 | - | |
| Reverse transfer capacitance | $C_{rss}$ | | - | 585 | - | |
| Total gate charge | $Q_g$ | $V_{DS}$ = -6 V, $V_{GS}$ = -8 V, $I_D$ = -13 A | - | 52 | 80 | nC |
| | | $V_{DS}$ = -6 V, $V_{GS}$ = -4.5 V, $I_D$ = -13 A | - | 31 | 47 | |
| Gate-source charge | $Q_{gs}$ | | - | 4.2 | - | |
| Gate-drain charge | $Q_{gd}$ | | - | 7.8 | - | |
| Gate resistance | $R_g$ | f = 1 MHz | 0.8 | 4.3 | 8.6 | Ω |
| Turn-on delay time | $t_{d(on)}$ | $V_{DD}$ = -6 V, $R_L$ = 0.6 Ω<br>$I_D$ ≅ -10 A, $V_{GEN}$ = -4.5 V, $R_g$ = 1 Ω | - | 30 | 60 | ns |
| Rise time | $t_r$ | | - | 30 | 60 | |
| Turn-off delay time | $t_{d(off)}$ | | - | 60 | 120 | |
| Fall time | $t_f$ | | - | 25 | 50 | |
| Turn-on delay time | $t_{d(on)}$ | $V_{DD}$ = -6 V, $R_L$ = 0.6 Ω<br>$I_D$ ≅ -10 A, $V_{GEN}$ = -8 V, $R_g$ = 1 Ω | - | 12 | 25 | |
| Rise time | $t_r$ | | - | 10 | 20 | |
| Turn-off delay time | $t_{d(off)}$ | | - | 65 | 130 | |
| Fall time | $t_f$ | | - | 20 | 40 | |
| **Drain-Source Body Diode Characteristics** | | | | | | |
| Continuous source-drain diode current | $I_S$ | $T_C$ = 25 °C | - | - | -12 | A |
| Pulse diode forward current | $I_{SM}$ | | - | - | -50 | |
| Body diode voltage | $V_{SD}$ | $I_S$ = -10 A, $V_{GS}$ = 0 V | - | -0.8 | -1.2 | V |
| Body diode reverse recovery time | $t_{rr}$ | $I_F$ = -10 A, di/dt = 100 A/µs,<br>$T_J$ = 25 °C | - | 25 | 50 | ns |
| Body diode reverse recovery charge | $Q_{rr}$ | | - | 7.5 | 15 | nC |
| Reverse recovery fall time | $t_a$ | | - | 8 | - | ns |
| Reverse recovery rise time | $t_b$ | | - | 17 | - | |

**Notes**

a. Pulse test; pulse width ≤ 300 µs, duty cycle ≤ 2%

b. Guaranteed by design, not subject to production testing

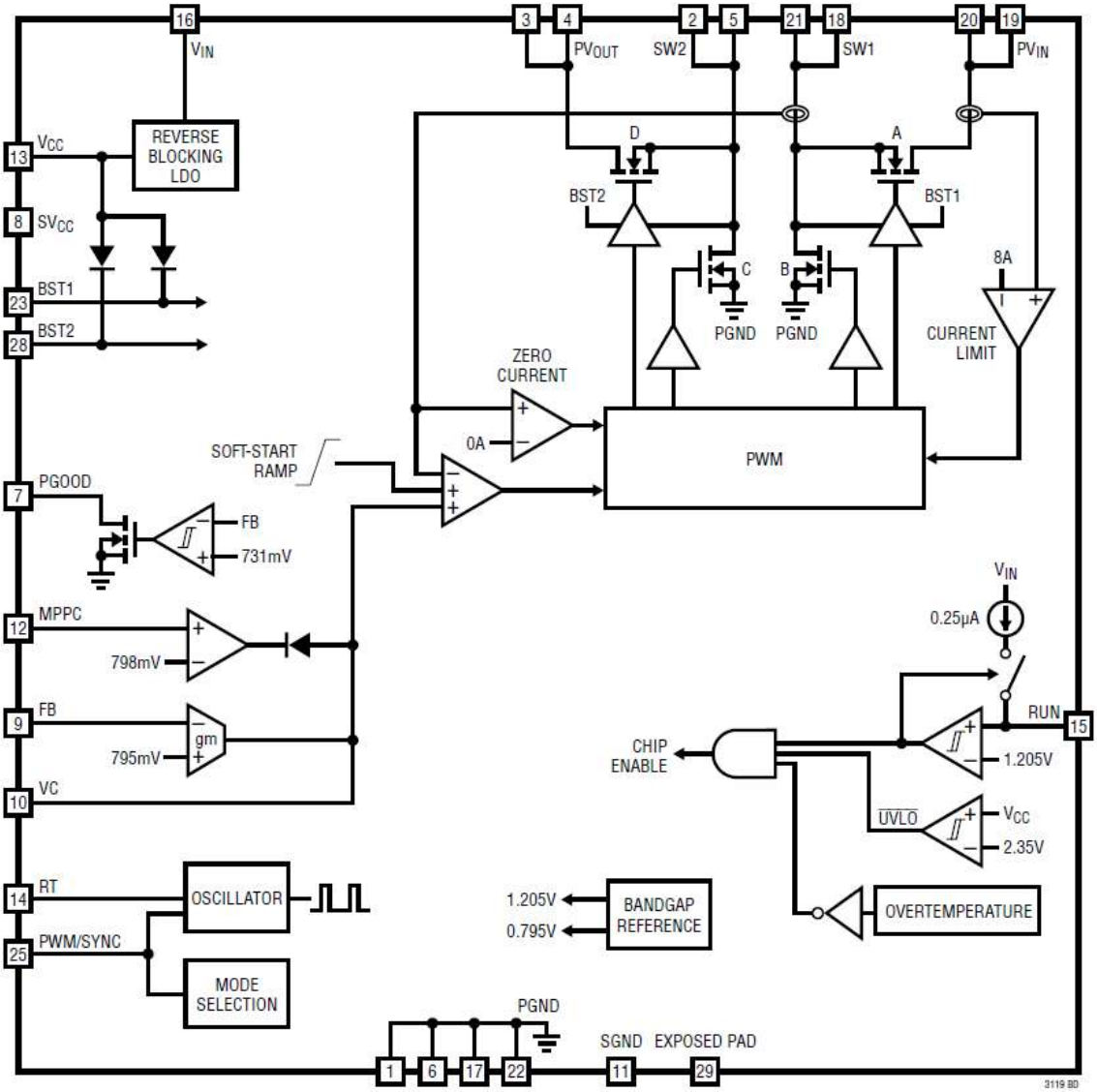
*Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability*

| SPECIFICATIONS ($T_J$ = 25 °C, unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **TEST CONDITIONS** | **MIN.** | **TYP.** | **MAX.** | **UNIT** |
| **Static** | | | | | | |
| Drain-Source Breakdown Voltage | $V_{DS}$ | $V_{GS}$ = 0 V, $I_D$ = 250 µA | 20 | - | - | V |
| $V_{DS}$ Temperature Coefficient | $\Delta V_{DS}/T_J$ | $I_D$ = 250 µA | - | 22 | - | mV/°C |
| $V_{GS(th)}$ Temperature Coefficient | $\Delta V_{GS(th)}/T_J$ | | - | -3 | - | |
| Gate-Source Threshold Voltage | $V_{GS(th)}$ | $V_{DS}$ = $V_{GS}$ , $I_D$ = 250 µA | 0.4 | - | 1 | V |
| Gate-Source Leakage | $I_{GSS}$ | $V_{DS}$ = 0 V, $V_{GS}$ = ± 8 V | - | - | ± 100 | nA |
| Zero Gate Voltage Drain Current | $I_{DSS}$ | $V_{DS}$ = 20 V, $V_{GS}$ = 0 V | - | - | 1 | µA |
| | | $V_{DS}$ = 20 V, $V_{GS}$ = 0 V, $T_J$ = 55 °C | - | - | 10 | |
| On-State Drain Current [a] | $I_{D(on)}$ | $V_{DS}$ ≥ 5 V, $V_{GS}$ = 10 V | 20 | - | - | A |
| Drain-Source On-State Resistance [a] | $R_{DS(on)}$ | $V_{GS}$ = 4.5 V, $I_D$ = 10 A | - | 0.0135 | 0.0164 | Ω |
| | | $V_{GS}$ = 2.5 V, $I_D$ = 9 A | - | 0.0160 | 0.0200 | |
| | | $V_{GS}$ = 1.8 V, $I_D$ = 8.2 A | - | 0.0190 | 0.0240 | |
| Forward Transconductance [a] | $g_{fs}$ | $V_{DS}$ = 10 V, $I_D$ = 10 A | - | 47 | - | S |
| **Dynamic** [b] | | | | | | |
| Input Capacitance | $C_{iss}$ | $V_{DS}$ = 10 V, $V_{GS}$ = 0 V, f = 1 MHz | - | 1220 | - | pF |
| Output Capacitance | $C_{oss}$ | | - | 180 | - | |
| Reverse Transfer Capacitance | $C_{rss}$ | | - | 80 | - | |
| Total Gate Charge | $Q_g$ | $V_{DS}$ = 15 V, $V_{GS}$ = 8 V, $I_D$ = 10 A | - | 21 | 32 | nC |
| | | $V_{DS}$ = 15 V, $V_{GS}$ = 4.5 V, $I_D$ = 10 A | - | 12 | 18 | |
| Gate-Source Charge | $Q_{gs}$ | | - | 2 | - | |
| Gate-Drain Charge | $Q_{gd}$ | | - | 1.3 | - | |
| Gate Resistance | $R_g$ | f = 1 MHz | - | 1.8 | 3.6 | Ω |
| Turn-On Delay Time | $t_{d(on)}$ | $V_{DD}$ = 10 V, $R_L$ = 1.25 Ω<br>$I_D$ ≅ 8 A, $V_{GEN}$ = 4.5 V, $R_g$ = 1 Ω | - | 10 | 15 | ns |
| Rise Time | $t_r$ | | - | 10 | 15 | |
| Turn-Off Delay Time | $t_{d(off)}$ | | - | 35 | 55 | |
| Fall Time | $t_f$ | | - | 10 | 15 | |
| Turn-On Delay Time | $t_{d(on)}$ | $V_{DD}$ = 10 V, $R_L$ = 1.25 Ω<br>$I_D$ ≅ 8 A, $V_{GEN}$ = 8 V, $R_g$ = 1 Ω | - | 10 | 15 | |
| Rise Time | $t_r$ | | - | 10 | 15 | |
| Turn-Off Delay Time | $t_{d(off)}$ | | - | 25 | 40 | |
| Fall Time | $t_f$ | | - | 10 | 15 | |
| **Drain-Source Body Diode Characteristics** | | | | | | |
| Continuous Source-Drain Diode Current | $I_S$ | $T_C$ = 25 °C | - | - | 19 | A |
| Pulse Diode Forward Current | $I_{SM}$ | | - | - | 40 | |
| Body Diode Voltage | $V_{SD}$ | $I_S$ = 8 A, $V_{GS}$ = 0 V | - | 0.81 | 1.2 | V |
| Body Diode Reverse Recovery Time | $t_{rr}$ | $I_F$ = 8 A, dI/dt = 100 A/µs, $T_J$ = 25 °C | - | 20 | 30 | ns |
| Body Diode Reverse Recovery Charge | $Q_{rr}$ | | - | 15 | 25 | nC |
| Reverse Recovery Fall Time | $t_a$ | | - | 12.5 | - | ns |
| Reverse Recovery Rise Time | $t_b$ | | - | 7.5 | - | |

**Notes**

a. Pulse test; pulse width ≤ 300 µs, duty cycle ≤ 2 %

b. Guaranteed by design, not subject to production testing.




**Block Diagram of DCDC convertor**

# Implementation plan extracted from JAXA's checklist for BIRDS-5

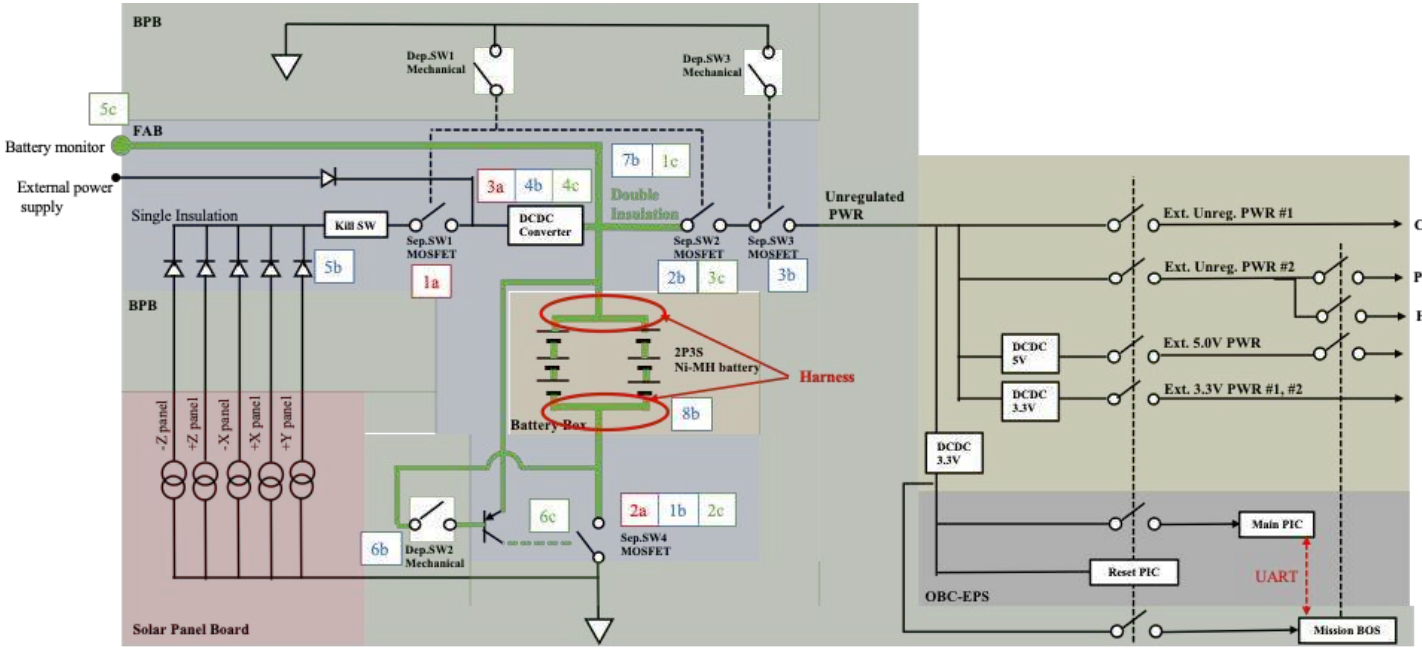
| Items for DFMR | Implementation plan extracted from JAXA's checklist | Implementation result for BIRDS-5 |
|---|---|---|
| (1) Inhibits on the positive AND negative current paths. (supply and return). This allows failure of one inhibit and still safe operations. | No chassis connection on the supply or return path between the battery and the inhibits. Any chassis connection must have an inhibit between the chassis and the battery to mitigate the effects of battery short circuits to chassis. | • See page 2 |
| , (2) Design to have no smart short between hot side and return side | Wiring:<br>• Proper insulation (double insulation)<br>• Wires connected to the battery positive and wires connected to the battery negative should not be in the same cable bundle. Or Additional insulation layers | • See *1 Wiring on page 3, 5, 7 |
| | Battery:<br>• Electrical connections to battery cells (tabs) should be insulated or covered with an insulating layer or cover.<br>• Battery tabs and conductive surfaces covered with insulation to prevent shorting from conductive debris. | • See *2 Battery on page 3, 5, 7 |
| | Circuit board:<br>• RTV and Kapton tape to prevent shorting from FOD, etc. | • See *3 Board on page 4, 6, 8 |
| | Connectors:<br>• Either separate connectors for positive and negative conductors, or pin spacing on connectors between positive and negative sufficient to prevent shorting between positive and negative conductors if a connector pin is bent. | • See *4 Connector on page 4, 6, 8 |

# Implementation plan extracted from JAXA's checklist for BIRDS-5

No chassis connection on the supply path between the battery, DCDC convertor, Deployment SW2, Separation SW2 and the battery monitor terminal.

No chassis connection on the return path between the battery and Separation SW4.

The Inhibit Schematic is shown on the right.




| Hazard | | Hazard Control #1 | Hazard Control #2 | Hazard Control #3 |
|---|---|---|---|---|
| Over-charge | | SepSW1[1a] | SepSW4[2a] | DCDC converter[3a] |
| Over-discharge | Load side | SepSW4[1b] | SepSW2[2b] | SepSW3[3b] |
| | Solar cell side | | DCDC converter[4b] | Diode[5b] |
| | DepSW2 side | DepSW2 [6b] | Proper Insulation[7b] | Proper Charging [8b] |
| External short | Load side | Proper Insulation[1c] | SepSW4[2c] | SepSW2[3c] |
| | Solar cell side | | | DCDC converter[4c] |
| | External power supply side | | | DCDC converter[4c] |
| | Battery monitor side | | | Proper Insulation[5c] |
| | DepSW2 side | | | Proper Insulation[6c] |

Note: Proper insulation (double isolation is shown by green line in figure above, single isolation is black line)
All wires and components between the battery and the first power functions are assembled as double insulation. .
The DCDC converter (LTC3119) used for inhibit is a buck-boost converter and its internal FET configuration prevents reverse current.

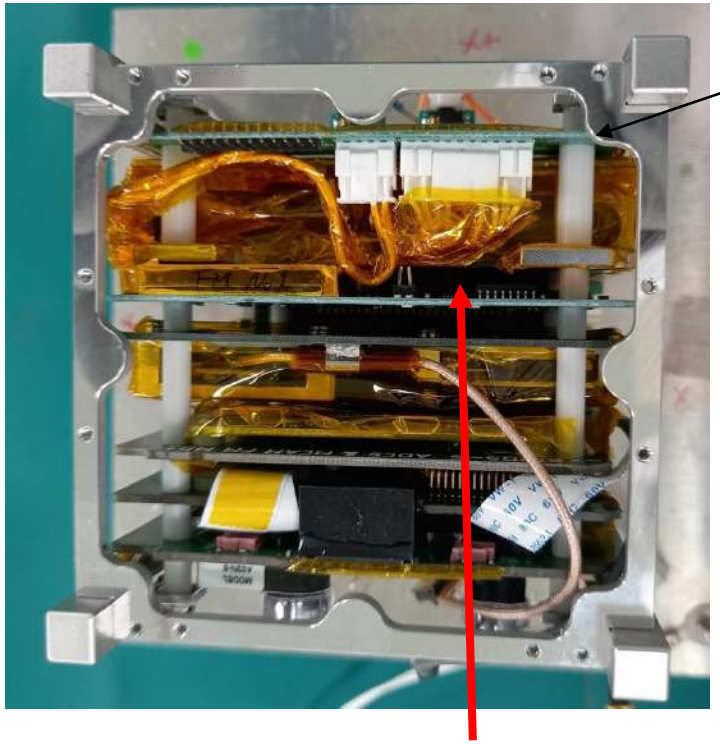
Inhibit Schematic

# Implementation plan extracted from JAXA's checklist for BIRDS-5

## PEARLAFRICASAT-1

### *1 Wiring

The power wires connecting the battery to FAB are double insulated (wire insulation and covering by Kapton tape).

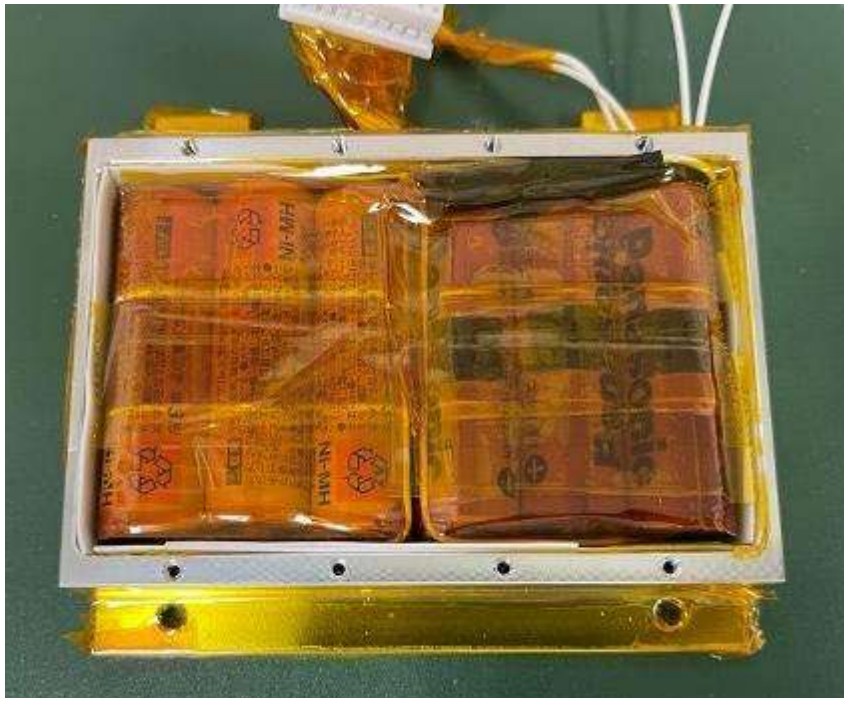






### *2 Battery

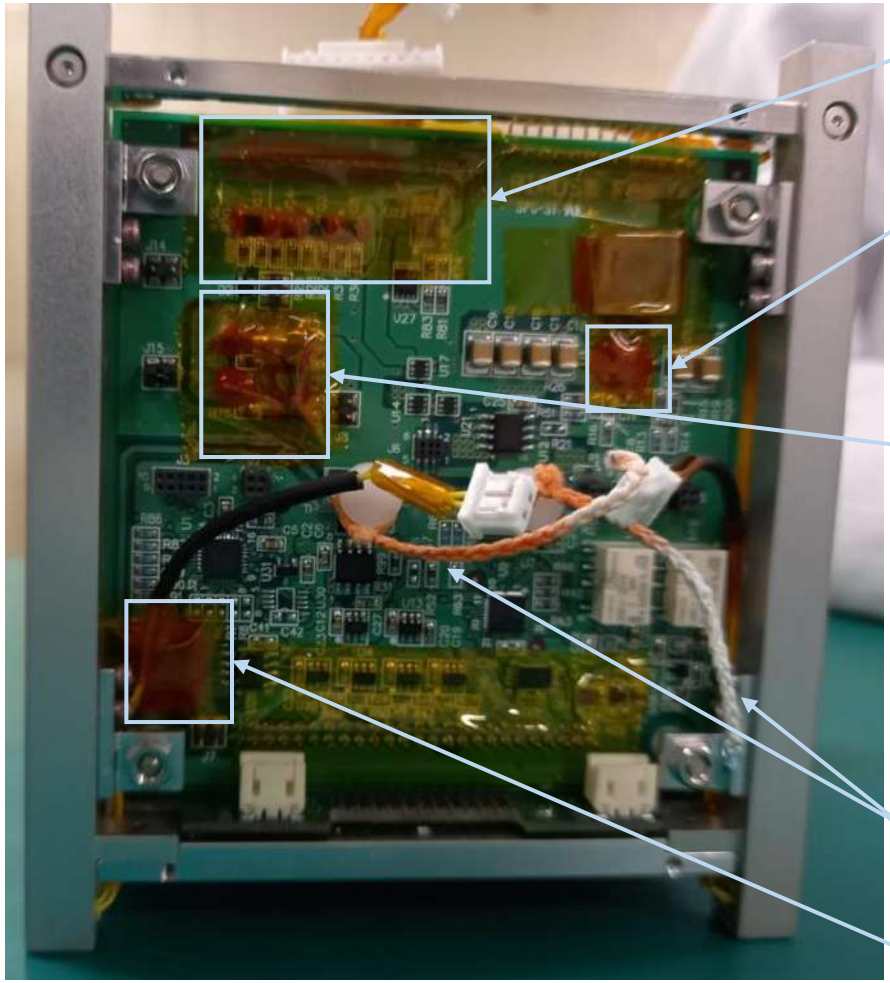
Electrical connections to battery cells (tabs) are covered with Kapton tape.
Battery tabs and conductive surfaces are covered with Kapton tape.



# Implementation plan extracted from JAXA's checklist for BIRDS-5

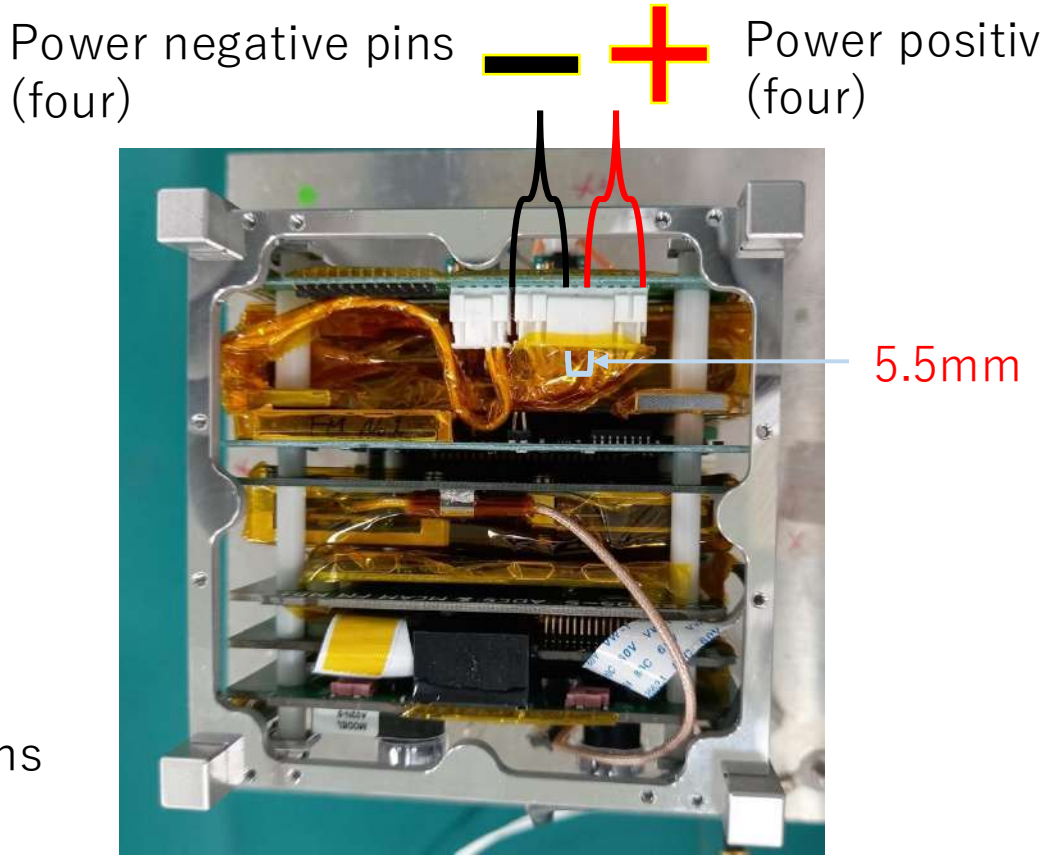
## *3 Circuit board

The terminals of the electronic elements (DCDC convertor and Diode) from the battery to the first inhibit are covered with RTV. Moreover they were covered with Kapton tape.




## *4 Connectors

The pins spacing on connectors between positive and negative sufficient.

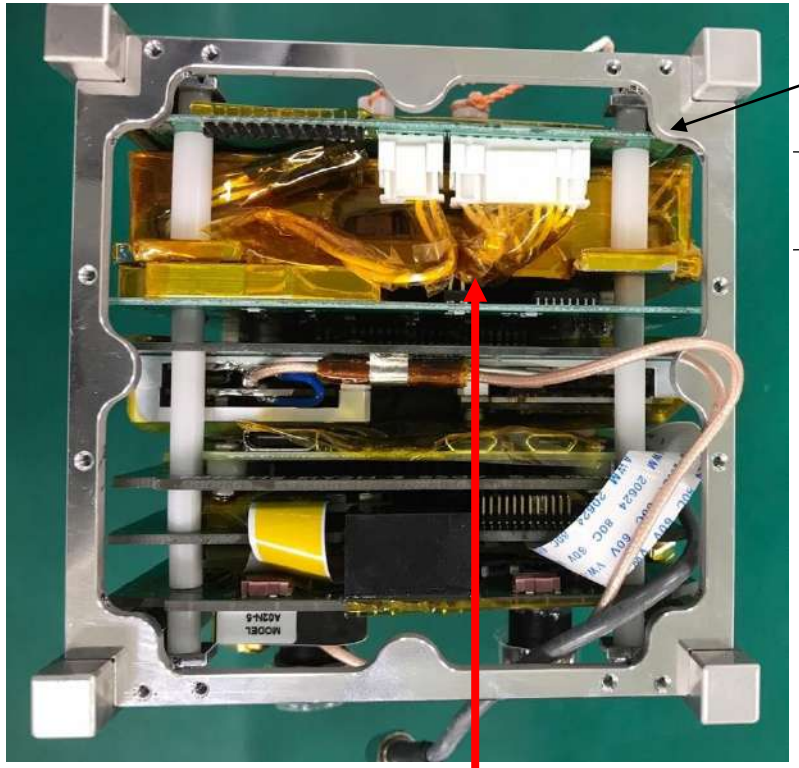


# Implementation plan extracted from JAXA's checklist for BIRDS-5

## ZIMSAT-1

### *1 Wiring

The power wires connecting the battery to FAB are double insulated (wire insulation and covering by Kapton tape).

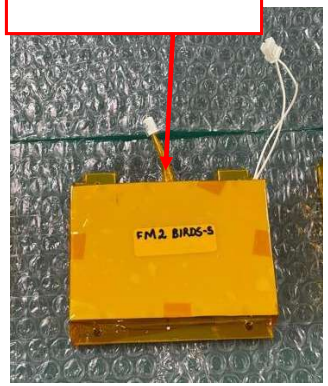


FAB
(Front Access Board)

Battery box

Power wires

The HOT and GND cables separated and individually covered with Kapton tape and finally covered with Kapton tape.




### *2 Battery

Electrical connections to battery cells (tabs) are covered with Kapton tape.
Battery tabs and conductive surfaces are covered with Kapton tape.



# Implementation plan extracted from JAXA's checklist for BIRDS-5

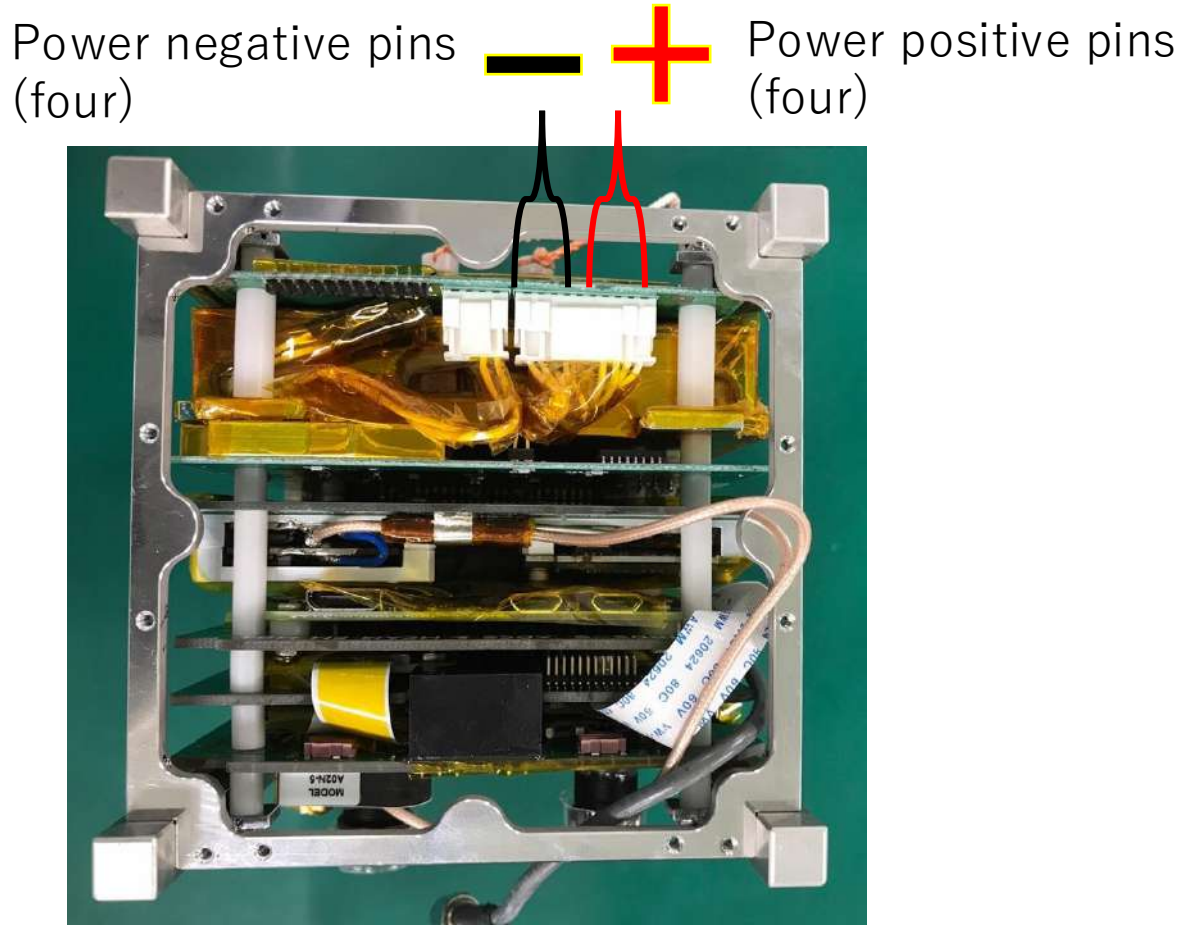
## *3 Circuit board

The terminals of the electronic elements (DCDC convertor and Diode) from the battery to the first inhibit are covered with RTV. Moreover they were covered with Kapton tape.



## *4 Connectors

The pins spacing on connectors between positive and negative sufficient.

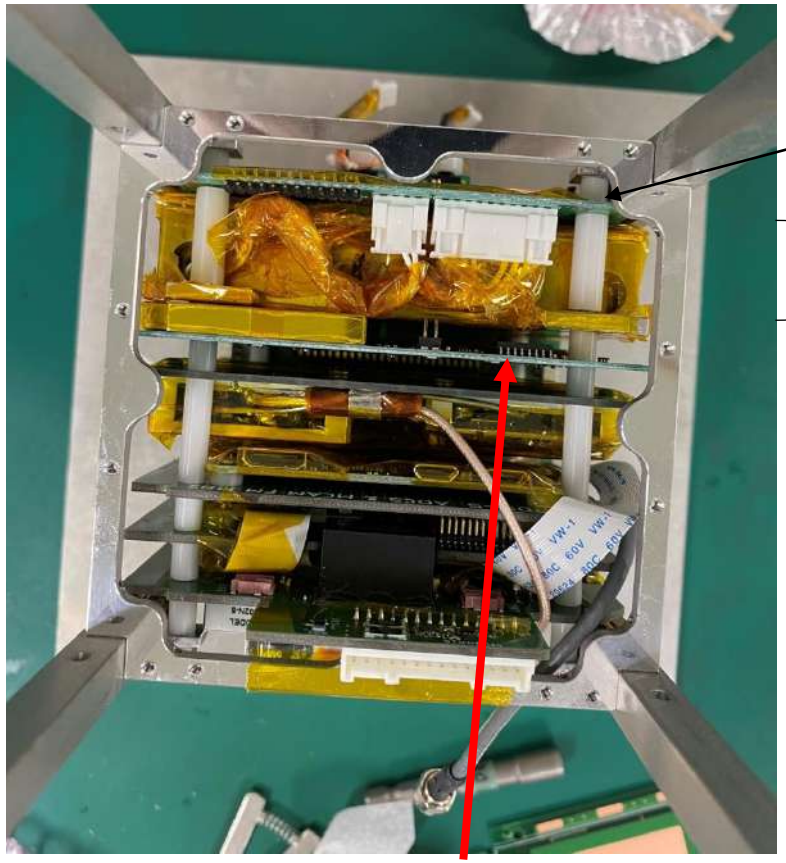


# Implementation plan extracted from JAXA's checklist for BIRDS-5

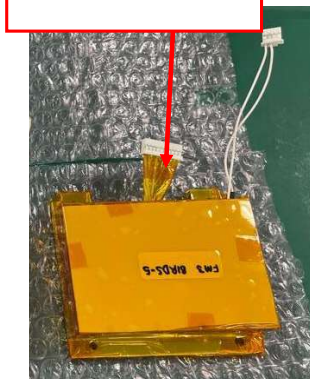
**TAKA**

## *1 Wiring

The power wires connecting the battery to FAB are double insulated (wire insulation and covering by Kapton tape).

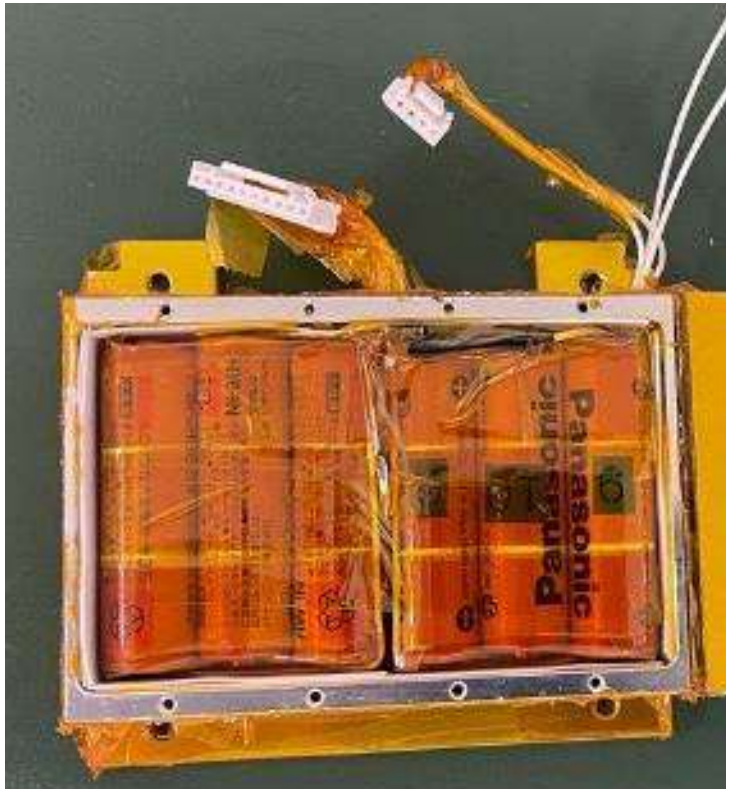



The HOT and GND cables separated and individually covered with Kapton tape and finally covered with Kapton tape.



## *2 Battery

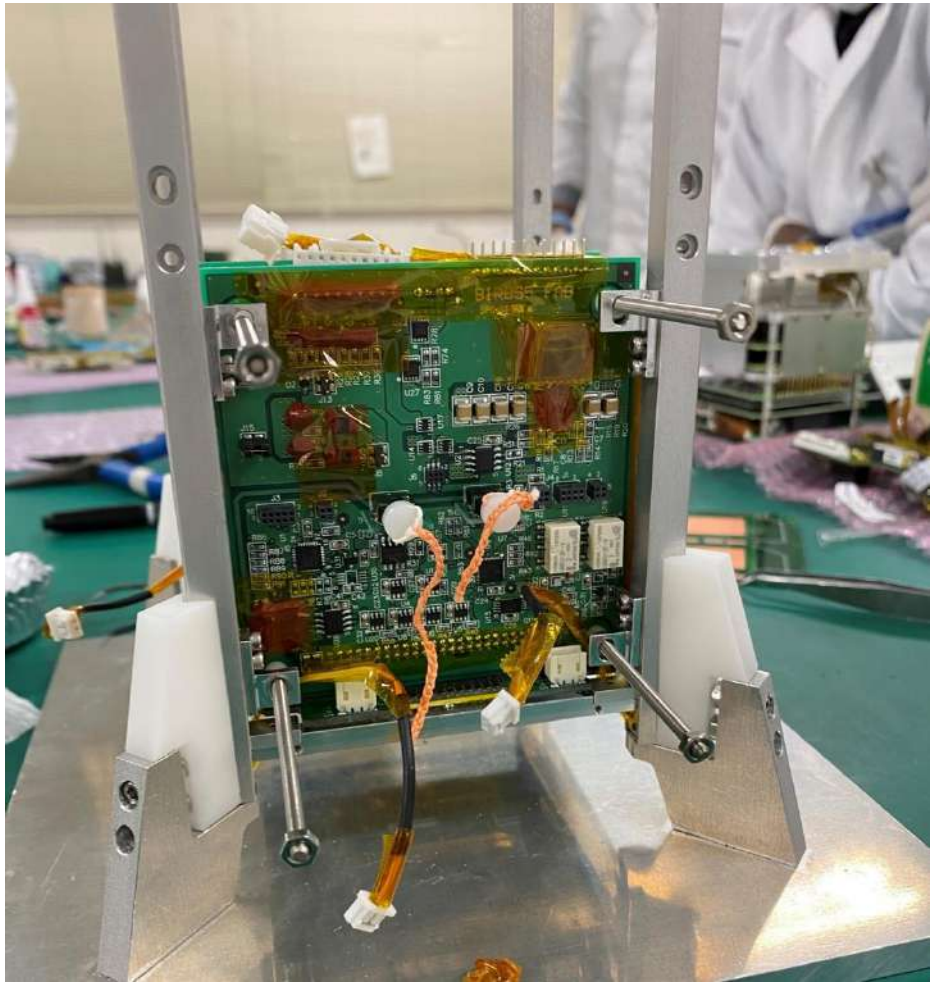
Electrical connections to battery cells (tabs) are covered with Kapton tape.
Battery tabs and conductive surfaces are covered with Kapton tape.



# Implementation plan extracted from JAXA's checklist for BIRDS-5

## *3 Circuit board

The terminals of the electronic elements (DCDC convertor and Diode) from the battery to the first inhibit are covered with RTV. Moreover they were covered with Kapton tape.



## *4 Connectors

The pins spacing on connectors between positive and negative sufficient.

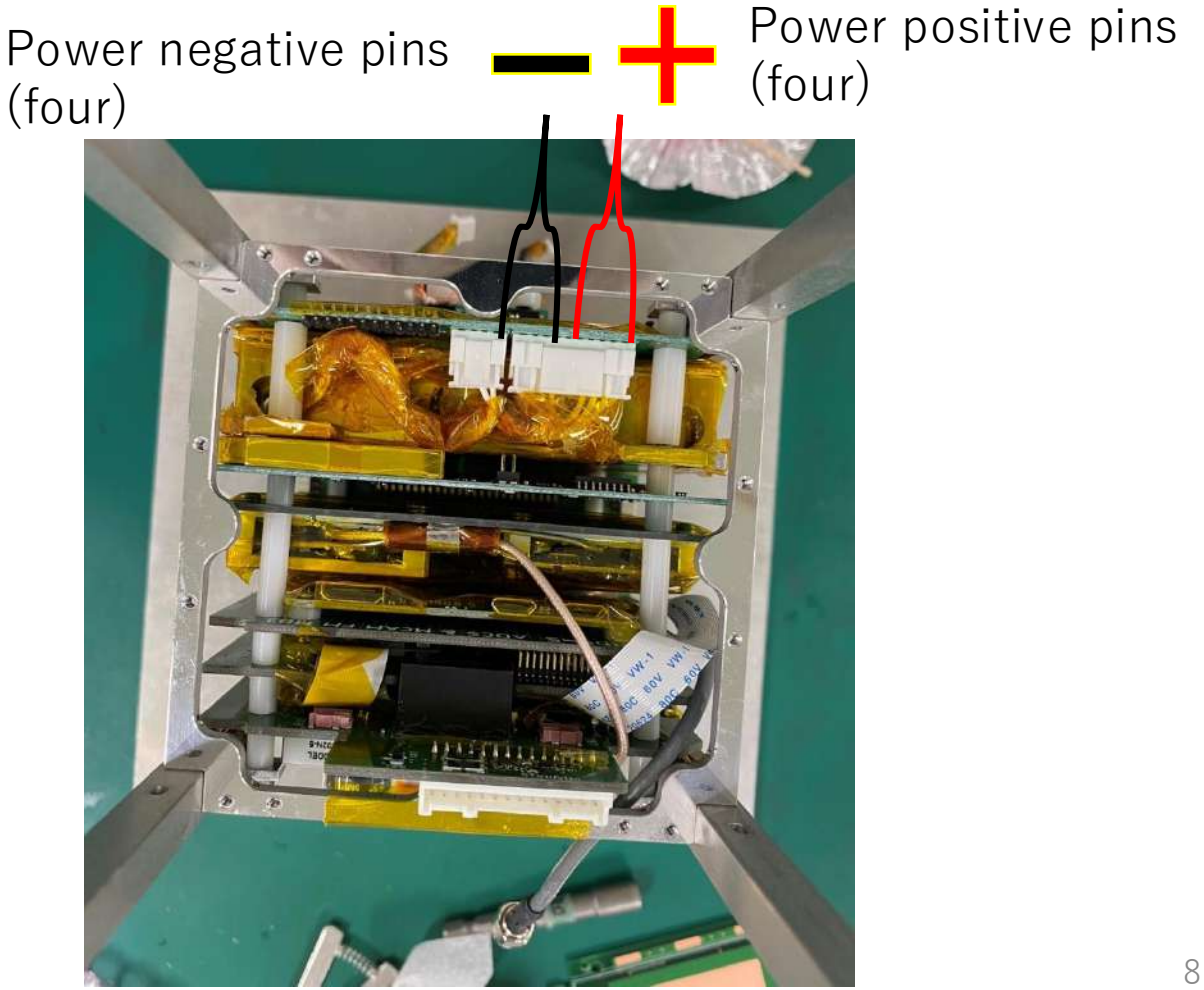


Table STD-10 Selection of Wires and Circuit Protection Devices

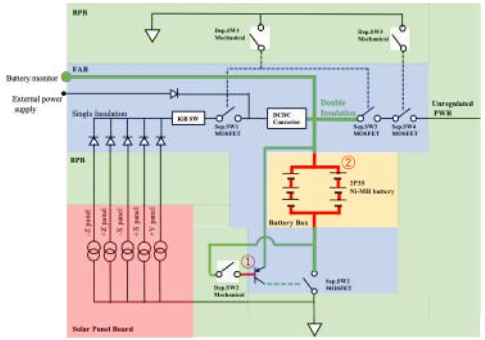
| No. | Power Line Wire | Max. Applied Load(A) | Max. Blow Current of Protection Device (A) | Crew accessible Wire/Cables (Yes/No) | Downstream Wire | | | | | | | | | Compatibility for Wire Derating (Yes/No) | Compatibility for Touch Temperature (Yes/No) | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Size (AWG) | Rated. Temp (°C) | Current Carrying Capacity of a Single Wire (A) | Current limit to meet touch temperature limited (A)※2 | Quantity; Hot/Return | Bundled (Yes/No) | Bundle Factor | Derating Current (A) | Smart Short Current (A) | | | |
| 1 | DepSW2 signal line | $185.4x10^{-6}$ | N/A | No | AWG30 | 200 | 1.3※1 | - | 1 | No | N/A | 1.3 | N/A | Yes | - | |
| 2 | Battery cable | $379.9x10^{-6}$ | N/A | No | AWG22 | 200 | 6.5※3 | - | 6 / 6 | No | N/A | 6.5 | N/A | Yes | - | |
| | | | | | | | | | | | | | | | | |

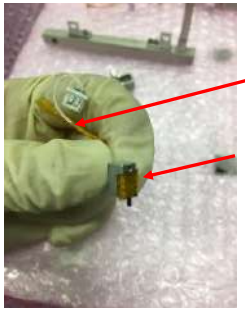
Note: Ambient Temperature +22.2°C, Ambient Pressure: $10^{-6}$ torr)

※1: Based on JAXA-JERG-2-212 5.2.1 Table 5.2-1 RECOMMENDED MAXIMUM CURRENT FOR SINGLE LINE as there was no indicattion of wire size to be used for SSP51721 4.3.1.2 Table 4.3.1.2-1.

※2: N/A because the target wire cannot be touched by the crew

※3: Based on SSP51721 4.3.1.2 Table 4.3.1.2-1 WIRE SIZE DERATING AND CIRCUIT PROTECTION ColumnA









① Wire

Alpha Wire | 711 Lidgerwood Avenue, Elizabeth, NJ 07207
Tel: 1-800-52 ALPHA (25742), Web: www.alphawire.com

# Customer Specification
# PART NO. 2841/7

## Construction

| | | Diameters (In) |
|---|---|---|
| 1) Component 1 | 1 X 1 HOOKUP | |
| a) Conductor | 30 (7/38) AWG SPC | 0.012 |
| b) Insulation | 0.006" Wall, Nom. PTFE | 0.024+/- 0.002 |
| (1) Color(s) | WHITE, BLACK, RED, GREEN, YELLOW, BLUE, BROWN ORANGE, SLATE, VIOLET | |

## Applicable Specifications

| | | |
|---|---|---|
| 1) Military | MIL-W-16878/6 (Type ET) | |
| 2) Other | NEMA HP3-ETXBBB | |

## Environmental

| | |
|---|---|
| 1) EU Directive 2011/65/EU(RoHS2): | |
| | All materials used in the manufacture of this part are in compliance with European Directive 2011/65/EU regarding the restriction of use of certain hazardous substances in electrical and electronic equipment. Consult Alpha Wire's web site for RoHS C of C. |
| 2) REACH Regulation (EC 1907/2006): | |
| | This product does not contain Substances of Very High Concern (SVHC) listed on the European Union's REACH candidate list in excess of 0.1% mass of the item. For up-to-date information, please see Alpha's REACH SVHC Declaration. |
| 3) California Proposition 65: | The outer surface materials used in the manufacture of this part meet the requirements of California Proposition 65. |

## Properties

| Physical & Mechanical Properties | |
|---|---|
| 1) Temperature Range | -60 to 200°C |
| 2) Bend Radius | 10X Cable Diameter |
| 3) Pull Tension | 0.88 Lbs, Maximum |
| Electrical Properties | (For Engineering purposes only) |
| 1) Voltage Rating | 250 $V_{RMS}$ |
| 2) Inductance | 0.06 µH/ft, Nominal |
| 3) Conductor DCR | 94 Ω/1000ft @20°C, Nominal |

## Other

| Packaging | Flange x Traverse x Barrel (inches) |
|---|---|
| a) 1000 FT | 3.5 x 3 x 1.125 Max. 3 separate pieces; Min length/piece 100 FT. |
| b) 100 FT | 2.75 x 1 x 1.125 Continuous length |
| | *[Spool dimensions may vary slightly]* |
| Notes: | |
| a) Certain colors and put-up combinations may only be available by special order, minimums may apply. | |

www.alphawire.com
Alpha Wire | 711 Lidgerwood Avenue, Elizabeth, NJ 07207

Tel: 1-800-52 ALPHA (25742)

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# EU/China ROHS CERTIFICATE OF COMPLIANCE

To Whom It May Concern:

Alpha Wire Part Number: 2841/7

2841/7, RoHS-Compliant Commencing With 3/1/2005 Production

Note: all colors and put-ups

This document certifies that the Alpha part number cited above is manufactured in accordance with Directive 2011/65/EU of the European Parliament, better known as the RoHS Directive (commonly known as RoHS 2), with regards to restrictions of the use of certain hazardous substances used in the manufacture of electrical and electronic equipment. This certification extends to amending Directive 2015/863/EU which expanded the list of restricted substances to 10 items (commonly known as RoHS 3) The reader is referred to these Directives for the specific definitions and extents of the Directives. **No Exemptions are required for RoHS Compliance on this item**. Additionally, Alpha certifies that the listed part number is in compliance with China RoHS "Marking for Control of Pollution by Electronic Information Products" standard SJ/T 11364-2014.

| Substance | Maximum Control Value |
|---|---|
| Lead | 0.1% by weight (1000 ppm) |
| Mercury | 0.1% by weight (1000 ppm) |
| Cadmium | 0.01% by weight (100 ppm) |
| Hexavalent Chromium | 0.1% by weight (1000 ppm ) |
| Polybrominated Biphenyls (PBB) | 0.1% by weight (1000 ppm) |
| Polybrominated Diphenyl Ethers (PBDE) , | |
| Including Deca-BDE | 0.1% by weight (1000 ppm) |
| Bis(2-ethylhexyl) phthalate (DEHP) | 0.1% by weight (1000 ppm) |
| Butyl benzyl phthalate (BBP) | 0.1% by weight (1000 ppm) |
| Dibutyl phthalate (DBP) | 0.1% by weight (1000 ppm) |
| Diisobutyl phthalate (DIBP) | 0.1% by weight (1000 ppm) |

The information provided in this document and disclosure is correct to the best of Alpha Wire's knowledge, information and belief at the date of its release. The information provided is designed only as a general guide for the safe handling, storage, and any other operation of the product itself or the one that it will become part of. The intent of this document is not to be considered a warranty or quality specification. Regulatory information is for guidance purposes only. Product users are responsible for determining the applicability of legislation and regulations based on their individual usage of the product.

Authorized Signatory for the Alpha Wire:



Dave Watson, Director of Engineering & QA 9/29/2020

Alpha Wire

711 Lidgerwood Ave.

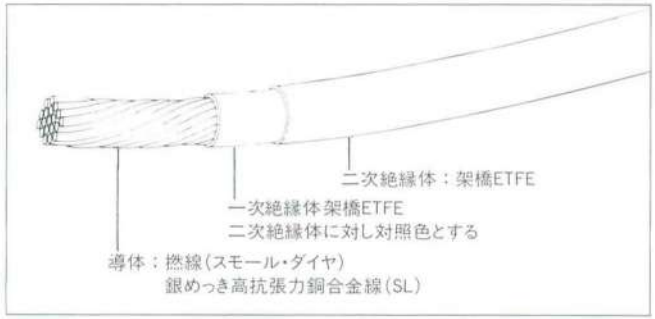
Elizabeth, NJ 07207

Tel: 1-908-925-8000

② Wire

## 航空宇宙機器用MIL規格ジュンフロン架橋ETFE(XL ETFE)電線

個別仕様書番号：MIL-W-22759/35C　定格：600V、200℃　認定番号：M-22759-82-0451B




導　体：MIL-W-22759に適合する銀めっき高抗張力銅合金撚線(SL)です。
絶縁体：架橋ETFEを導体上に均一の厚さに被覆し、絶縁体とします。
色　別：色別は、絶縁体によって行ないMIL-STD-104に適合する黒、茶、赤、橙、黄、緑、青、紫、灰、および白の10色です。

| 部品番号[注] | 導体 AWG | 導体 構成 No.×AWG(本／mm) | 導体 外径 最小 inch(mm) | 導体 外径 最大 inch(mm) | 絶縁体外径 inch | 絶縁体外径 mm | 導体抵抗 20℃、最大 Ω／km (Ω／1000ft) | 質量 最大 kg／km (lbs／1000ft) |
|---|---|---|---|---|---|---|---|---|
| M22759/35-26-※ | 26 | 19×38(19/0.102) | .018(0.457) | .020(0.508) | .040±.002 | 1.016±0.051 | 147 (44.8) | 2.58(1.7) |
| M22759/35-24-※ | 24 | 19×36(19/0.127) | .023(0.584) | .025(0.635) | .045±.002 | 1.143±0.051 | 93.2(28.4) | 3.48(2.3) |
| M22759/35-22-※ | 22 | 19×34(19/0.160) | .029(0.737) | .032(0.813) | .050±.002 | 1.270±0.051 | 57.4(17.5) | 4.97(3.3) |
| M22759/35-20-※ | 20 | 19×32(19/0.203) | .037(0.940) | .040(1.016) | .058±.002 | 1.473±0.051 | 35.1(10.7) | 7.21(4.8) |

注)部品番号の末尾の※は、二次絶縁体色記号に置き換えます。　インパルス耐電圧：8000V、絶縁抵抗(初期値)：1524MΩ・km(5000MΩ・1000ft)(最小)
高周波スパーク耐電圧：5700V

**最低受注数量**：100m(20m以上で3条長以内の組み合わせになる場合があります。)

# Appendix B-2

# Unique Hazard Report

Provider Name:

| Hazard Report #: BIRDS5-UNQ-01 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Structural Failure | | |

| System: BIRDS-5 | Sub-Subsystem: Structure |
|---|---|
| **Flight/Increment Applicability:**<br>HTV-X / Cygnus / Dragon<br>Inc.(TBD) and subsequent stages. | **Mission Phases:**<br>☐Launch Processing: ☒Launch: ☐Rendezvous / Docking:<br>☒Deployment: ☐Orbital Assembly & Checkout: ☒On-orbit Operation:<br>☐On-orbit Maintenance: ☐Descent / Landing: |
| **Scope:**<br>☒Payload: ☐JEM - PM: ☐ JEM - EF:<br>☐ Other( ): | **Interfaces:**<br>☐JEM-PM ☐JEM-EF ☒JEM-AIRLOCK ☒JEMRMS<br>☐ Other( ): |

**Hazardous Condition Description:**

If a structure failure of the CubeSat occurs in the Satellite Install Case of the J-SSOD, the CubeSat may not be appropriately released from the J-SSOD along with the expected contact to the Satellite Install Case. Consequently, the released satellite may collide against the ISS Structure. The release of shatterable material such as glass particles may cause injury to crew.

[Note] This UNQ HR addresses glass of solar cells, camera lens and filter.

- This UNQ HR addresses glass of solar cells, camera lens and filter.
- The IVA/EVA crew applied load is excluded because the aperture surface (+Z) of the J-SSOD or J-SSOD-R Launch Case is identified as NO Touch Area for the sharp edge hazard by J-SSOD Upgrade Unique Hazard Report (No. J-SSOD-UG-03) of JMX-2016127 and J-SSOD-R Unique Hazard Report (No. J-SSOD-R-03) of JMX-2017181. The other surfaces of the satellites are covered by the J-SSOD or J-SSOD-R Launch Case.

**Cause Summary:**

Cause 1. Structure failure of the CubeSat

Cause 2. Release of fragments from shatterable material. (Refer to Figure 1)

**Remarks:**

| Submittal Concurrence: | | Safety Review Panel Approval | |
|---|---|---|---|
| **Signature** | **Date** | **Signature** | **Date** |
| T.Yamauchi | 2026.9.15 | | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

Provider Name:

| Hazard Report #: BIRDS5-UNQ-01 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Structural Failure | | |

| Cause Number: | Cause Title: |
|---|---|
| 1 | Structure failure of the CubeSat |

**Hazard Cause Description:**

1.1 Inadequate structural strength for launch, ascent, On-orbit load (excluding crew applied load) and depressurization.
1.2 Improper material selection and processing, including usage of stress corrosion sensitive materials.
1.3 Material fatigue or propagation of inherent cracks or internal flaws.
1.4 Use of counterfeit fasteners
1.5 Loosening of fasteners during launch and on-orbit
1.6 Improper manufacturing and/or assembly

| Severity: | Likelihood: |
|---|---|
| ☒I (Catastrophic) ☐II (Critical) | ☐A (Probable) ☐B (Infrequent) ☒C (Remote) ☐D (Improbable) |

| Controls: | Verification Method and Status: |
|---|---|
| 1.1-1. Design to meet the applicable requirements for soft-stowed items for launch | 1.1-1(1). Inspection to verify that the CubeSat is installed per the approved packing requirement.<br>[Status] Closed to VTL BIRDS5-VTL-01<br><br>1.1-1(2). Mass measurement to verify compatibility with JX-ESPC-101132E (Japanese) / 101133E (English).<br>[Status] Closed:<br>BIRDS5-IVR-01, BIRDS-5 Interface Verification Record (2022/02/21) |
| 1.1-2. Safety critical structure shall be designed with positive margins of safety.<br>For the launch environment, following factors of safety as defined in JMX-2011303E are used;<br>For HTV launch, 1.5 for yield and 1.875 for ultimate<br>For Dragon and Cygnus, 1.25 for yield and 2.0 for ultimate<br>For the on-orbit loads, following factors of safety as defined in JMX-2011303E are used; 1.25 for yield and 2.0 for ultimate | 1.1-2. Structural Analysis to verify structural integrity according to JMX-2011303E<br>[Status] Closed: BIRDS5-SR-01, BIRDS-5 Structural Analysis Report (2021/12/09) |

| Hazard Report #: BIRDS5-UNQ-01 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Structural Failure | | |

| | |
|---|---|
| 1.2. For safety critical structures, materials shall be selected according to CR-99117K, “JAXA Space Station Program Requirements for Materials and Processes”. | 1.2. Review of Materials used will be listed in Material Identification and Usage List (MIUL) by JAXA.<br>[Status] Closed: BIRDS5_MIUL_01, BIRDS-5 MIUL (2022/03/11) |
| 1.3 For safety critical structures, fracture control shall be implemented according to JAXA approved structure verification and fracture control plan (JMX-2011303E), of which scope covers JBX-97160B, JEM Payload Fracture Control Plan. | 1.3 Fracture Control Evaluation Form is submitted to and approved by JAXA.<br>[Status] Closed: BIRDS5-FCE-01, BIRDS-5 Fracture Control Evaluation Form for Phase 0/I/II (2022/03/14)<br><br>[Status] Closed: BIRDS5-FCE-02, BIRDS-5 Fracture Control Evaluation Form for Phase III (2023/10/06) |
| 1.4 For safety critical fasteners, fastener control shall be implemented according to JAXA approved structure verification and fracture control plan (JMX-2011303E), of which scope covers JBX-97159, JEM Payload Fastener Control Plan. | 1.4(1). Fracture Control Evaluation Form is submitted to and approved by JAXA.<br>[Status] Closed: BIRDS5-FCE-01, BIRDS-5 Fracture Control Evaluation Form for Phase 0/I/II (2022/03/14)<br><br>[Status] Closed: BIRDS5-FCE-02, BIRDS-5 Fracture Control Evaluation Form for Phase III (2023/10/06)<br><br>1.4(2). The random vibration test at the flight level is performed and the inspection is also performed to verify if there are any damages after the test.<br>[Status] Closed: BIRDS5-VT-01, BIRDS-5 Vibration Test Report (2022/02/18) |
| 1.5 For safety critical fasteners, liquid locking compound shall be used to prevent back off. | 1.5(1). Review of assembly procedures directing liquid locking compound.<br>[Status] Closed: BIRDS5-AP-01, BIRDS-5 Assembly Procedure (2021/12/09) |

| Hazard Report #: BIRDS5-UNQ-01 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Structural Failure | | |

| | |
|---|---|
| 1.6 Manufacture according to approved drawings | 1.5(2). Inspection of as-built hardware per the assembly procedures which directs application of the liquid locking compound.<br>[Status] Closed: BIRDS5-AR-01, BIRDS-5 Assembly Record (2022/02/18)<br><br>1.5(3). Inspection of no loosening of fasteners after vibration test.<br>[Status] Closed: BIRDS5-VT-01, BIRDS-5 Vibration Test Report (2022/02/18)<br><br>1.6(1). Review of design drawings complying with JX-ESPC-101132 E(Japanese) / 101133E (English).<br>[Status] Closed: BIRDS5-AD-01, BIRDS-5 Assembly Drawing (2021/12/09)<br>1.6(2). Inspection of as-built hardware per the drawing.<br>[Status] Closed: BIRDS5-AR-01, BIRDS-5 Assembly Record (2022/02/18) |

**Safety Requirements:**

SSP51721
SSP52005F, Payload Flight Equipment Requirements and Guidelines for Safety-Critical Structures
SSP50835E, ISS Pressurized Volume Hardware Common Interface Requirements Document

| Detection and Warning Methods: | Additional Safety Features: |
|---|---|
| - | - |

**Cause Remarks:**

-

| Mission Phases: | Point of Contact |
|---|---|
| ☐**Launch Processing:** ☒**Launch:** ☐**Rendezvous / Docking:**<br>☒**Deployment:** ☒**Orbital Assembly & Checkout:** ☒**On-orbit Operation:**<br>☐**On-orbit Maintenance:** ☐**Descent / Landing:** | |

Provider Name:

| Hazard Report #: BIRDS5-UNQ-01 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Structural Failure | | |

| Cause Number: 2 | Cause Title: Release of fragments from shatterable material. |
|---|---|

**Hazard Cause Description:**

2.1 Broken glass parts due to launch, ascent and on-orbit load (excluding crew applied load).

| Severity: | Likelihood: |
|---|---|
| ☒I (Catastrophic) ☐II (Critical) | ☐A (Probable) ☐B (Infrequent) ☒C (Remote) ☐D (Improbable) |

| Controls: | Verification Method and Status: |
|---|---|
| 2.1.1. Shatterable materials are non-stressed (no delta pressure), recessed, and supervised by the crew when in exposed use. | 2.1.1. Verification result for IVA and EVA are followings.<br><For IVA><br>The CubeSat is placed in protected storage (J-SSOD or J-SSOD-R Launch Case). Therefore, cover glass of solar cells is non-stressed, recessed and not exposed to IVA crew.<br><br><For EVA><br>J-SSOD Hazard Report (No. J-SSOD-03) and associated NCR-JAXA-JSSOD-03 are already approved with the aperture plane of the J-SSOD, the +Z plane of the satellite, as EVA No Touch Area (NTA). The other surfaces of the satellites are covered by the J-SSOD. |
| 2.1.2. Shatterable materials have passed a vibration test at flight levels/post-test visual inspection. | 2.1.2(1). The random vibration test at the flight level is performed and the visual inspection is also performed to verify if there are any damages to the shatterable materials after the test.<br>[Status] Closed: BIRDS5-VT-01, BIRDS-5 Vibration Test Report (2022/02/18)<br><br>2.1.2(2). The visual inspection is performed to verify that there are no damages to the shatterable materials or appropriate repairs are performed after all tests.<br>[Status] Closed: BIRDS5-VT-01, BIRDS-5 Vibration Test Report (2022/02/18) |

**Safety Requirements:**

Provider Name:

| Hazard Report #: BIRDS5-UNQ-01 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Structural Failure | | |

SSP51721
SSP52005F, Payload Flight Equipment Requirements and Guidelines for Safety-Critical Structures

| Detection and Warning Methods: | Additional Safety Features: |
|---|---|
| - | - |

**Cause Remarks:**

-

| Mission Phases: | Point of Contact |
|---|---|
| ☐Launch Processing: ☒Launch: ☐Rendezvous / Docking:<br>☒Deployment: ☒Orbital Assembly & Checkout: ☒On-orbit Operation:<br>☐On-orbit Maintenance: ☐Descent / Landing: | |

## Attachment BIRDS5-UNQ-01, Summary of Structural Analysis Results for 1U

### Table 1 Structural Analysis Results for 1U (Acceleration Load along X axis)

| Part | Material | Max Stress (Smax) (MPa) | Yield strength (MPa) | Ultimate Strength, Ftu(MPa) | MS (Yield) FS=1.5 | MS (Ultimate) FS=2 | Smax/Ftu [%] <30 |
|---|---|---|---|---|---|---|---|
| Structure-01 | A6061-T6 | 37.9 | 276 | 310 | 3.9 | 3.1 | 12.2 |
| Structure-02 | A6061-T6 | 78.6 | 276 | 310 | 1.3 | 1.0 | 25.4 |
| Structure-03 | A6061-T6 | 40.4 | 276 | 310 | 3.6 | 2.8 | 13.0 |
| Structure-04 | A6061-T6 | 50.3 | 276 | 310 | 2.7 | 2.1 | 16.2 |
| Structure-05 | A6061-T6 | 38.7 | 276 | 310 | 3.7 | 3.0 | 12.5 |
| Structure-06 | A6061-T6 | 30.3 | 276 | 310 | 5.1 | 4.1 | 9.8 |
| Structure-07 | A6061-T6 | 25.3 | 276 | 310 | 6.3 | 5.1 | 8.2 |
| Structure-08 | A6061-T6 | 21.1 | 276 | 310 | 7.7 | 6.4 | 6.8 |
| Structure-09 | A6061-T6 | 26.5 | 276 | 310 | 5.9 | 4.8 | 8.6 |
| Structure-10 | A6061-T6 | 13.5 | 276 | 310 | 12.6 | 10.4 | 4.4 |
| Plus_Z panel | FR4 | 11.3 | 320 | 450 | 10.4 | 9.1 | - |
| Minus_Z panel | FR4 | 13.4 | 320 | 450 | 14.9 | 15.8 | - |
| Plus_X panel | FR4 | 23.0 | 320 | 450 | 8.3 | 8.8 | - |
| Minus_X panel | FR4 | 9.4 | 320 | 450 | 21.7 | 22.9 | - |
| Plus_Y panel | FR4 | 9.2 | 320 | 450 | 22.2 | 23.5 | - |
| Minus_Y panel | FR4 | 9.8 | 320 | 450 | 20.8 | 22.0 | - |

*1:Margin of Safety, *2:Factor of Safety

## Table 2 Structural Analysis Results for 1U (Acceleration Load along Y axis)

| Part | Material | Max Stress (Smax) (MPa) | Yield strength (MPa) | Ultimate Strength, Ftu(MPa) | MS (Yield) FS=1.5 | MS (Ultimate) FS=2 | Smax/Ftu [%] <30 |
|---|---|---|---|---|---|---|---|
| Structure-01 | A6061-T6 | 44.6 | 276 | 310 | 3.1 | 2.5 | 14.4 |
| Structure-02 | A6061-T6 | 79.6 | 276 | 310 | 1.3 | 0.9 | 25.7 |
| Structure-03 | A6061-T6 | 40.4 | 276 | 310 | 3.6 | 2.8 | 13.0 |
| Structure-04 | A6061-T6 | 50.4 | 276 | 310 | 2.7 | 2.1 | 16.3 |
| Structure-05 | A6061-T6 | 37.4 | 276 | 310 | 3.9 | 3.1 | 12.1 |
| Structure-06 | A6061-T6 | 30.4 | 276 | 310 | 5.1 | 4.1 | 9.8 |
| Structure-07 | A6061-T6 | 26.9 | 276 | 310 | 5.8 | 4.8 | 8.7 |
| Structure-08 | A6061-T6 | 20.9 | 276 | 310 | 7.8 | 6.4 | 6.7 |
| Structure-09 | A6061-T6 | 27.0 | 276 | 310 | 5.8 | 4.7 | 8.7 |
| Structure-10 | A6061-T6 | 14.0 | 276 | 310 | 12.1 | 10.0 | 4.5 |
| Plus_Z panel | FR4 | 11.5 | 320 | 450 | 10.2 | 8.9 | - |
| Minus_Z panel | FR4 | 13.4 | 320 | 450 | 15.0 | 15.9 | - |
| Plus_X panel | FR4 | 23.2 | 320 | 450 | 8.2 | 8.7 | - |
| Minus_X panel | FR4 | 9.5 | 320 | 450 | 21.5 | 22.7 | - |
| Plus_Y panel | FR4 | 9.2 | 320 | 450 | 22.1 | 23.4 | - |
| Minus_Y panel | FR4 | 10.3 | 320 | 450 | 19.7 | 20.8 | - |

*1:Margin of Safety, *2:Factor of Safety

## Table 3 Structural Analysis Results for 1U (Acceleration Load along Z axis)

| Part | Material | Max Stress (Smax) (MPa) | Yield strength (MPa) | Ultimate Strength, Ftu(MPa) | MS (Yield) FS=1.5 | MS (Ultimate) FS=2 | Smax/Ftu [%] <30 |
|---|---|---|---|---|---|---|---|
| Structure-01 | A6061-T6 | 53.6 | 276 | 310 | 2.4 | 1.9 | 17.3 |
| Structure-02 | A6061-T6 | 80.3 | 276 | 310 | 1.3 | 0.9 | 25.9 |
| Structure-03 | A6061-T6 | 39.8 | 276 | 310 | 3.6 | 2.9 | 12.8 |
| Structure-04 | A6061-T6 | 50.3 | 276 | 310 | 2.7 | 2.1 | 16.2 |
| Structure-05 | A6061-T6 | 50.9 | 276 | 310 | 2.6 | 2.0 | 16.4 |
| Structure-06 | A6061-T6 | 30.2 | 276 | 310 | 5.1 | 4.1 | 9.7 |
| Structure-07 | A6061-T6 | 29.9 | 276 | 310 | 5.1 | 4.2 | 9.7 |
| Structure-08 | A6061-T6 | 21.6 | 276 | 310 | 7.5 | 6.2 | 7.0 |
| Structure-09 | A6061-T6 | 27.3 | 276 | 310 | 5.7 | 4.7 | 8.8 |
| Structure-10 | A6061-T6 | 13.7 | 276 | 310 | 12.4 | 10.3 | 4.4 |
| Plus_Z panel | FR4 | 11.7 | 320 | 450 | 10.0 | 8.7 | - |
| Minus_Z panel | FR4 | 13.4 | 320 | 450 | 15.0 | 15.8 | - |
| Plus_X panel | FR4 | 9.3 | 320 | 450 | 22.0 | 23.3 | - |
| Minus_X panel | FR4 | 9.3 | 320 | 450 | 22.0 | 23.3 | - |
| Plus_Y panel | FR4 | 9.4 | 320 | 450 | 21.8 | 23.1 | - |
| Minus_Y panel | FR4 | 10.9 | 320 | 450 | 18.6 | 19.6 | |

*1:Margin of Safety, *2:Factor of Safety

Provider Name:

| Hazard Report #: BIRDS5-UNQ-01 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Structural Failure | | |

**Attachment BIRDS5-UNQ-01, Shatterable materials used for 1U satellite**

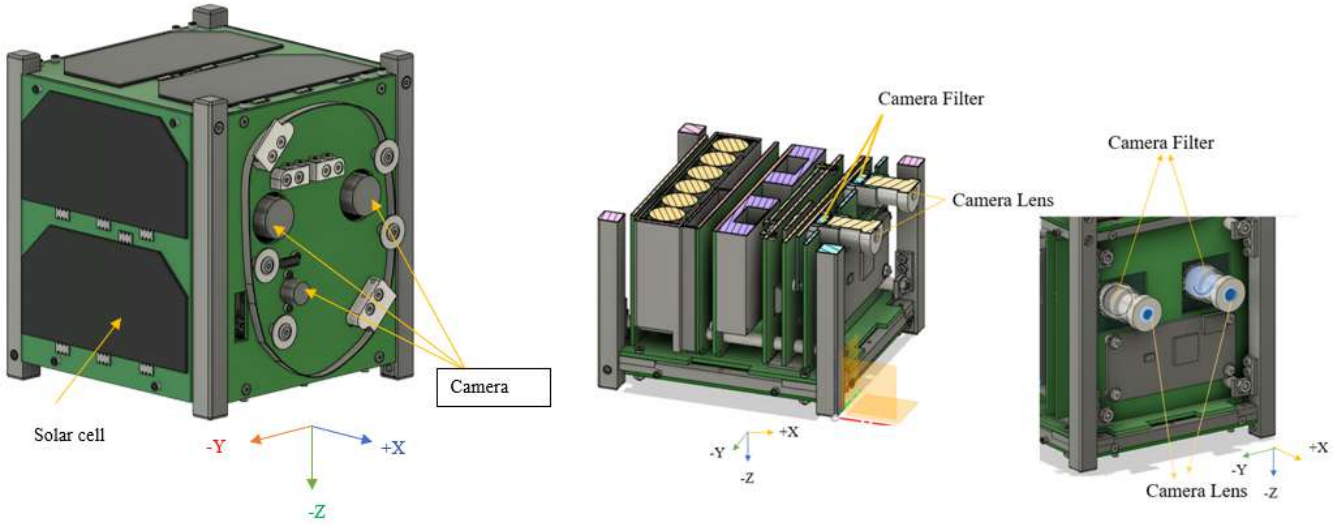



**Figure1 1U Shatter able materials**

## Attachment BIRDS5-UNQ-01, Summary of Structural Analysis Results for 2U

### Table 4 Structural Analysis Results for 2U (Acceleration Load along X axis)

| Part | Material | Max Stress (Smax) (MPa) | Yield strength (MPa) | Ultimate Strength, Ftu(MPa) | MS (Yield) FS=1.5 | MS (Ultimate) FS=2 | Smax/Ftu [%] <30 |
|---|---|---|---|---|---|---|---|
| Structure-01 | A6061-T6 | 78.1 | 276 | 310 | 1.4 | 1.0 | 25.2 |
| Structure-02 | A6061-T6 | 42.0 | 276 | 310 | 3.4 | 2.7 | 13.5 |
| Structure-03 | A6061-T6 | 39.2 | 276 | 310 | 3.7 | 3.0 | 12.6 |
| Structure-04 | A6061-T6 | 64.2 | 276 | 310 | 1.9 | 1.4 | 20.7 |
| Structure-05 | A6061-T6 | 28.5 | 276 | 310 | 5.5 | 4.4 | 9.2 |
| Structure-06 | A6061-T6 | 53.9 | 276 | 310 | 2.4 | 1.9 | 17.4 |
| Structure-07 | A6061-T6 | 61.0 | 276 | 310 | 2.0 | 1.5 | 19.7 |
| Structure-08 | A6061-T6 | 37.7 | 276 | 310 | 3.9 | 3.1 | 12.2 |
| Structure-09 | A6061-T6 | 40.8 | 276 | 310 | 3.5 | 2.8 | 13.2 |
| Structure-10 | A6061-T6 | 11.8 | 276 | 310 | 14.6 | 12.1 | 3.8 |
| Plus_Z plate | A5052P | 61.6 | 193 | 228 | 1.1 | 0.9 | 27.0 |
| Minus_Z panel | FR4 | 14.2 | 320 | 450 | 14.0 | 14.8 | - |
| Plus_X panel | FR4 | 18.6 | 320 | 450 | 10.5 | 11.1 | - |
| Minus_X panel | FR4 | 8.5 | 320 | 450 | 24.1 | 25.5 | - |
| Plus_Y panel | FR4 | 15.5 | 320 | 450 | 12.8 | 13.5 | - |
| Minus_Y panel | FR4 | 20.9 | 320 | 450 | 9.2 | 9.8 | - |

*1:Margin of Safety, *2:Factor of Safety

## Table 5 Structural Analysis Results for 2U (Acceleration Load along Y axis)

| Part | Material | Max Stress (Smax) (MPa) | Yield strength (MPa) | Ultimate Strength, Ftu(MPa) | MS (Yield) FS=1.5 | MS (Ultimate) FS=2 | Smax/Ftu [%] <30 |
|---|---|---|---|---|---|---|---|
| Structure-01 | A6061-T6 | 47.6 | 276 | 310 | 2.9 | 2.3 | 15.4 |
| Structure-02 | A6061-T6 | 38.6 | 276 | 310 | 3.8 | 3.0 | 12.5 |
| Structure-03 | A6061-T6 | 35.4 | 276 | 310 | 4.2 | 3.4 | 11.4 |
| Structure-04 | A6061-T6 | 64.4 | 276 | 310 | 1.9 | 1.4 | 20.8 |
| Structure-05 | A6061-T6 | 28.8 | 276 | 310 | 5.4 | 4.4 | 9.3 |
| Structure-06 | A6061-T6 | 53.3 | 276 | 310 | 2.5 | 1.9 | 17.2 |
| Structure-07 | A6061-T6 | 57.1 | 276 | 310 | 2.2 | 1.7 | 18.4 |
| Structure-08 | A6061-T6 | 36.0 | 276 | 310 | 4.1 | 3.3 | 11.6 |
| Structure-09 | A6061-T6 | 37.1 | 276 | 310 | 4.0 | 3.2 | 12.0 |
| Structure-10 | A6061-T6 | 9.9 | 276 | 310 | 17.6 | 14.7 | 3.2 |
| Plus_Z plate | A5052P | 57.1 | 193 | 228 | 1.3 | 1.0 | 25.0 |
| Minus_Z panel | FR4 | 14.4 | 320 | 450 | 13.8 | 14.6 | - |
| Plus_X panel | FR4 | 17.9 | 320 | 450 | 10.9 | 11.6 | - |
| Minus_X panel | FR4 | 10.3 | 320 | 450 | 19.7 | 20.8 | - |
| Plus_Y panel | FR4 | 14.0 | 320 | 450 | 14.2 | 15.1 | - |
| Minus_Y panel | FR4 | 15.6 | 320 | 450 | 12.7 | 13.4 | - |

*1:Margin of Safety, *2:Factor of Safety

## Table 6 Structural Analysis Results for 2U (Acceleration Load along Z axis)

| Part | Material | Max Stress (Smax) (MPa) | Yield strength (MPa) | Ultimate Strength, Ftu(MPa) | MS (Yield) FS=1.5 | MS (Ultimate) FS=2 | Smax/Ftu [%] <30 |
|---|---|---|---|---|---|---|---|
| Structure-01 | A6061-T6 | 50.3 | 276 | 310 | 2.7 | 2.1 | 16.2 |
| Structure-02 | A6061-T6 | 38.7 | 276 | 310 | 3.8 | 3.0 | 12.5 |
| Structure-03 | A6061-T6 | 35.4 | 276 | 310 | 4.2 | 3.4 | 11.4 |
| Structure-04 | A6061-T6 | 64.4 | 276 | 310 | 1.9 | 1.4 | 20.8 |
| Structure-05 | A6061-T6 | 28.8 | 276 | 310 | 5.4 | 4.4 | 9.3 |
| Structure-06 | A6061-T6 | 54.3 | 276 | 310 | 2.4 | 1.9 | 17.5 |
| Structure-07 | A6061-T6 | 57.3 | 276 | 310 | 2.2 | 1.7 | 18.5 |
| Structure-08 | A6061-T6 | 31.8 | 276 | 310 | 4.8 | 3.9 | 10.3 |
| Structure-09 | A6061-T6 | 37.5 | 276 | 310 | 3.9 | 3.1 | 12.1 |
| Structure-10 | A6061-T6 | 10.6 | 276 | 310 | 16.4 | 13.6 | 3.4 |
| Plus_Z plate | A5052P | 52.2 | 193 | 228 | 1.5 | 1.2 | 22.9 |
| Minus_Z panel | FR4 | 14.5 | 320 | 450 | 13.7 | 14.5 | - |
| Plus_X panel | FR4 | 17.9 | 320 | 450 | 10.9 | 11.6 | - |
| Minus_X panel | FR4 | 8.4 | 320 | 450 | 24.4 | 25.8 | - |
| Plus_Y panel | FR4 | 14.0 | 320 | 450 | 14.2 | 15.1 | - |
| Minus_Y panel | FR4 | 15.5 | 320 | 450 | 12.8 | 13.5 | - |

*1:Margin of Safety, *2:Factor of Safety

## Attachment BIRDS5-UNQ-01, Shatterable materials used for 2U satellite

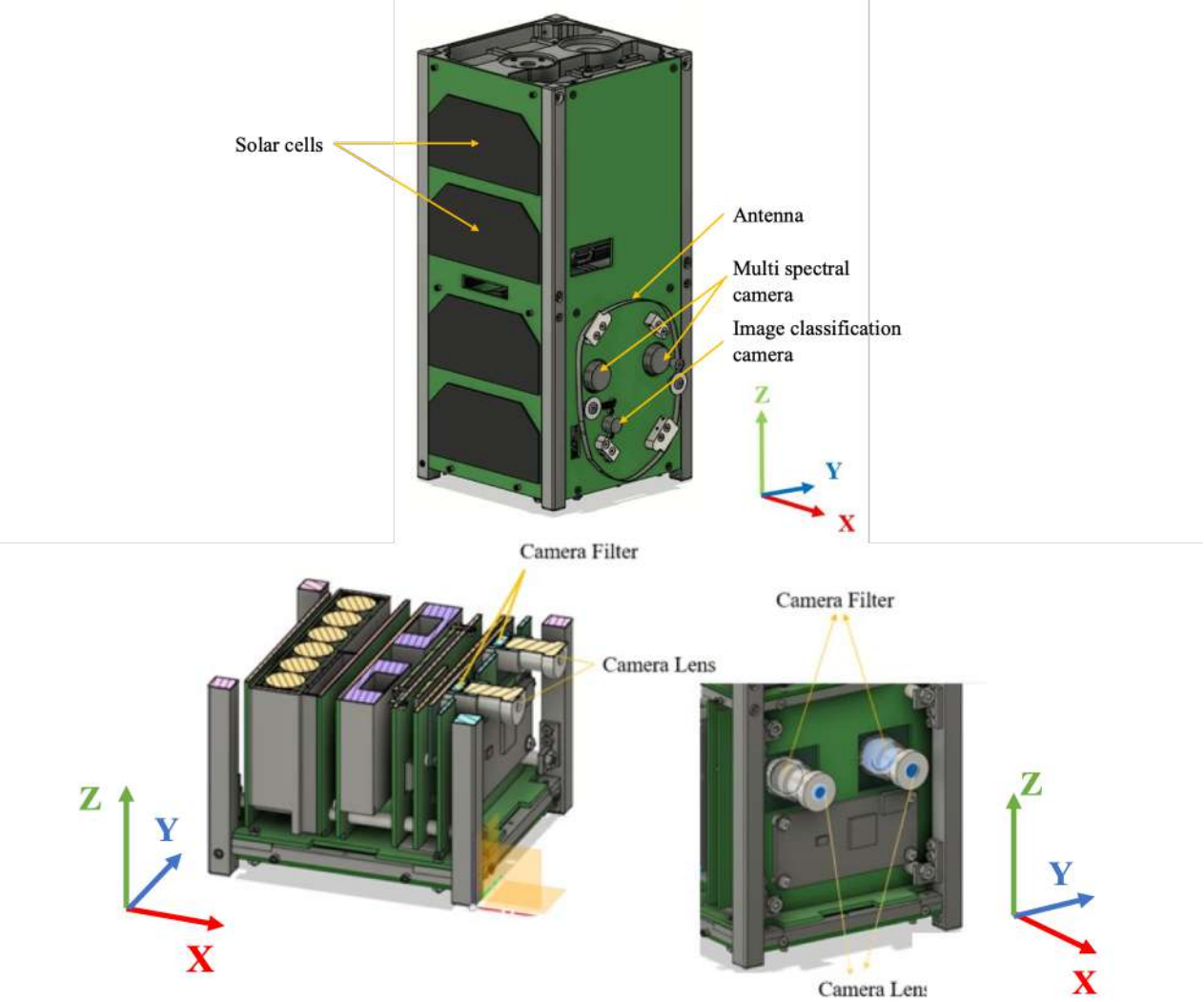



**Figure2 2U Shatter able materials**

Provider Name:

| Hazard Report #: BIRDS5-UNQ--02 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|

**Title :** Battery Leakage/Rapture

| System: BIRDS-5 | Sub-Subsystem: Battery |
|---|---|
| **Flight/Increment Applicability:**<br>HTV-X / Cygnus / Dragon<br>Inc, (TBD) and subsequent stages. | **Mission Phases:**<br>☐ Launch Processing: ☒ Launch: ☐ Rendezvous / Docking:<br>☒ Deployment: ☒ Orbital Assembly & Checkout: ☒ On-orbit Operation:<br>☐ On-orbit Maintenance: ☐ Descent / Landing: |
| **Scope:**<br>☒ Payload: ☐ JEM - PM: ☐ JEM - EF:<br>☐ Other( ): | **Interfaces:**<br>☐ JEM-PM ☐ JEM-EF ☒ JEM-AIRLOCK ☒ JEMRMS<br>☐ Other( ): |

**Hazardous Condition Description:**

Leakage of electrolyte or rupture of battery will potentially lead to contamination, corrosion, injury to ISS crew, or damage to other equipment on ISS.

Note. CubeSat uses Ni-MH battery cells.
Refer to Appendix G-2, BIRDS-5 Battery Description Form (Ni-MH, Panasonic, BK-3MCC, configuration 3S2P)

**Cause Summary:**

Cause 1. Battery Failure of the CubeSat

**Remarks:**

| Submittal Concurrence: | | Safety Review Panel Approval | |
|---|---|---|---|
| **Signature** | **Date** | **Signature** | **Date** |
| T. Yamauchi | 2026.9.15 | | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

Provider Name:

| **Hazard Report #:** BIRDS5-UNQ--02 | **Revision Date:** 15 September, 2026 | **Review Level:** Phase III |
|---|---|---|
| **Title :** Battery Leakage/Rapture | | |

| **Cause Number:** | **Cause Title:** |
|---|---|
| 1 | Battery Failure of the CubeSat |

**Hazard Cause Description:**

1.1 Battery cell internal short
1.2 Cell/battery external short
1.3 Overcharging of battery
1.4 Over-discharging of battery
1.5 Thermal Extremes

| **Severity:** | **Likelihood:** |
|---|---|
| ☒**I (Catastrophic)** ☐**II (Critical)** | ☐**A (Probable)** ☐**B (Infrequent)** ☒**C (Remote)** ☐**D (Improbable)** |

| **Controls:** | **Verification Method and Status:** |
|---|---|
| 1.1. Screening by environment test such as vibration and vacuum tests are performed to check that there is no defect in the battery cells (Refer to Figure 3). | 1.1(1). Perform random vibration test at MSL (Minimum Screening Level) for each battery cells.<br>[Status] Closed: BIRDS5-BVR-01, BIRDS-5 Battery verification report (2022/02/22)<br><br>1.1(2). Perform vacuum leak test (less than 0.1[psia] for longer than 6 hours) for each battery cells.<br>[Status] Closed: BIRDS5-BVR-01, BIRDS-5 Battery verification report (2022/02/22)<br><br>1.1(3). Confirm there is no change in charge/discharge characteristics before and after the environmental tests.<br>[Status] Closed: BIRDS5-BVR-01, BIRDS-5 Battery verification report (2022/02/22)<br><br>1.1(4). Perform the trend measurement of the open circuit voltage of the total battery, after the environment test of the satellite system.<br>[Status] Closed: BIRDS5-BVR-01, BIRDS-5 Battery verification report (2022/02/22) |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

Provider Name:

| Hazard Report #: BIRDS5-UNQ--02 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Battery Leakage/Rapture | | |

| | |
|---|---|
| 1.2-1. Battery pack is designed / assembled with appropriate electrical isolation between each cell and wiring (Refer to Figure 2). | 1.2-1. Inspection of flight hardware to ensure there is appropriate electrical isolation between each cell and wiring.<br>[Status] Closed: BIRDS5-IFTR-01, BIRDS-5 Inhibit Function Test Report (2022/02/17) |
| 1.2-2. Protection devices for short circuit are equipped to protect from external short. The protection devices are DCDC converter, SepSW2, SepSW4 and Isolation (Refer to Figure 1). | 1.2-2(1). Confirm the protective design to prevent external short for the battery.<br>[Status] Closed: BIRDS5-EP-01, BIRDS-5 Battery Description Form (2023/09/15)<br>1.2-2(2). Perform short circuit protection function test.<br>[Status] Closed: BIRDS5-IFTR-01, BIRDS-5 Inhibit Function Test Report (2022/02/17) |
| 1.3. Protection devices are equipped to protect from overcharging.<br>The protection devices are SepSW1, SepSW4 and DCDC converter. (Refer to Figure 1).<br>Note: The battery cells would not be charged even if a fluorescent light shines on the solar cells while on ISS. | 1.3(1). Confirm the protective design to prevent overcharging for the battery.<br>[Status] Closed: BIRDS5-EP-01, BIRDS-5 Battery Description Form (2023/09/15)<br>1.3(2). Perform overcharging protection function test.<br>[Status] Closed: BIRDS5-IFTR-01, BIRDS-5 Inhibit Function Test Report (2022/02/17) |
| 1.4-1. Protection device is equipped to protect from over-discharging.<br>The protection devices are DepSW2, SepSW2, SepSW3, SepSW4, DCDC converter, Diode and Isolation. (Refer to Figure 1). | 1.4-1(1). Confirm the protective design to prevent over-discharging for the battery.<br>[Status] Closed: BIRDS5-EP-01, BIRDS-5 Battery Description Form (2023/09/15)<br>1.4-1(2). Perform over-discharging protection function test.<br>[Status] Closed: BIRDS5-IFTR-01, BIRDS-5 Inhibit Function Test Report (2022/02/17)<br>1.4-1(3). Confirm the battery monitor terminals are covered with Kapton tape<br>[Status] Close to VTL: BIRDS5-VTL-01 |

| Hazard Report #: BIRDS5-UNQ--02 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Battery Leakage/Rapture | | |

| | |
|---|---|
| 1.4-2. Proper charge the battery at the time of delivery as a measure against leakage current from signal lines. | 1.4-2(1). Evaluate the leakage current of signal lines.<br>[Status] Closed: BIRDS5-UNQ-02_Leak analysis (2023/09/15)<br><br>1.4-2(2). Confirm that the battery voltage is in a state of final charge (SOC > 80%) at the time of delivery.<br>[Status] Close to VTL: BIRDS5-VTL-01 |
| 1.5. Battery cells which meet the thermal environment condition defined in JX-ESPC-101132E(Japanese)/101133E(English) JPAH Vol.8 Small Satellite Deployment ICD, are selected. | 1.5. Perform thermal vacuum test covering the J-SSOD thermal environment condition for the battery cells from same lot with the flight cells.<br>[Status] Closed: BIRDS5-BVR-01, BIRDS-5 Battery verification report (2022/02/22) |

**Safety Requirements:**

SSP51721, ISS Safety Requirements Document

| **Detection and Warning Methods:** | **Additional Safety Features:** |
|---|---|
| — | — |

**Cause Remarks:**

—

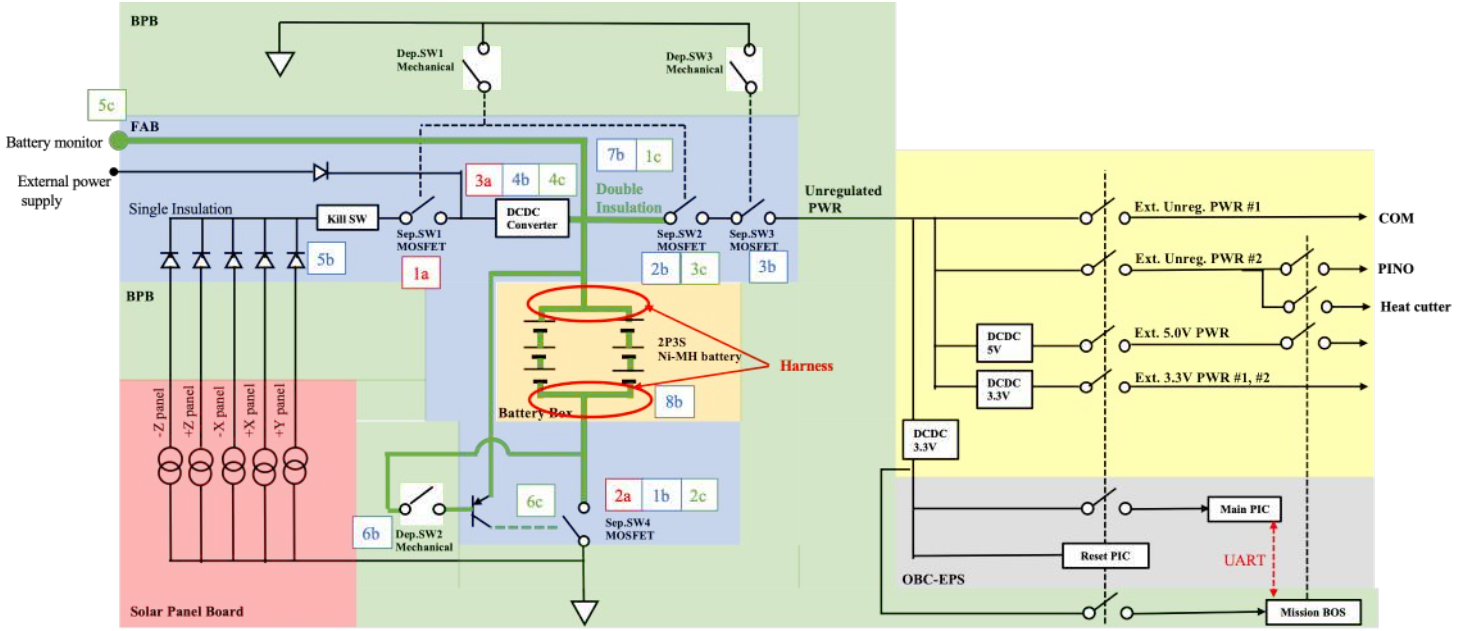
| **Mission Phases:** | **Point of Contact** |
|---|---|
| ☐ **Launch Processing:** ☒ **Launch:** ☐ **Rendezvous / Docking:**<br>☒ **Deployment:** ☒ **Orbital Assembly & Checkout:** ☒ **On-orbit Operation:**<br>☐ **On-orbit Maintenance:** ☐ **Descent / Landing:** | |

**Attachment BIRDS5-UNQ-02, Battery Description (1/3)**

Figure 1 shows inhibit schematic. There are three deployment switches (DepSW1, DepSW2 and DepSW3)

Figure 2 shows battery connectivity.

Normal electricity consumption is ~1W on orbit. BIRDS-5 does not use the wet Electrolytic Capacitors in EPS.




| Hazard | | Hazard Control #1 | Hazard Control #2 | Hazard Control #3 |
|---|---|---|---|---|
| Over-charge | | SepSW1[1a] | SepSW4[2a] | DCDC converter[3a] |
| Over-discharge | Load side | SepSW4[1b] | SepSW2[2b] | SepSW3[3b] |
| | Solar cell side | | DCDC converter[4b] | Diode[5b] |
| | DepSW2 side | DepSW2 [6b] | Proper Insulation[7b] | Proper Charging [8b] |
| External short | Load side | Proper Insulation[1c] | SepSW4[2c] | SepSW2[3c] |
| | Solar cell side | | | DCDC converter[4c] |
| | External power supply side | | | DCDC converter[4c] |
| | Battery monitor side | | | Proper Insulation[5c] |
| | DepSW2 side | | | Proper Insulation[6c] |

Note: Proper insulation (double isolation is shown by green line in figure above, single isolation is black line)
All wires and components between the battery and the first power functions are assembled as double insulation. .
The DCDC converter (LTC3119) used for inhibit is a buck-boost converter and its internal FET configuration prevents reverse current.

**Figure 1. Electrical Power Schematic**

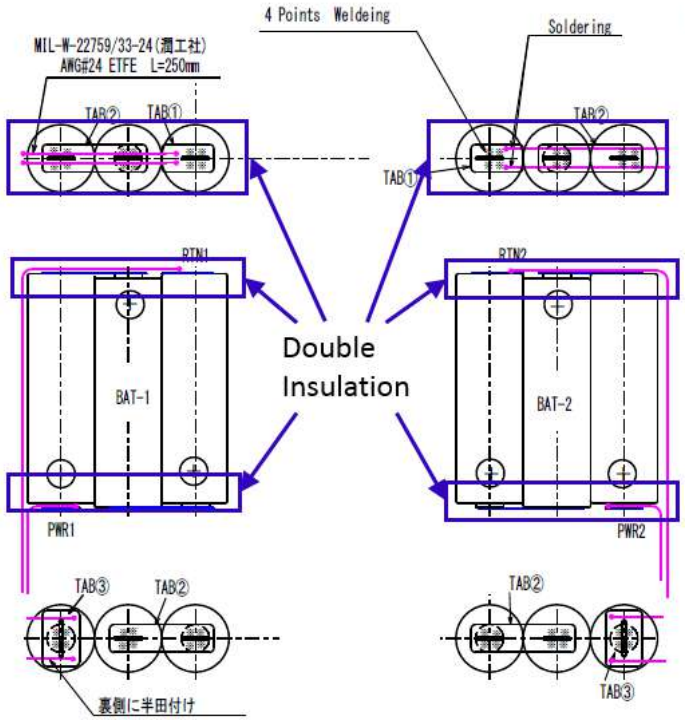
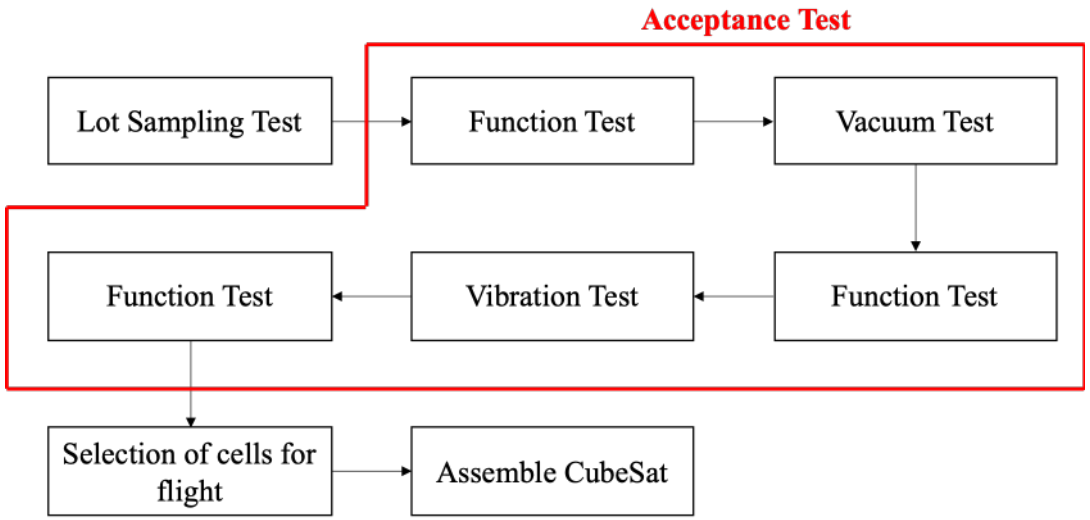



**Figure 2. Battery Connectivity**




**Function Test**

- Visual Inspection (scratches, misaligned seals, electrolyte leakage etc.)
- Measurement mass
- Measurement open circuit voltage
- Measurement charge and discharge characteristic / Temperature characteristic for discharge
- Measurement capacity

**Figure 3. Flowchart for screening test**

**Leakage current evaluation of power supply circuits**

The inhibit circuit of the BIRDS BUS has the circuit configuration shown in Figure 1. The MOSFET (SepSW4) on the battery GND side is operated by turning the transistor ON and OFF with DepSW2. Considering the bias resistance inside this transistor, etc., closing DepSW2 in Figure 2, causes current to flow through the bias resistance, etc. We analyzed the leakage current from the battery when two inhibit switches fail.

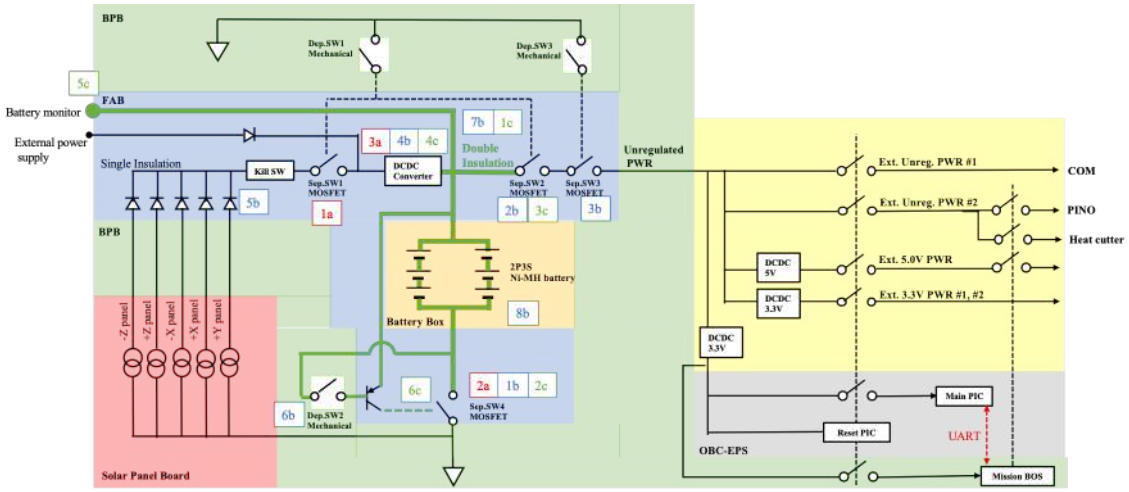



**Figure1 Inhibit diagram**

**Calculate the discharge current when only DepSW2 is turned on**

The following assumptions are made in the calculation.

・The main consumption currents are assumed to be $I_1$ to $I_5$.

・The bias resistance inside the transistor ranges from 33kΩ to 66kΩ, but as a worst-case scenario, 33kΩ is assumed.

・R17=100kΩ, R21=430kΩ

・The battery voltage is assumed to be constant at 3.6V.

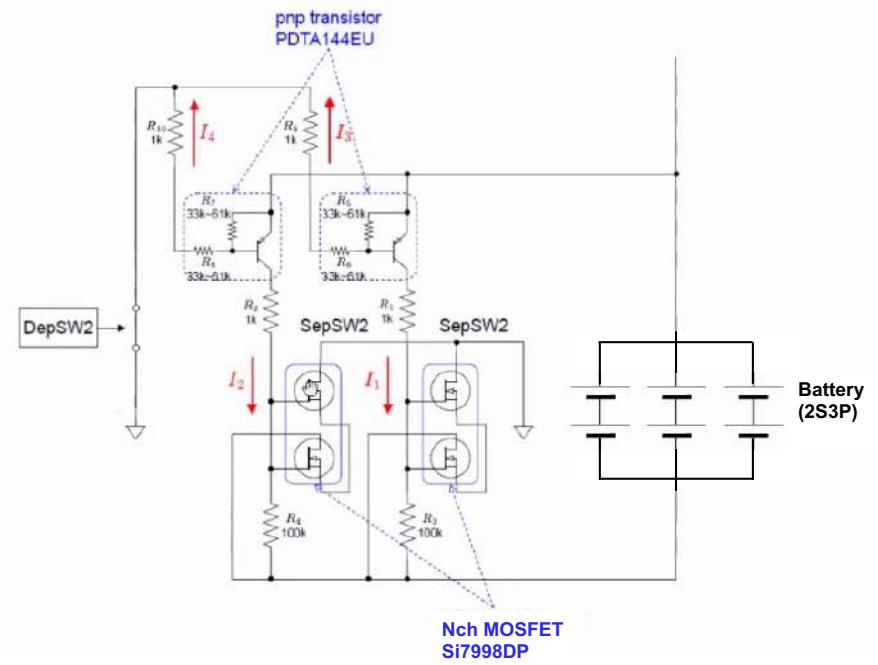



**Figure2 Detail of the separation switch on GND side**

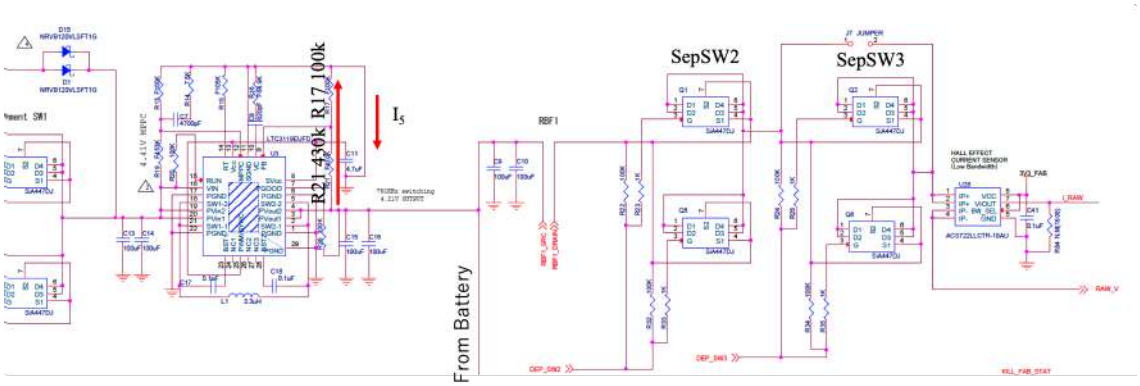



**Figure3 Resistor between battery and DCDC**

$$I_1 = \frac{V_{Bat}}{R_1 + R_3} = \frac{3.6}{10^3 + 10^5} = 35.6 \times 10^{-6}$$

$$I_2 = \frac{V_{Bat}}{R_2 + R_4} = \frac{3.6}{10^3 + 10^5} = 35.6 \times 10^{-6}$$

$$I_3 = \frac{V_{Bat}}{R_5 + R_6 + R_9} = \frac{3.6}{33 \times 10^3 + 33 \times 10^3 + 10^3} = 53.7 \times 10^{-6}$$

$$I_4 = \frac{V_{Bat}}{R_7 + R_8 + R_{10}} = \frac{3.6}{33 \times 10^3 + 33 \times 10^3 + 10^3} = 53.7 \times 10^{-6}$$

$$I_5 = \frac{V_{Bat}}{R17 + R21} = \frac{3.6}{100 \times 10^3 + 430 \times 10^3} = 6.8 \times 10^{-6}$$

$$I_{total} = I_1 + I_2 + I_3 + I_4 + I_5 = 185.4 \times 10^{-6} \text{ [A]}$$

With only DepSW2 on, a current of about 185.4 μA flows from the battery. The battery cell capacity is 2,000 mAh. Since the battery is a 3S2P configuration, the capacity at full charge is 4,000 mAh. Assuming that the battery is discharged from a full charge, the following number of days are required to use up the battery capacity.

$$t = \frac{4000}{0.1854} = 2.16 \times 10^4 [h] \approx 900 \, [day]$$

**If DepSW1 and DepSW2 have an ON fault**

Calculate the discharge current when DepSW1 and DepSW2 are turned on.
When DepSW1 and DepSW2 are turned on and the battery is connected to GND, leakage current flows through the resistors placed upstream of SepSW2 and the FETs in the DCDC convertor.
The following assumptions are made for the calculation:

・The main current consumption is assumed to be the currents $I_1$ through $I_7$.

・The internal bias resistor of the transistor are in the range of 33kΩ to 66kΩ. As a worst-case condition, R5, R6, R7 and R8 are assumed to be 33 kΩ.

・R1, R2, R9 and R10 are assumed to be 1 kΩ, R3, R4, R22, R32 and R17 are assumed to be 100 kΩ, and R21 is assumed to be 430 kΩ.

・The battery voltage is assumed to remain constant at 3.6 V.

・The leakage current of the FETs in the DCDC converter is assumed to be 10 μA based on the specifications shown in Table1.

**Table1 Electrical characteristics of DCDC convertor**

LTC3119

**ELECTRICAL CHARACTERISTICS** The ● denotes the specifications which apply over the specified operating junction temperature range, otherwise specifications are at $T_A$ = 25°C. $V_{IN}$ = $PV_{IN}$ = 12V, $PV_{OUT}$ = 5V, $R_T$ = 76.8k unless otherwise stated.

| PARAMETER | CONDITIONS | | MIN | TYP | MAX | UNITS |
|---|---|---|---|---|---|---|
| N-Channel Switch Resistance | Switch A ($PV_{IN}$ to SW1)<br>Switch B (SW1 to PGND)<br>Switch C (SW2 to PGND)<br>Switch D (SW2 to $PV_{OUT}$) | | | 30<br>30<br>30<br>30 | | mΩ<br>mΩ<br>mΩ<br>mΩ |
| N-Channel Switch Leakage | $PV_{IN}$ = $PV_{OUT}$ = 18V, SW1 = SW2 = 0V, 18V | | | 1 | 10 | µA |
| $V_{CC}$ Regulation Voltage | | ● | 3.55 | 3.70 | 3.85 | V |
| $V_{CC}$ Dropout Voltage | $V_{CC}$ Current = 50mA, $V_{IN}$ = 3V | | | 90 | | mV |
| $V_{CC}$ Current Limit | | | 180 | | | mA |
| $V_{CC}$ Reverse Current | $V_{CC}$ = 5V, $V_{IN}$ = 3V | | | 5 | | µA |

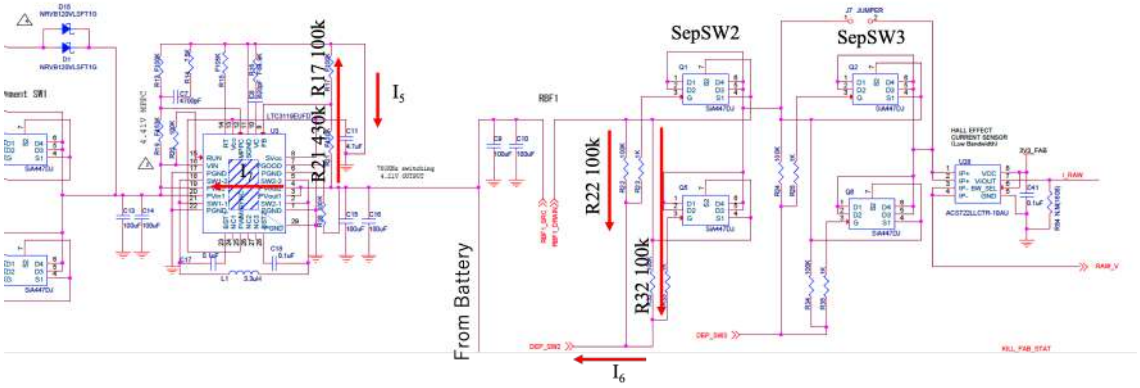



**Figure3 Detail of the separation switch on HOT side**

$$I_6 = \frac{V_{Bat}(R_{22} + R_{32})}{R_{22}R_{32}} = \frac{3.6(100 \times 10^3 + 100 \times 10^3)}{100 \times 10^3 \times 100 \times 10^3} = 72 \times 10^{-6}$$

$$I_7 = 10 \times 10^{-6}$$

$$I_{total} = I_1 + I_2 + I_3 + I_4 + I_5 + I_6 + I_7 = 267.4 \times 10^{-6}[A]$$

The total leakage current $I_{total}$ is about 0.267 mA. The capacity at which the battery reaches over-discharge is 4,000 mAh, so if the battery is discharged from a full charge, it takes the following number of days for the battery to become over-discharged.

$$t = \frac{4000}{0.267} = 14.98 \times 10^3[h] \approx 624[day]$$

This number of days is well more than one year from satellite delivery to release. The calculations assume a fully charged battery at rated capacity. In an actual battery, the time until over-discharge is expected to be somewhat shorter than this calculation result, but it is still considered to be sufficient.

Provider Name:

| Hazard Report #: BIRDS5-UNQ-03 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|

**Title :** Exposure of the ISS to Excessive Levels of EMI radiation and RF radiation

| System: BIRDS-5 | Sub-Subsystem: Non-ionizing Radiation |
|---|---|
| **Flight/Increment Applicability:**<br>HTV / Cygnus / Dragon<br>Inc.(TBD) and subsequent stages. | **Mission Phases:**<br>☐Launch Processing: ☒Launch: ☐Rendezvous / Docking:<br>☒Deployment: ☒Orbital Assembly & Checkout: ☒On-orbit Operation:<br>☐On-orbit Maintenance: ☐Descent / Landing: |
| **Scope:**<br>☒Payload: ☐JEM - PM: ☐JEM - EF:<br>☐Other( ): | **Interfaces:**<br>☐JEM-PM ☐JEM-EF ☒JEM-AIRLOCK ☒JEMRMS<br>☐Other( ): |

**Hazardous Condition Description:**

Inadvertent RF radiation and/or EMI from VHF ([redacted] MHz) in the habitable volume induce hazardous effects on ISS avionics/circuitry, other payloads, and/or visiting vehicles.

| Freq. [MHz] | Transmit Power [W] | Antenna Gain [dBi] | Max. Radiation Power [W] | Electrical Field Strength [V/m] (*1) | Power Density [W/m$^2$] | Criteria(*2) | |
|---|---|---|---|---|---|---|---|
| | | | | | | Max. Radiation Power [W] | Electrical Field Strength [V/m] |
| [redacted] | 0.5 | 2 | 0.79 | 4.88 | -39 | 0.075 | 1.58 |

(*1) at 1[m] away from the source. (*2) OE-14-002.

[Note1]: Regarding the power supply from the solar cell, the amount of light in the ISS is about 1,400Lm / m$^2$ (= 2.1W / m$^2$). The solar cells used on the outer surface of CubeSat are two in series on each side, and are attached to five sides for a total of ten solar cells for 1U CubeSat and a fourteen solar cells for 2U CubeSat. The area of each cell is 0.003mm2, and the total energy that can be generate is about 42.2mW for 1U CubeSat and 59.1mW for 2U CubeSat. The minimum power required to operate the Reset PIC, which controls the power supply to the OBC and COM, is 380 mW. Therefore, OBC and ResetPIC cannot be started with the power from the solar cell, and power cannot be supplied to the COM device.

**Cause Summary:**

Cause 1. Exposure of the ISS to Excessive Levels of EMI radiation and RF radiation

**Remarks:**

| Submittal Concurrence: | | Safety Review Panel Approval | |
|---|---|---|---|
| **Signature** | **Date** | **Signature** | **Date** |
| T.Yamauchi | 2026.9.15 | | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

Provider Name:

| Hazard Report #: BIRDS5-UNQ-03 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Exposure of the ISS to Excessive Levels of EMI radiation and RF radiation | | |

| Cause Number: 1 | Cause Title: Exposure of the ISS to Excessive Levels of EMI radiation and RF radiation |
|---|---|

**Hazard Cause Description:**
1.1 Electromagnetic Radiation (Non-ionizing) causing injury to crew or interference with ISS systems

| Severity: | Likelihood: |
|---|---|
| ☒I (Catastrophic) ☐II (Critical) | ☐A (Probable) ☐B (Infrequent) ☒C (Remote) ☐D (Improbable) |

| Controls: | Verification Method and Status: |
|---|---|
| 1.1. The transmitter is designed to provide adequate controls of the RF radiation hazard by using three separation switches. (Refer to Figure 1)<br><br>[Inhibits for RF transmission]<br>(1) Separation Switch #2<br>(2) Separation Switch #3<br>(3) Separation Switch #4 | 1.1. Perform function test of the protection devices.<br>[Status] Closed: BIRDS5-IFTR-01, BIRDS-5 Inhibit Function Test Report (2022/02/17) |

**Safety Requirements:**
SSP51721, ISS Safety Requirements Document

| Detection and Warning Methods: | Additional Safety Features: |
|---|---|
| — | It does not emit RF with the electromotive force of Solar Array |

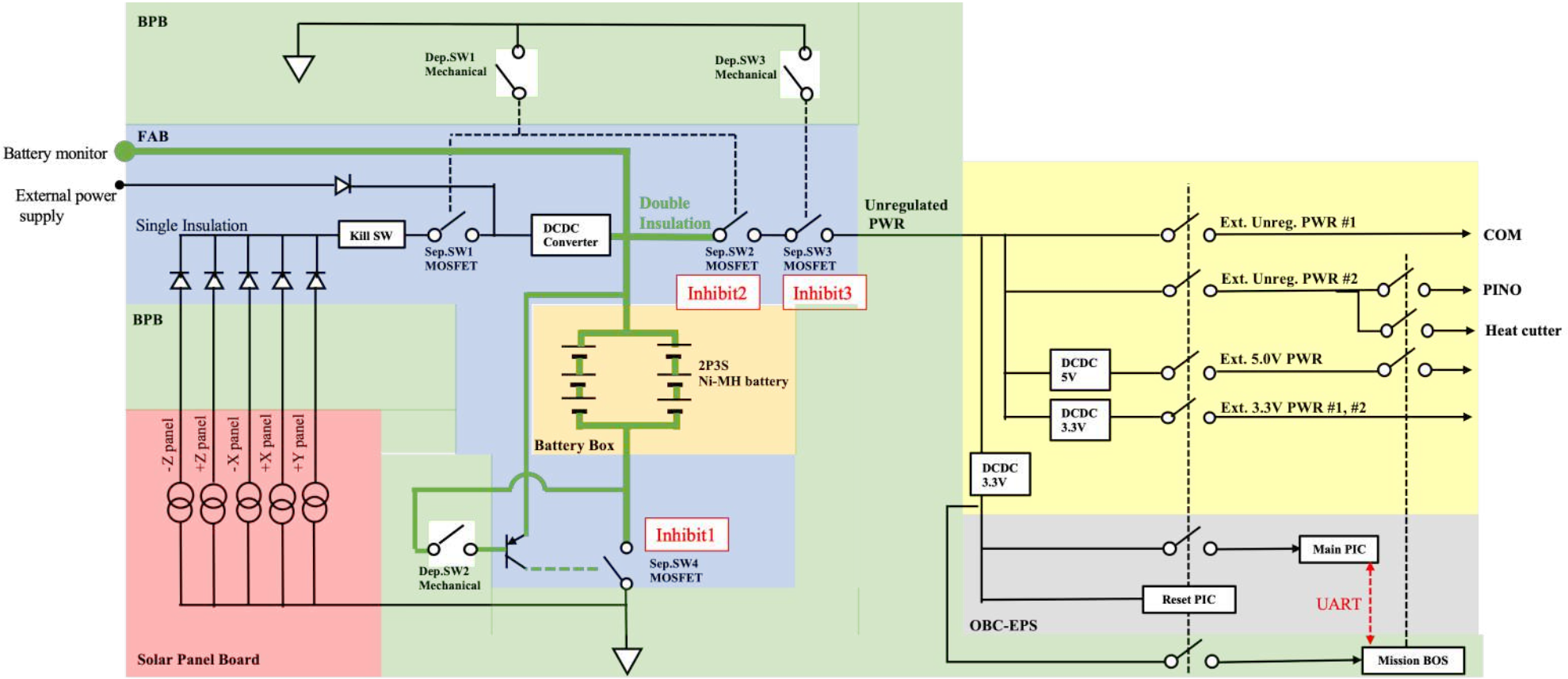
**Cause Remarks:**
—

| Mission Phases: | Point of Contact |
|---|---|
| ☐Launch Processing: ☒Launch: ☐Rendezvous / Docking:<br>☒Deployment: ☒Orbital Assembly & Checkout: ☒On-orbit Operation:<br>☐On-orbit Maintenance: ☐Descent / Landing: | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

**Attachment-1 of BIRDS5-UNQ-03, BIRDS-5 Inhibit Description (1/6)**

When at least one of the three separation switches are turned off, current from battery to the communication system located at Load is cut by disconnecting circuit.




| | Inhibit1 | Inhibit2 | Inibit3 |
|---|---|---|---|
| Deactivate COM system | SepSW4 | SepSW2 | SepSW3 |

**Figure 1. Inhibit Schematic**

## Attachment BIRDS5-UNQ-03, BIRDS-5 RF sequence Description (2/6)

**The sequence of RF radiation is as follows.**

1. DepSW1 keeps SepSW1 and SepSW2, DepSW2 keeps SepSW4 and DepSW3 keeps SepSW3 open until the satellite is released.
2. When the DepSW1, DepSW2, and DepSW3 are closed after the satellite is released from the ISS, the PIC microcomputer (Reset PIC) is worked.
3. After the Reset PIC counts for 2,000 seconds, it starts energizing the Main PIC, COM.
4. The Main PIC gives instructions to the Mission BOS PIC.
5. The Mission BOS PIC receives the instruction and turns on the power of the heat cutter.
6. Antennas are deployed, and communication begins.

**Attachment BIRDS5-UNQ-03, BIRDS-5 Verification of solar cell power (3/6)**

Since walls of J-SSOD-R launch case are transparent, if there are not enough inhibits and all inhibits located on the closed circuit from solar cell to load are accidentally closed, there is possibility to generate power with solar cells and activate some function of satellite which may cause hazard. To prove this function are not activated with the power generated by light inside JEM, it is required to show the power for activating the function is larger than the generated power with solar cells.

[Presupposition]

All we got from JAXA are shown below.

1. Graph of Relative Intensity of 3 mode of GLA (General Luminaire Assembly) (Figure 5)
2. Value of Max Illuminance at 1m from deck*: 1400[lx]

*There are 8 GLA in every 1m on each standoff of AFT/OVHD and FWD/OVHD inside JEM PM as shown in Figure 6. It is little difficult to calculate the distance from all GLA to work space, so this value was calculated by considering the work space surrounded by 2 GLA which has 8 times power of real GLA for each are located on AFT/OVHD and FWD/OVHD. Thus, it is very conservative value.

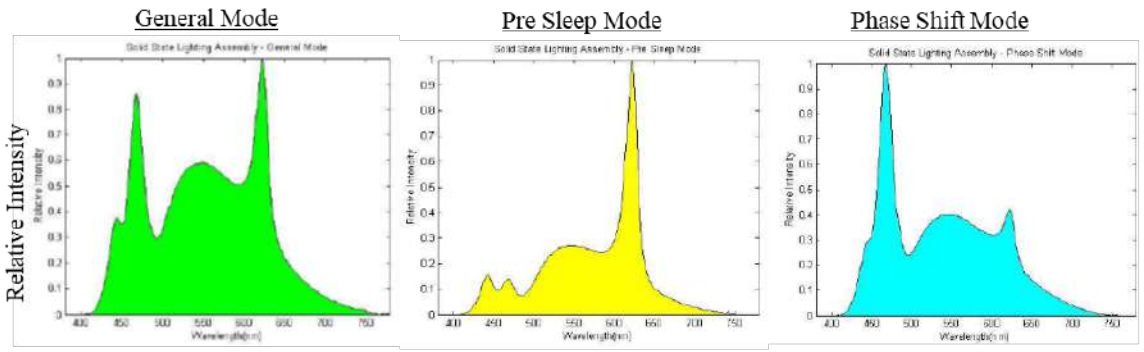



Figure 5: Relative Intensity

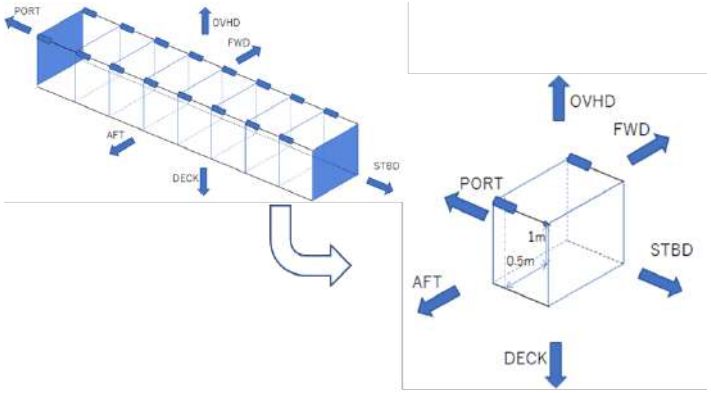



Figure 6: Illuminance inside JEM

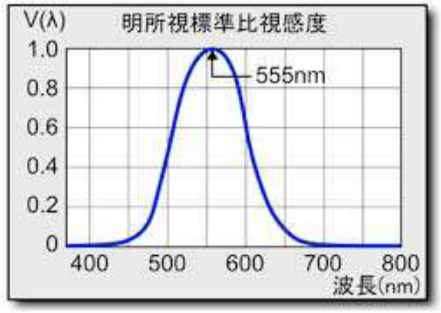
## Attachment BIRDS5-UNQ-03, BIRDS-5 Verification of solar cell power (4/6)

Illuminance is calculated by the following formula.

Illuminance[lx] = Σ (683[lm/W] x Luminosity Function @each wave length x Irradiance [W/m2)@each wave length)

= Σ (683 x A x Relative Intensity @each wave length x Irradiance [W/m2)@each wave length)

683: Max Luminous Sensitivity. , Luminosity Function: shown right. It is defined as CIE1924., A: Absolute Irradiance/Relative Intensity




To get total irradiance [W/m2] of all wavelengths of the GLA, we have to integrate Irradiance @each wavelength. And, to get absolute Irradiance@ each wavelength, factor A (Absolute Irradiance/Relative Intensity) is needed. With the value read from each relative intensity graph, luminosity function and Max Illuminance 1400[lx], the whole irradiance was calculated, and the result are shown below.

General Mode

A: 3.32 x 10^-2

Irradiance: 4.47 [W/m2]

Pre-Sleep Mode

A: 7.26 x 10^-2

Irradiance: 4.19 [W/m2]

Phase Shift Mode

A: 5.32 x 10^-2

Irradiance: 4.69 [W/m2]

**Attachment BIRDS5-UNQ-03, BIRDS-5 Verification of solar cell power (5/6)**

## ・1U CubeSat case

## [Generated Power]

Spec, number of solar cells of BIRDS-5 is shown below.

Area of each solar cell: 0.003 [m2]

Number of solar cells: 10

Conversion Efficiency: 30 [%] *  *Exact value of this solar cells is unknown. So, theoretical max value is used.

The generated power on each mode is calculated as shown below.

General Mode

Generated Power : 4.47 x 0.003 x 10 x 0.3 = 0.0402[W]

Pre-Sleep Mode

Generated Power : 4.19 x 0.003 x 10 x 0.3 = 0.0377[W]

Phase Shift Mode

Generated Power : 4.69 x 0.003 x 10 x 0.3 = 0.0422[W]

## ・2U CubeSat case

## [Generated Power]

Spec, number of solar cells of BIRDS-5 is shown below.

Area of each solar cell: 0.003[m2]

Number of solar cell: 14

Conversion Efficiency: 30[%]*  *Exact value of this solar cells are unknown. So, theoretical max value is used.

**Attachment BIRDS5-UNQ-03, BIRDS-5 Verification of solar cell power (6/6)**

The generated power on each mode is calculated as shown below.

General Mode

Generated Power : 4.47 x 0.003 x 14 x 0.3 = 0.0563[W]

Pre-Sleep Mode

Generated Power : 4.19 x 0.003 x 14 x 0.3 = 0.0528 [W]

Phase Shift Mode

Generated Power : 4.69 x 0.003 x 14 x 0.3 = 0.0591[W]

## [Required Power for Activate Function]

The function of the BRIDS-5 whose accidental activation may cause hazard are a battery heater and a RF radiation.

The battery heater is activated by the OBC.

The minimum power required to operate the Reset PIC, which controls the power supply to the OBC and COM is shown below.

Activation of Reset PIC: 0.380 [W]

## Conclusion

The required power for activating the battery heater and the COM is very larger than power generated with solar cells with light inside ISS, even Reset PIC cannot be activated with power generated with solar cells. Thus, there is no possibility to cause hazard by this function activation.

Provider Name:

| Hazard Report #: BIRDS5-UNQ-04 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Impact / Collision to ISS due to inappropriate CubeSat deployment from J-SSOD by inadvertently-deployment | | |

| System: BIRDS-5 | Sub-Subsystem: Mechanical |
|---|---|
| **Flight/Increment Applicability:**<br>HTV / Cygnus / Dragon<br>Inc.(TBD) and subsequent stages. | **Mission Phases:**<br>☐Launch Processing: ☒Launch: ☐Rendezvous / Docking:<br>☒Deployment: ☒Orbital Assembly & Checkout: ☒On-orbit Operation:<br>☐On-orbit Maintenance: ☐Descent / Landing: |
| **Scope:**<br>☒Payload: ☐JEM - PM: ☐JEM - EF:<br>☐Other( ): | **Interfaces:**<br>☐JEM-PM ☐JEM-EF ☒JEM-AIRLOCK ☒JEMRMS<br>☐Other( ): |

**Hazardous Condition Description:**

Inadvertent deployment of antenna inside J-SSOD will potentially cause collision with ISS structure due to sticking of the CubeSat antenna into any gap of the J-SSOD inner surface or adjacent satellites. Also, inappropriate design and/or manufacturing of the satellite may lead to inappropriate satellite deployment from J-SSOD.

**Cause Summary:**

Cause 1. Inadvertently Deployed of the CubeSat deployment

**Remarks:**

| Submittal Concurrence: | | Safety Review Panel Approval | |
|---|---|---|---|
| **Signature** | **Date** | **Signature** | **Date** |
| T.Yamauchi | 2026.9.15 | | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

Provider Name:

| Hazard Report #: BIRDS5-UNQ-04 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Impact / Collision to ISS due to inappropriate CubeSat deployment from J-SSOD by inadvertently-deployment | | |

| Cause Number: | Cause Title: |
|---|---|
| 1 | Inadvertently Deployed of the CubeSat deployment |

**Hazard Cause Description:**

1.1 Sticking due to inadvertent deployment inside J-SSOD.
1.2 Inappropriate design or manufacturing of the satellite.

[Note 1]. Unintentional deployment of the antennas is controlled by three deployment switches. These three deployment switches are released after deploying the CubeSat from the Satellite Install Case, then the CubeSat activates, and the deployment items are deployed after 30 minutes.

[Note 2]: Regarding the power supply from the solar cell, the amount of light in the ISS is about 1,400Lm / $m^2$ (= 2.1W / $m^2$). The solar cells used on the outer surface of CubeSat are two in series on each side, and are attached to five sides for a total of ten solar cells for 1U CubeSat and a fourteen solar cells for 2U CubeSat. The area of each cell is 0.003mm2, and the total energy that can be generate is about 42.2mW for 1U CubeSat and 59.1mW for 2U CubeSat. The minimum power required to operate the Reset PIC, which controls the power supply to the OBC and COM, is 380 mW. Therefore, OBC and Reset PIC cannot be started with the power from the solar cell, and power cannot be supplied to the COM device.

| Severity: | Likelihood: |
|---|---|
| ☒I (Catastrophic) ☐II (Critical) | ☐A (Probable) ☐B (Infrequent) ☒C (Remote) ☐D (Improbable) |

| Controls: | Verification Method and Status: |
|---|---|
| 1.1-1 Protection devices are equipped to deactivate deployment mechanism before deployment. The protection devices are three deployment switches. | 1.1-1(1) Perform function test of the protection devices.<br>[Status] Closed: BIRDS5-IFTR-01, BIRDS5 Inhibit Function Test report (2022/02/17)<br><br>1.1-1(2) Inspection of the proper insulation.<br>[Status] Closed: BIRDS5-IFTR-01, BIRDS5 Inhibit Function Test report (2022/02/17) |
| 1.1-2 The CubeSat shall be implemented according to JAXA approved structure verification and control plan (JMX-2011303E), of which scope covers JBX-97160C, JEM Payload Fracture Control Plan. | 1.1-2 Fracture Control Evaluation Form is submitted to and approved by JAXA.<br>[Status] Closed: BIRDS5-FCE-01, BIRDS-5 Fracture Control Evaluation Form for Phase 0/I/II (2023/09/15) |

Provider Name:

| **Hazard Report #:** BIRDS5-UNQ-04 | **Revision Date:** 15 September, 2026 | **Review Level:** Phase III |
|---|---|---|
| **Title :** Impact / Collision to ISS due to inappropriate CubeSat deployment from J-SSOD by inadvertently-deployment | | |

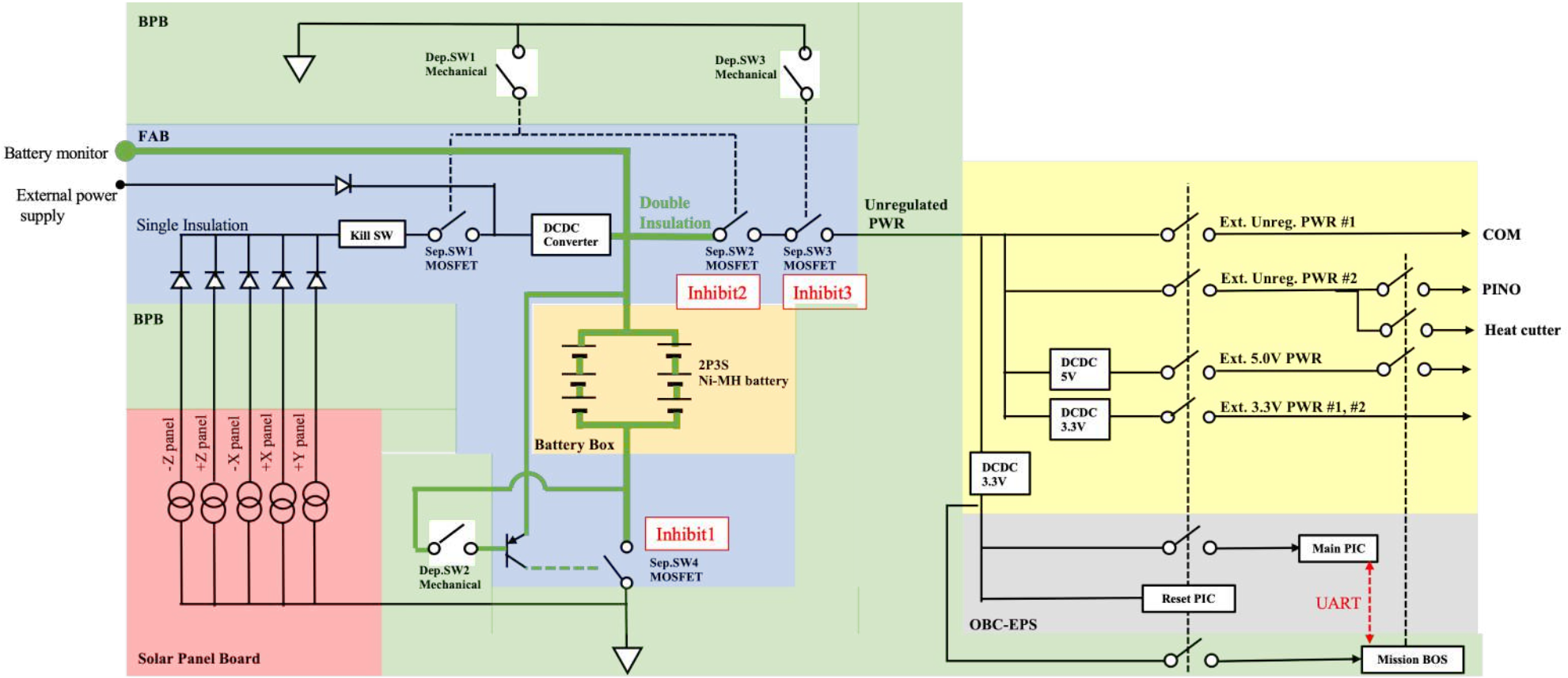
| | |
|---|---|
| 1.2 Design and Manufacture the satellite in accordance with the JX-ESPC-101132E(Japanese) / 101133E(English) dimension and mass requirements. | [Status] Closed: BIRDS5-FCE-02, BIRDS-5 Fracture Control Evaluation Form for Phase III (2023/10/06)<br><br>1.2 Inspection of as-built hardware.<br>[Status] Closed: BIRDS5-AD-01, BIRDS-5 FM Assembly Drawing (2021/12/09)<br>[Status] Closed: BIRDS5-AP-01, BIRDS-5 Assembly Procedure (2021/12/09)<br>[Status] Closed: BIRDS5-AR-01, BIRDS-5 Assembly Record (2022/02/18) |
| **Safety Requirements:**<br>SSP51721, ISS Safety Requirements Document<br>JX-ESPC-101132 E (Japanese) / 101133E (English), JEM Payload Accommodation Handbook Vol.8 | |
| **Detection and Warning Methods:**<br>— | **Additional Safety Features:**<br>— |
| **Cause Remarks:**<br>— | |
| **Mission Phases:**<br>☐ **Launch Processing:** ☒ **Launch:** ☐ **Rendezvous / Docking:**<br>☒ **Deployment:** ☒ **Orbital Assembly & Checkout:** ☒ **On-orbit Operation:**<br>☐ **On-orbit Maintenance:** ☐ **Descent / Landing:** | **Point of Contact** |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

**Attachment-1 of BIRDS5-UNQ-04 (1/8)**

When at least one of the three deployment switches are pushed in, current from battery to the heat cutter is cut by disconnecting circuit.

Furthermore, a sufficient power is not expected to be generated by solar panels to deploy the antenna since they are covered by J-SSOD's outer structure.




| | Inhibit1 | Inhibit2 | Inibit3 |
|---|---|---|---|
| Deactivate the heat cutter circuit | SepSW4 | SepSW2 | SepSW3 |

**Figure 1. Inhibit Schematic with Deployment System**

## Attachment BIRDS5-UNQ-05 Deployment System sequence Description (2/8)

**The sequence of deployment system is as follows.**

1. DepSW1 keeps SepSW1 and SepSW2, DepSW2 keeps SepSW4 and DepSW3 keeps SepSW3 open until the satellite is released.
2. When the DepSW1, DepSW2, and DepSW3 are closed after the satellite is released from the ISS, the PIC microcomputer (Reset PIC) is worked.
3. After the Reset PIC counts for 2,000 seconds, it starts energizing the Main PIC, COM.
4. The Main PIC gives instructions to the Mission BOS PIC.
5. The Mission BOS PIC receives the instruction and turns on the power of the heat cutter.
6. Antennas are deployed, and communication begins.
7. Received PINO system start command from GS.
8. Main PIC gives an instruction to start PINO system to Mission BOS PIC.
9. Mission BOS PIC receives instructions and starts supplying power to PINO system.

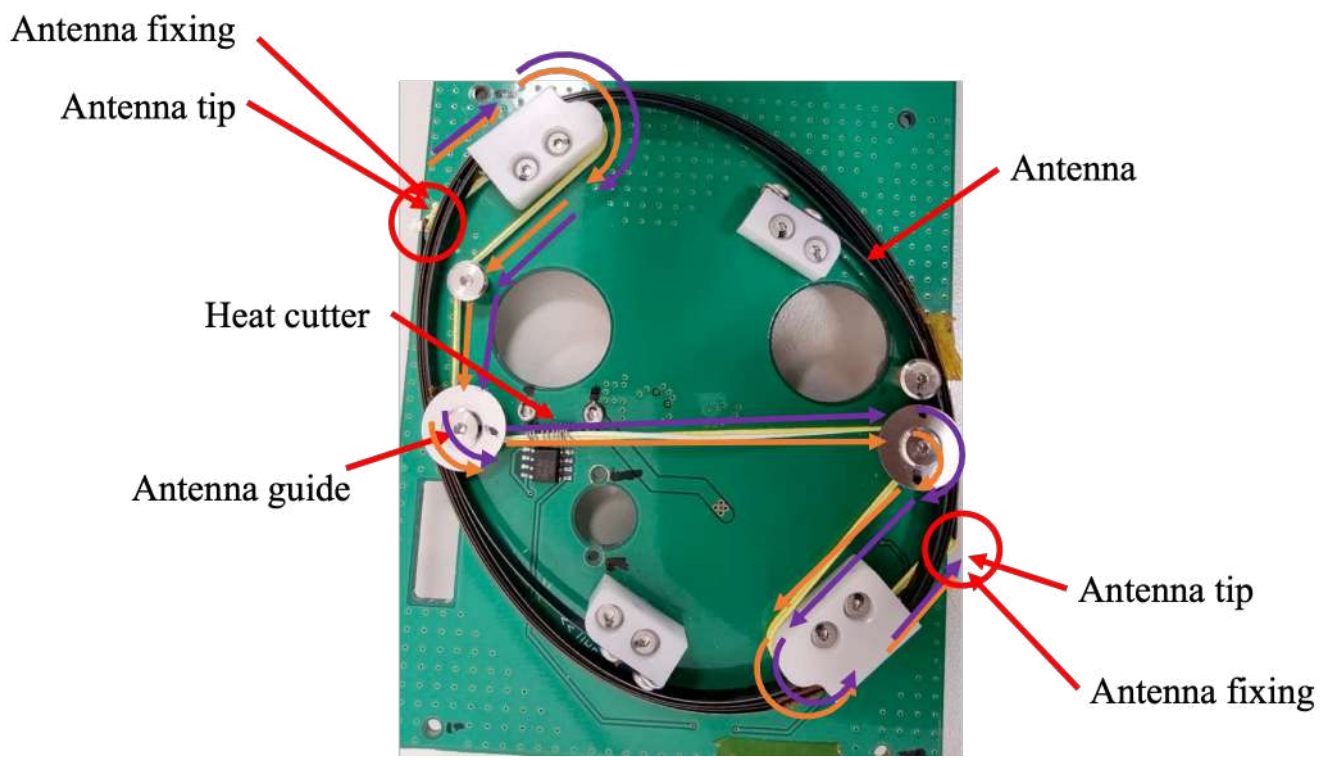



**Figure 2 Stowed Antenna Configuration**

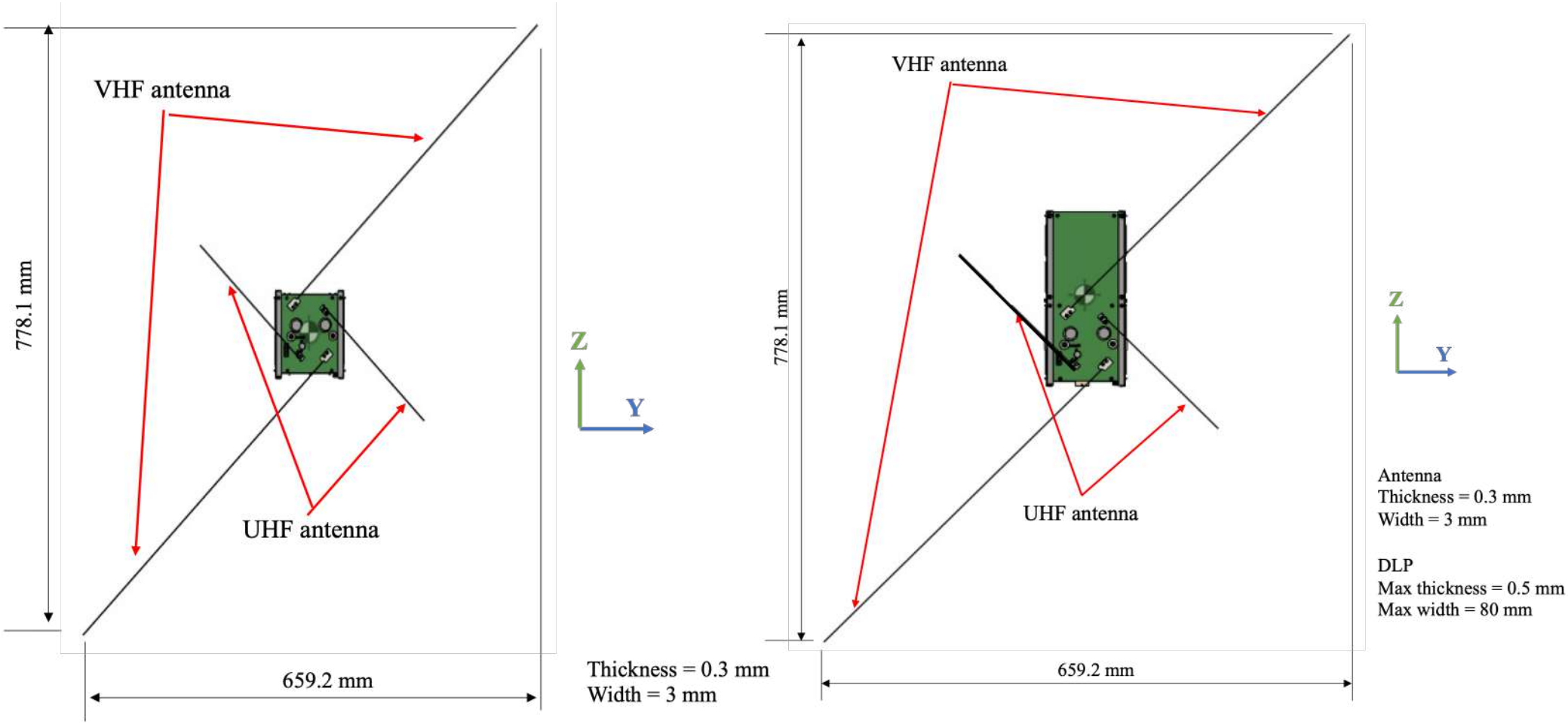



**Figure 3 Deployed Antenna Configuration**

## Attachment BIRDS5-UNQ-04, BIRDS-5 Verification of solar cell power (5/8)

Since walls of J-SSOD-R launch case are transparent, if there are not enough inhibits and all inhibits located on the closed circuit from solar cell to load are accidentally closed, there is possibility to generate power with solar cells and activate some function of satellite which may cause hazard. To prove this function are not activated with the power generated by light inside JEM, it is required to show the power for activating the function is larger than the generated power with solar cells.

[Presupposition]

All we got from JAXA are shown below.

1. Graph of Relative Intensity of 3 mode of GLA (General Luminaire Assembly) (Figure 5)
2. Value of Max Illuminance at 1m from deck*: 1400[lx]

*There are 8 GLA in every 1m on each standoff of AFT/OVHD and FWD/OVHD inside JEM PM as shown in Figure 6. It is little difficult to calculate the distance from all GLA to work space, so this value was calculated by considering the work space surrounded by 2 GLA which has 8 times power of real GLA for each are located on AFT/OVHD and FWD/OVHD. Thus, it is very conservative value.

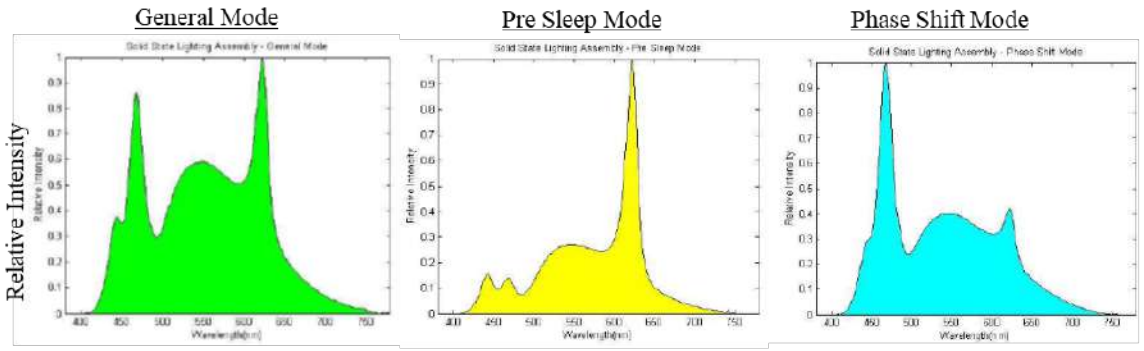



Figure 5: Relative Intensity

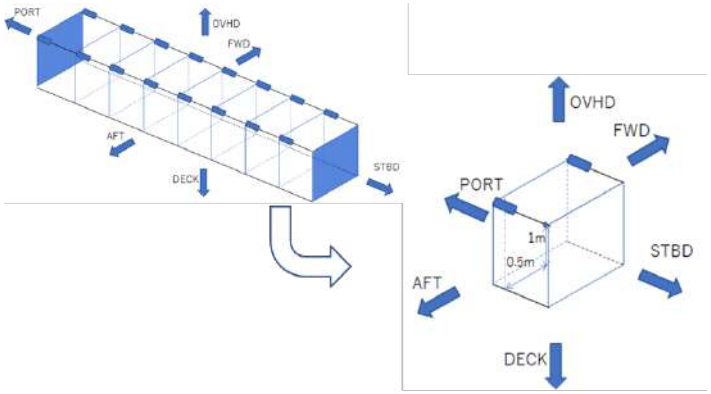



Figure 6: Illuminance inside JEM

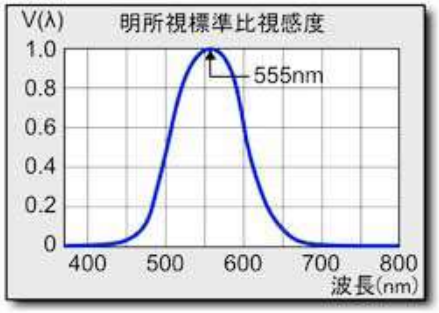
## Attachment BIRDS5-UNQ-04, BIRDS-5 Verification of solar cell power (6/8)

Illuminance is calculated by the following formula.

Illuminance[lx] = Σ (683[lm/W] x Luminosity Function @each wave length x Irradiance [W/m2)@each wave length)

= Σ (683 x A x Relative Intensity @each wave length x Irradiance [W/m2)@each wave length)

683: Max Luminous Sensitivity. , Luminosity Function: shown right. It is defined as CIE1924., A: Absolute Irradiance/Relative Intensity




To get total irradiance [W/m2] of all wavelengths of the GLA, we have to integrate Irradiance @each wavelength. And, to get absolute Irradiance@ each wavelength, factor A (Absolute Irradiance/Relative Intensity) is needed. With the value read from each relative intensity graph, luminosity function and Max Illuminance 1400[lx], the whole irradiance was calculated, and the result are shown below.

General Mode

A: 3.32 x 10^-2

Irradiance: 4.47 [W/m2]

Pre-Sleep Mode

A: 7.26 x 10^-2

Irradiance: 4.19 [W/m2]

Phase Shift Mode

A: 5.32 x 10^-2

Irradiance: 4.69 [W/m2]

**Attachment BIRDS5-UNQ-04, BIRDS-5 Verification of solar cell power (7/8)**

## ・1U CubeSat case

## [Generated Power]

Spec, number of solar cells of BIRDS-5 is shown below.

Area of each solar cell: 0.003 [m2]

Number of solar cells: 10

Conversion Efficiency: 30 [%] *  *Exact value of this solar cells is unknown. So, theoretical max value is used.

The generated power on each mode is calculated as shown below.

General Mode

Generated Power : 4.47 x 0.003 x 10 x 0.3 = 0.0402[W]

Pre-Sleep Mode

Generated Power : 4.19 x 0.003 x 10 x 0.3 = 0.0377[W]

Phase Shift Mode

Generated Power : 4.69 x 0.003 x 10 x 0.3 = 0.0422[W]

## ・2U CubeSat case

## [Generated Power]

Spec, number of solar cells of BIRDS-5 is shown below.

Area of each solar cell: 0.003[m2]

Number of solar cell: 14

Conversion Efficiency: 30[%]*  *Exact value of this solar cells are unknown. So, theoretical max value is used.

The generated power on each mode is calculated as shown below.

General Mode

Generated Power : 4.47 x 0.003 x 14 x 0.3 = 0.0563[W]

Pre-Sleep Mode

Generated Power : 4.19 x 0.003 x 14 x 0.3 = 0.0528 [W]

Phase Shift Mode

Generated Power : 4.69 x 0.003 x 14 x 0.3 = 0.0591[W]

## [Required Power for Activate Function]

The function of the BRIDS-5 whose accidental activation may cause hazard are a Heat cutter, a RF radiation and PINO.

The heat cutter is activated by the OBC.

The minimum power required to operate the Reset PIC, which controls the power supply to the OBC and COM is shown below.

Activation of Reset PIC: 0.380 [W]

## Conclusion

The required power for activating the heat cutter, the COM and the PINO is very larger than power generated with solar cells with light inside ISS, even Reset PIC cannot be activated with power generated with solar cells. Thus, there is no possibility to cause hazard by this function activation.

Provider Name:

| Hazard Report #: BIRDS5-UNQ-05 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Electric Shock | | |

| System : BIRDS-5 | Sub-Subsystem: Power Line |
|---|---|
| **Flight/Increment Applicability:**<br>HTV / Cygnus / Dragon<br>Inc.(TBD) and subsequent stages. | **Mission Phases:**<br>☐**Launch Processing:** ☒**Launch:** ☐**Rendezvous / Docking:**<br>☒**Deployment:** ☒**Orbital Assembly & Checkout:** ☒**On-orbit Operation:**<br>☐**On-orbit Maintenance:** ☐**Descent / Landing:** |
| **Scope:**<br>☒**Payload:** ☐**JEM - PM:** ☐ **JEM - EF:**<br>☐ **Other( ):** | **Interfaces:**<br>☐**JEM-PM** ☐**JEM-EF** ☒**JEM-AIRLOCK** ☒**JEMRMS**<br>☐ **Other( ):** |

**Hazardous Condition Description:**

BIRDS-5 2U "TAKA" has the electrical board generating High voltage* for the radiation measurement of precipitation of high-energy electrons in the Van-Allen radiation belt inside PINO module, which may cause electrical shock of ISS crew. To prevent electrical shock of crew, three deployment switches are equipped independently to cut the power.

*2kV maximum for PINO (Particle Instrument for Nano-satellites) module.

[Note1]: Regarding the power supply from the solar cell, the amount of light in the ISS is about 1400Lm / $m^2$ (= 2.1W / $m^2$). The solar cells used on the outer surface of CubeSat are two in series on each side, and are attached to five sides for a total of ten solar cells for 1U CubeSat and a fourteen solar cells for 2U CubeSat. The area of each cell is 0.003mm2, and the total energy that can be generate is about 42.2mW for 1U CubeSat and 59.1mW for 2U CubeSat. The minimum power required to operate the Reset PIC, which controls the power supply to the OBC and COM, is 380 mW. Therefore, OBC and ResetPIC cannot be started with the power from the solar cell, and power cannot be supplied to the COM device.

**Cause Summary:**

Cause 1. Contact to activating high voltage circuit

**Remarks:**

| Submittal Concurrence: | | Safety Review Panel Approval | |
|---|---|---|---|
| **Signature** | **Date** | **Signature** | **Date** |
| T.Yamauchi | 2026.9.15 | | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

Provider Name:

| Hazard Report #: | BIRDS5-UNQ-05 | Revision Date: | 15 September, 2026 | Review Level: | Phase III |
|---|---|---|---|---|---|
| Title : | Electric Shock | | | | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

Provider Name:

| Hazard Report #: BIRDS5-UNQ-05 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Electric Shock | | |

| Cause Number: | Cause Title: |
|---|---|
| 1 | Contact to activating high voltage circuit |

**Hazard Cause Description:**

1.1 Contact to activating high voltage circuit

| Severity: | Likelihood: |
|---|---|
| ☒**I (Catastrophic)** ☐**II (Critical)** | ☐**A (Probable)** ☐**B (Infrequent)** ☒**C (Remote)** ☐**D (Improbable)** |

| Controls: | Verification Method and Status: |
|---|---|
| 1.1-1 Protection device for activating high voltage circuit is equipped. The protection device is Separation Switch 4 by Deployment Switch 2 shown in Figure 1. | 1.1-1. Perform protection function test to check satellite is deactivated by pushing the Deployment Switch 3 after FM vibration test.<br>[Status] Closed: BIRDS5-IFTR-01, Inhibit Function Test report (2022/02/17) |
| 1.1-2 Protection device for activating high voltage circuit is equipped. The protection device is Separation Switch 2 by Deployment Switch 1 shown in Figure 1. | 1.1-2. Perform protection function test to check satellite is deactivated by pushing Deployment Switch 2 after FM vibration test.<br>[Status] Closed: BIRDS5-IFTR-01, Inhibit Function Test report (2022/02/17) |
| 1.1-3 Protection device for activating high voltage circuit is equipped. The protection device is Separation Switch 3 by Deployment Switch 3 shown in Figure 1.<br><br>[NOTE] The circuit that goes the line of the Solar cells and Load (not through the battery line) in Figure 1 has two independent inhibits of three separation switches (Sep SW 1, 2 and 3). Even if these three separation switches are closed, the satellite sustains the condition of deactivation. | 1.1-3. Perform protection function test to check satellite is deactivated by pushing Deployment Switch 1 after FM vibration test.<br>[Status] Closed: BIRDS5-IFTR-01, Inhibit Function Test report (2022/02/17) |

**Safety Requirements:**

Provider Name:

| Hazard Report #: BIRDS5-UNQ-05 | Revision Date: 15 September, 2026 | Review Level: Phase III |
|---|---|---|
| Title : Electric Shock | | |

JSC Form 1298, 10.
SSP51721, ISS Safety Requirements Document

| Detection and Warning Methods: | Additional Safety Features: |
|---|---|
| - | - |

**Cause Remarks:**

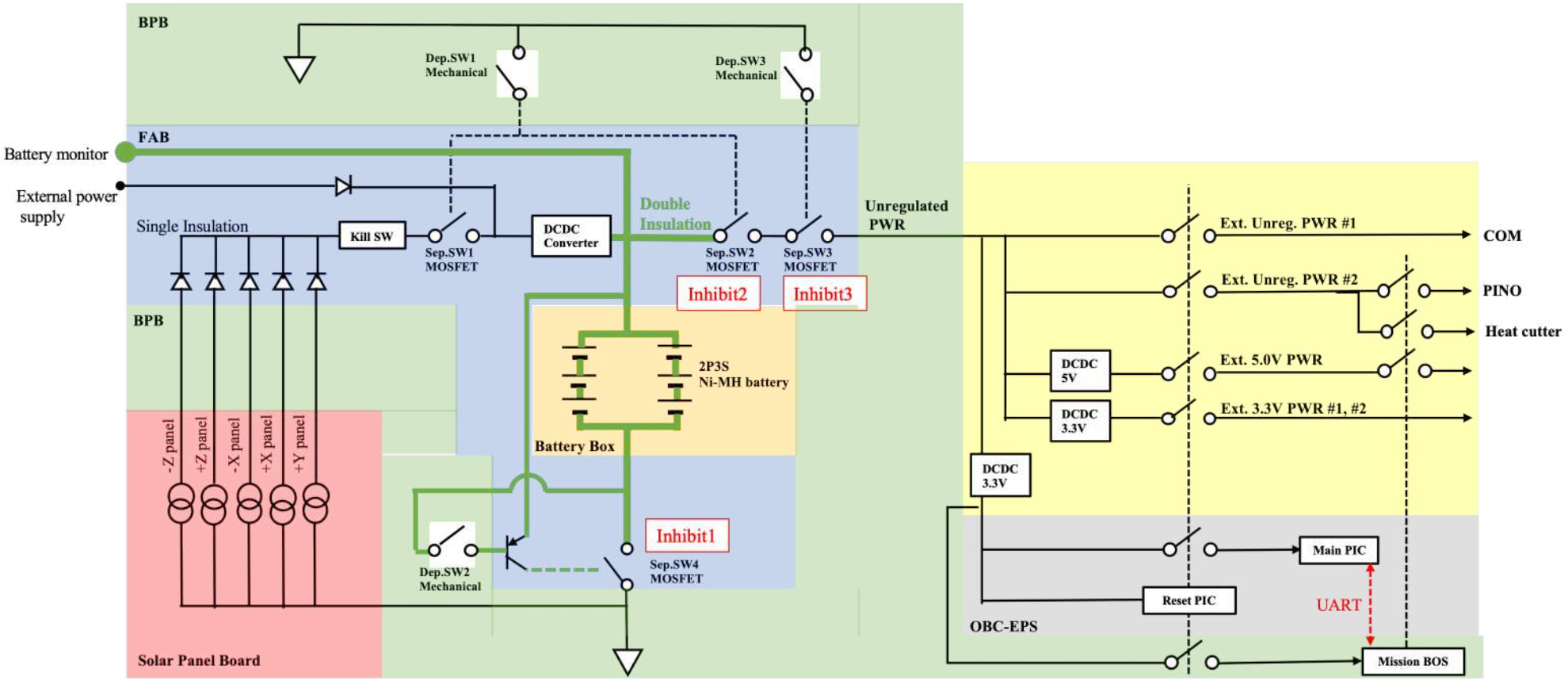
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| Mission Phases: | Point of Contact |
|---|---|
| ☐**Launch Processing:** ☒**Launch:** ☐**Rendezvous / Docking:** ☒**Deployment:** ☒**Orbital Assembly & Checkout:** ☒**On-orbit Operation:** ☐**On-orbit Maintenance:** ☐**Descent / Landing:** | |

ISS_OE_851
Export Control Classification:
Proprietary Statement (if required):

**Attachment of BIRDS5-UNQ-05 (1/7)**

When at least one of the three separation switches are turned off, current from battery to the PINO system located at Load is cut by disconnecting circuit.




| | Inhibit1 | Inhibit2 | Inibit3 |
|---|---|---|---|
| Deactivate PINO system | SepSW4 | SepSW2 | SepSW3 |

**Figure 1. Inhibit Schematic with Deployment System**

## Attachment BIRDS5-UNQ-05 PINO sequence Description (2/7)

**The sequence of PINO system is as follows.**

1. DepSW1 keeps SepSW1 and SepSW2, DepSW2 keeps SepSW4 and DepSW3 keeps SepSW3 open until the satellite is released.
2. When the DepSW1, DepSW2, and DepSW3 are closed after the satellite is released from the ISS, the PIC microcomputer (Reset PIC) is worked.
3. After the Reset PIC counts for 2,000 seconds, it starts energizing the Main PIC, COM.
4. The Main PIC gives instructions to the Mission BOS PIC.
5. The Mission BOS PIC receives the instruction and turns on the power of the heat cutter.
6. Antennas are deployed, and communication begins.
7. Received PINO system start command from GS.
8. Main PIC gives an instruction to start PINO system to Mission BOS PIC.
9. Mission BOS PIC receives instructions and starts supplying power to PINO system.

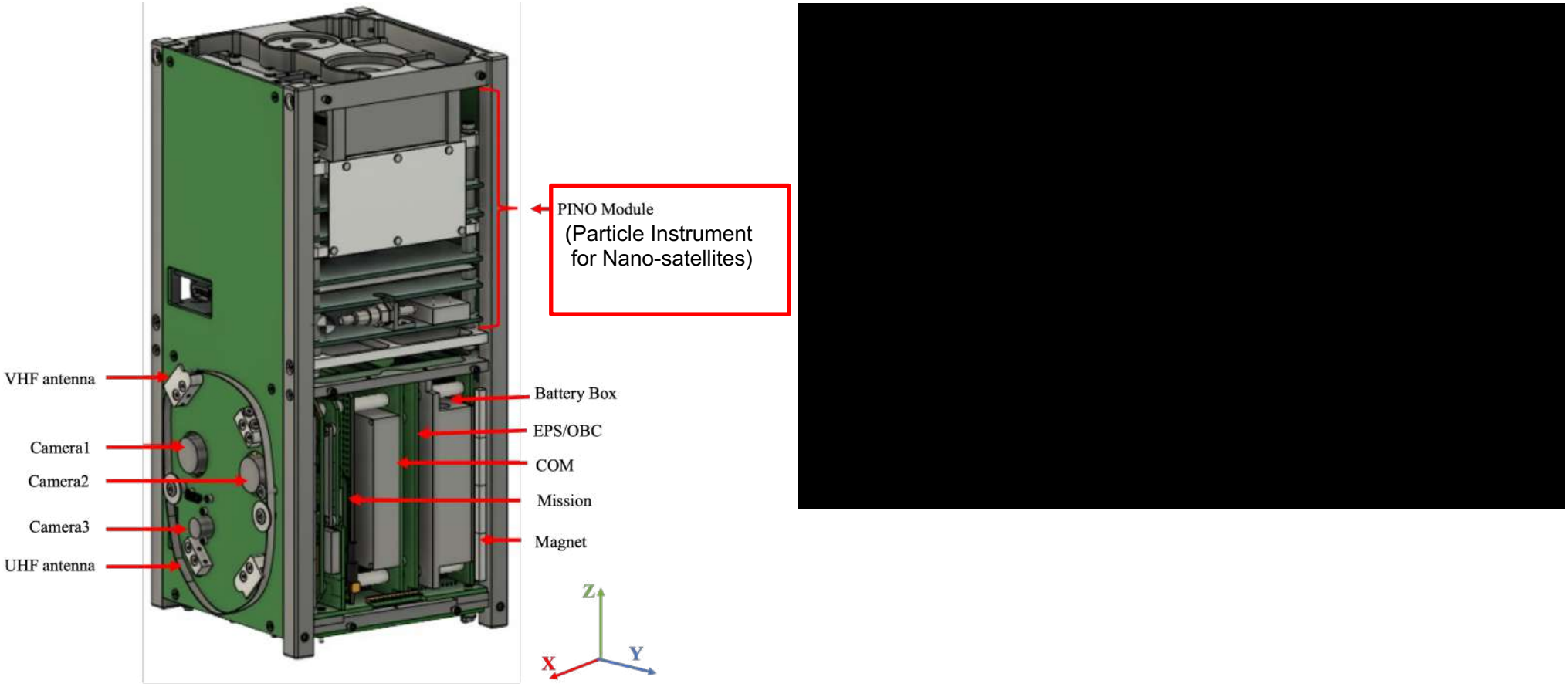



**Figure 2 PINO module and High voltage parts in the sectional view**

**Attachment BIRDS5-UNQ-05, BIRDS-5 Verification of solar cell power (4/7)**

Since walls of J-SSOD-R launch case are transparent, if there are not enough inhibits and all inhibits located on the closed circuit from solar cell to load are accidentally closed, there is possibility to generate power with solar cells and activate some function of satellite which may cause hazard. To prove this function are not activated with the power generated by light inside JEM, it is required to show the power for activating the function is larger than the generated power with solar cells.

[Presupposition]
All we got from JAXA are shown below.

1. Graph of Relative Intensity of 3 mode of GLA (General Luminaire Assembly) (Figure 5)
2. Value of Max Illuminance at 1m from deck*: 1400[lx]

*There are 8 GLA in every 1m on each standoff of AFT/OVHD and FWD/OVHD inside JEM PM as shown in Figure 6. It is little difficult to calculate the distance from all GLA to work space, so this value was calculated by considering the work space surrounded by 2 GLA which has 8 times power of real GLA for each are located on AFT/OVHD and FWD/OVHD. Thus, it is very conservative value.

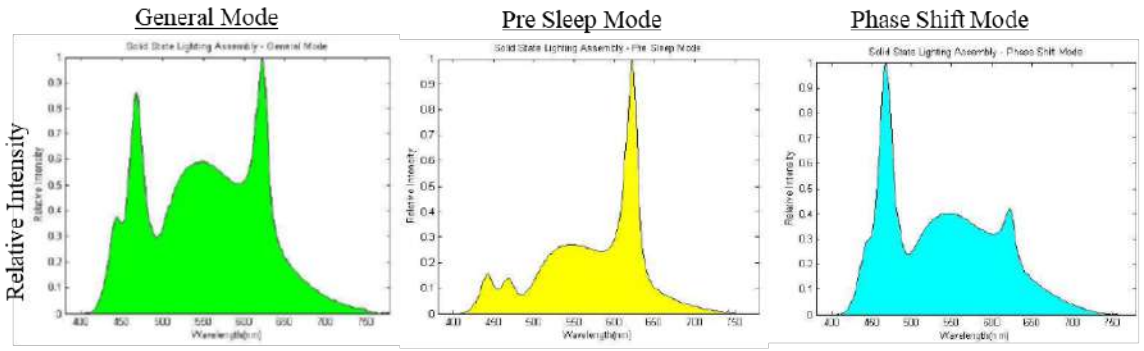



Figure 5: Relative Intensity

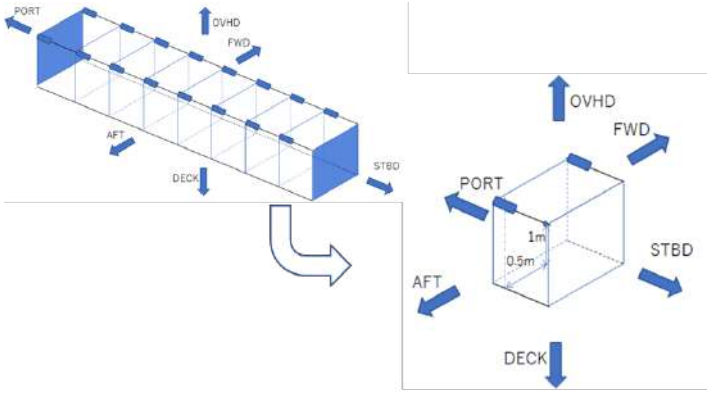
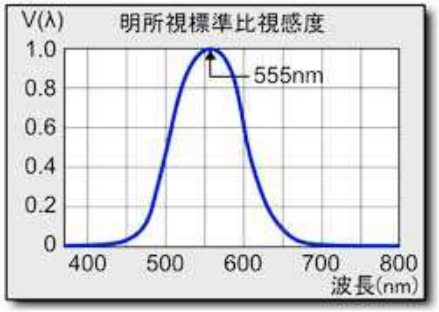



Figure 6: Illuminance inside JEM

**Attachment BIRDS5-UNQ-05, BIRDS-5 Verification of solar cell power (5/7)**

Illuminance is calculated by the following formula.
Illuminance[lx] = Σ (683[lm/W] x Luminosity Function @each wave length x Irradiance [W/m2)@each wave length)
= Σ (683 x A x Relative Intensity @each wave length x Irradiance [W/m2)@each wave length)
683: Max Luminous Sensitivity. , Luminosity Function: shown right. It is defined as CIE1924., A: Absolute Irradiance/Relative Intensity




To get total irradiance [W/m2] of all wavelengths of the GLA, we have to integrate Irradiance @each wavelength. And, to get absolute Irradiance@ each wavelength, factor A (Absolute Irradiance/Relative Intensity) is needed. With the value read from each relative intensity graph, luminosity function and Max Illuminance 1400[lx], the whole irradiance was calculated, and the result are shown below.

General Mode
A: 3.32 x 10^-2
Irradiance: 4.47 [W/m2]
Pre-Sleep Mode
A: 7.26 x 10^-2
Irradiance: 4.19 [W/m2]
Phase Shift Mode
A: 5.32 x 10^-2
Irradiance: 4.69 [W/m2]

## ・1U CubeSat case

## [Generated Power]

Spec, number of solar cells of BIRDS-5 is shown below.

Area of each solar cell: 0.003 [m2]
Number of solar cells: 10
Conversion Efficiency: 30 [%] * *Exact value of this solar cells is unknown. So, theoretical max value is used.

The generated power on each mode is calculated as shown below.

General Mode

Generated Power : 4.47 x 0.003 x 10 x 0.3 = 0.0402[W]

Pre-Sleep Mode

Generated Power : 4.19 x 0.003 x 10 x 0.3 = 0.0377[W]

Phase Shift Mode

Generated Power : 4.69 x 0.003 x 10 x 0.3 = 0.0422[W]

## ・2U CubeSat case

## [Generated Power]

Spec, number of solar cells of BIRDS-5 is shown below.

Area of each solar cell: 0.003[m2]
Number of solar cell: 14
Conversion Efficiency: 30[%]* *Exact value of this solar cells are unknown. So, theoretical max value is used.

**Attachment BIRDS5-UNQ-05, BIRDS-5 Verification of solar cell power (7/7)**

The generated power on each mode is calculated as shown below.

General Mode

Generated Power : 4.47 x 0.003 x 14 x 0.3 = 0.0563[W]

Pre-Sleep Mode

Generated Power : 4.19 x 0.003 x 14 x 0.3 = 0.0528 [W]

Phase Shift Mode

Generated Power : 4.69 x 0.003 x 14 x 0.3 = 0.0591[W]

## [Required Power for Activate Function]

The function of the BRIDS-5 whose accidental activation may cause hazard are a battery heater, a RF radiation and a PINO system.
The battery heater is activated by the OBC.
The minimum power required to operate the Reset PIC, which controls the power supply to the OBC and COM is shown below.

Activation of Reset PIC: 0.380 [W]

## Conclusion

The required power for activating the battery heater and the COM is very larger than power generated with solar cells with light inside ISS, even Reset PIC cannot be activated with power generated with solar cells. Thus, there is no possibility to cause hazard by this function activation.

# Appendix C

## Fire Detection and Suppression (FDS)

## Implementation Approach

# Appendix D

## Critical Services

Not Applicable

# Appendix E

# JAXA SR A/I Status

Not Applicable

# Appendix F

## Waiver and Deviation

Not Applicable

# Appendix G

# Support Documents

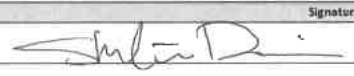
# G-1 Fracture Control Evaluation Form

## Structure Fracture Control Evaluation Form for Small Satellite deployed from J-SSOD

Satellite Name : BIRDS-5【Phase 3】

JAXA Structure and Fracture Control Board (JAXA SFCB)

Form Ver. March 23, 2018

| | Signature | DATE |
|---|---|---|
| JAXA SFCB Chair |  | 2022/04/11 |
| Satellite Project Manager | | 2022.4.4 |

If all of condition A) to G) are fulfilled or approved waiver, this form can be applicable instead of Fracture Control Summary Report.
If any of condition H) to M) are applicable, correspondent verifications must be implemented.

| Check | Title | Condition | Verification Document |
|---|---|---|---|
| ☑ | A) Soft Stowed Launch | Launched with stowed in softbag. | [Phase 012] Closed, Document No. :09_BIRDS5-SAR-01 Safety Assessment Report for Phase012 (2021/12/09) [Phase 3] Closed, Document No. :16_BIRDS5-SAR-02 Safety Assessment Report for Phase3 (2022/02/22) |
| ☑ | B) Mass | Less than 1.33kg per 1U. | [Phase 012] Closed, Document No. :09_BIRDS5-SAR-01 Safety Assessment Report for Phase012 (2021/12/09) [Phase 3] Closed, Document No. :16_BIRDS5-SAR-02 Safety Assessment Report for Phase3 (2022/02/22) |
| ☑ | C) Neither Pressure System nor Pressure Vessel | With neither Pressure System nor Pressure Vessel | [Phase 012] Closed, Document No. :09_BIRDS5-SAR-01 Safety Assessment Report for Phase012 (2021/12/09) [Phase 3] Closed, Document No. :16_BIRDS5-SAR-02 Safety Assessment Report for Phase3 (2022/02/22) |
| ☑ | D) No Hazardous Materials | No toxic or biological material except for electrolyte of battery. | [Phase 012] Closed, Document No. :09_BIRDS5-SAR-01 Safety Assessment Report for Phase012 (2021/12/09) [Phase 3] Closed, Document No. :16_BIRDS5-SAR-02 Safety Assessment Report for Phase3 (2022/02/22) |
| ☑ | E) Minimum Stress History | H/W is not exposed to the stress environment other than the followings:<br>1) Random vibration test<br>2) Ground Transportation<br>3) Launch(1 time) | [Phase 012] Closed, Document No. :09_BIRDS5-SAR-01 Safety Assessment Report for Phase012 (2021/12/09) [Phase 3] Closed, Document No. :16_BIRDS5-SAR-02 Safety Assessment Report for Phase3 (2022/02/22) |
| - | F) Low Risk Fracture Part | Outer structure meets the following criteria(as Low Risk Fracture Part) | — |
| ☑ | F-1) Maximum Stress | Total tensile stresses are no greater than 30% of ultimate tensile strength | [Phase012] Closed, Document No. :02_BIRDS5-SR-01 Structural Analysis Report (2022/03/15) |
| ☑ | F-2) Material | Aluminum alloy and A-rated in MSFC-HDBK-527/JSC09604/CR-99117 or Table-I in MSFC-STD-3029 | [Phase012] Closed, Document No. :03_BIRDS5-MIUL-01 MIUL (2021/08/03) |
| ☑ | F-3) Material Processing | Not using a process such as welding, forging, casting, or quenching heat treatment | [Phase012] Closed, Document No. :02_BIRDS5-SR-01 Structural Analysis Report (2022/03/15) |
| ☑ | F-4) Visual Inspection | No defects or surface damage is detected by visual inspection | [Phase3] Closed, Document No. :25_BIRDS5-VT-01 Vibration Test Report (2022/02/18) |
| ☑ | G) No Delta Pressure Hazard | No hazard is identified regarding the delta-pressure during launch, pressurization/ depressurization in airlock. | [Phase 012] Closed, Document No. :09_BIRDS5-SAR-01 Safety Assessment Report for Phase012 (2021/12/09) [Phase 3] Closed, Document No. :16_BIRDS5-SAR-02 Safety Assessment Report for Phase3 (2022/02/22) |

If the satellite consists of the following items, applicable section ( H) to N) ) should be fulfilled.

| ☑ applicable | H) Shatterable Structure | Shatterable Structure (Camera Lens, Solar Cell Cover etc.) | — |
|---|---|---|---|
| ☑ | H-1) Vibration Test | 1. Verified by visual inspection after vibration test under the condition specified in JX-ESPC-101132/101133 | [Phase3] Closed, Document No. :25_BIRDS5-VT-01 Vibration Test Report (2022/02/18) |
| ☑ applicable | J) Deployment Structure | Deployment restraint wire whose fracture could cause hazard. | |
| ☑ | J-1) Fail Safe Approach | Redundant wire | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☑ | J-2) Proof Test | Each wire is proof-tested and visual-inspected | [Phase 3] Closed, Document No. :27_BIRDS5-WTR-01 Wire Strength Test Report (2022/03/09) |
| ☑ | J-3) Assembly Procedure | Wire handling process is defined in assembly procedure. | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☑ | J-4) Round | (If any) The part touching the wire is rounded appropriately. | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☑ | J-5) Loosening Prevention | (If any) Loose prevention is provided on the tied portion. | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☑ applicable | K) Fail Safe Fastener | Fail Safe Fastener | — |
| ☑ | K-1) Fail Safe Analysis | K-1) Failsafety analysis shows MS >0. (F.S = 1.0) | [Phase012] Closed, Document No. :02_BIRDS5-SR-01 Structural Analysis Report (2022/03/15) |
| ☑ | K-2) Quality Control | Quality Control meets the condition (L-2) to (L-5) | Please refer from (L-2) to (L-5). |
| ☑ applicable | L) Safety Critical Fastener | Safety Critical Fastener | — |
| ☑ | L-1-1) Secondary Locking Feature | L-1-1b) Locking compound of which the application process MUA is approved. | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☑ | L-2-1) Certificates for fastener materials | L-2-1) Certificates for fastener materials | [Phase 012] Closed, Document No. :05_BIRDS5-AD-01 Assembly Drawing (2021/12/09) |
| ☑ | L-3) Torque mark inspection | L-3) Torque mark inspection | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) [Phase3] Closed, Document No. :25_BIRDS5-VT-01 Vibration Test Report (2022/02/18) |
| ☑ | L-4) Fastening torque control | L-4) Fastening torque control | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☑ | L-5) Fastener traceability | L-5) Fastener traceability | [Phase 012] Closed, Document No. :06_BIRDS5-AP-01 Assembly Procedure(2021/12/09) [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☐ applicable | M) Sealed Container | Sealed Container | — |
| ☐ | M-1) Container characteristic | Single, independent Container containing a non-hazardous substance. | |
| ☐ | M-2) Stored Energy | Contain less than 19,310 Joules(14,240 foot-pounds) of stored energy | |
| ☐ | M-3) Maximum delta pressure | Maximum delta pressure is less than 1.5atm(22psia, 1.5bars) | |
| ☑ applicable | N) Fracture Critical Part | Fracture Critical Part | — |
| ☑ | N-1) Design Verification | Verified by Structural Analysis with appropriate mechanical properties | [Phase012] Closed, Document No. :02_BIRDS5-SR-01 Structural Analysis Report (2022/03/15) |
| ☐ | N-2) Production Verification | Verified by appropriate material selection and production process | [Phase 3] Closed, Document No. :18_BIRDS5-AR-01 Assembly Record (2022/02/18) |
| ☑ | N-3) Product Verification | Verified by visual inspection after vibration test under the condition specified in JX-ESPC-101132/101133<br>Visual Inspection and NDE(Tapping test acceptable) before and after Tests | [Phase3] Closed, Document No. :25_BIRDS5-VT-01 Vibration Test Report (2022/02/18) |

# G-2 Battery Description Form

# Battery Description HR Attachment

**For the Phase 0/I Safety Data Package, provide the following information:**

| | |
|---|---|
| **1a. Hardware Point-of-Contact:** **(Name/Company/Phone/Fax/email)**<br>Yamauchi Takashi<br>Kyushu Institute of Technology<br>+81-93-884-3222<br>yamauchi.takashi819@mail.kyutech.jp | **1b. Hardware Name:** BIRDS5<br>**Hardware Part Number:** BIRDS5-FM-01<br>**Hardware Acronym:** BIRDS5<br>**Battery Name:** Ni-MH Battery |

**2a. Hardware / Battery Managing Group, Company, or Agency:**

Kyushu Institute of Technology

**2b. Hardware and Battery Environmental Requirements:**

**Thermal Environment (max, min, operational and non-operational ranges):**

-15 to +60 degC outside ISS

+16.7 to +29.4 degC inside ISS

[Note]: Battery specification: Storage: -20 to +50 degC

Since battery thermal specification does not cover the thermal environment, we will confirm by thermal test.

**Pressure Environment (EVA, IVA):**

Both EVA and IVA environments. Maximum pressure during launch and inside the ISS is as follows. A pressure inside JEM Airlock at depressurization and outboard is 0 Pa.

HTV, Cygnus: 104.8 kPa, Dragon: 102.7 kPa, Inside the ISS: 104.8 kPa

**Life (calendar/shelf, cycle/service):**

Duration 3 years / Product warranty of stowage is 5 years to keep more than 90% capacity from fully charged

**3a. Battery and Hardware Description:**

**Is the battery pack (including all components) Commercial-off-the-shelf (COTS)?** ☐Yes ☒ No

**Function/Operating modes (continuous, pulse, intermittent, clock backup, memory, etc.):**

No operation inside J-SSOD. Continuous operation after deployment.

**Battery/Cell crew access on-orbit?** ☐Yes ☒ No

**Short Description of Battery System: (Number of batteries to be flown; details if multiple batteries are required to power** Application, unique design information, etc.)

Battery cells, FAB and EPS are COTS.

**[Battery pack]**

Part Name: BIRDS5-Battery

Type: Ni-MH Battery

Part Number: BIRDS5-Battery-01

Specification: 14.4 Wh

Manufacturer: Kyushu Institute of Technology

**[FAB]**

Part Name: BIRDS5-FAB

Part Number: BIRDS5-FAB-01

Manufacturer: Sagami Tsushin

**[EPS]**

Part Name: BIRDS5-EPS

Part Number: BIRDS5-EPS-01

Manufacturer: Sagami Tsushin

**3b. Cell Description:**

**Chemistry (If li-ion, what type: NMC, NCA, FePO4, etc.):** Ni-MH

**Cell size:** 14.35 mm dia. x 50.4 mm

**Manufacture and Model:** Panasonic / BK-3MCC

**Nominal OCV:** 1.2 V **Maximum Voltage:** 1.6 V **Minimum Voltage:** 1.0 V

**Rated Capacity:** 2,000 mAh

**Maximum Rated OEM Discharge Current:** Nonpublic information

**Maximum Recommended OEM Charge Current:** 2,000 mA

**Minimum and Maximum OEM Discharge Temperatures:** 0 to +50 degC

**Minimum and Maximum OEM Charge Temperatures:** 0 to +40 degC

**Minimum and Maximum Storage Temperatures:** -20 to +40 degC

**Date of Manufacture (Mo/Yr):** 05/2020

**3c. Battery Information: (9 V COTS batteries shall be considered as single units)**

**Quantity of total cells:** 6 per satellite

**Cell connectivity (#P#S, #S#P):** 3S2P

**Operational Battery Environment Temperatures Range (Min/Max):** 0 to +40 degC

**Nominal OCV:** 3.6 V **Maximum Voltage:** 4.8 V **Minimum Voltage:** 3.0 V

**For the Phase II/III Safety Data Package, provide the following information:**

**Stowage location (launch and on-orbit use locations):**

Installed inside CubeSat. BIRDS-5 (1U and 2U CubeSat) will be launched to ISS by HTV-X, NG Cygnus or Space-X Dragon with being installed inside J-SSOD (JEM Small Satellite Orbital Deployer) satellite install case or a dedicated launch case, which is soft stowed inside a bag.

**Packaging and hardware approved by flammability group (reference requirement):**

08/03/2021

**Is the battery charged on orbit?, Yes:** ☐ **No:** ☒

**If yes, describe charge rate, charger hardware, and protections to prevent overcharge**

**Is the battery being discharged on-orbit? Yes: ☐ No: ☒ If yes what is the discharge rate(s):**

**Circuit Description and Electrical Schematic (attach electronically or reference location in HR):**

All of the cells are covered by TEFLON panels and metal box same as BIRDS-1: EP-J-17-001 (Figure 1).

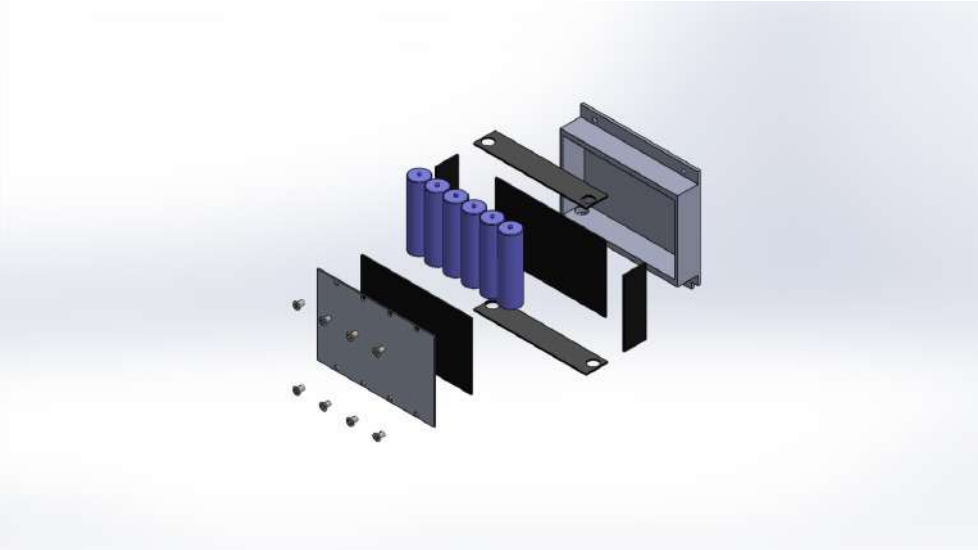


**Figure 1 Battery Box**

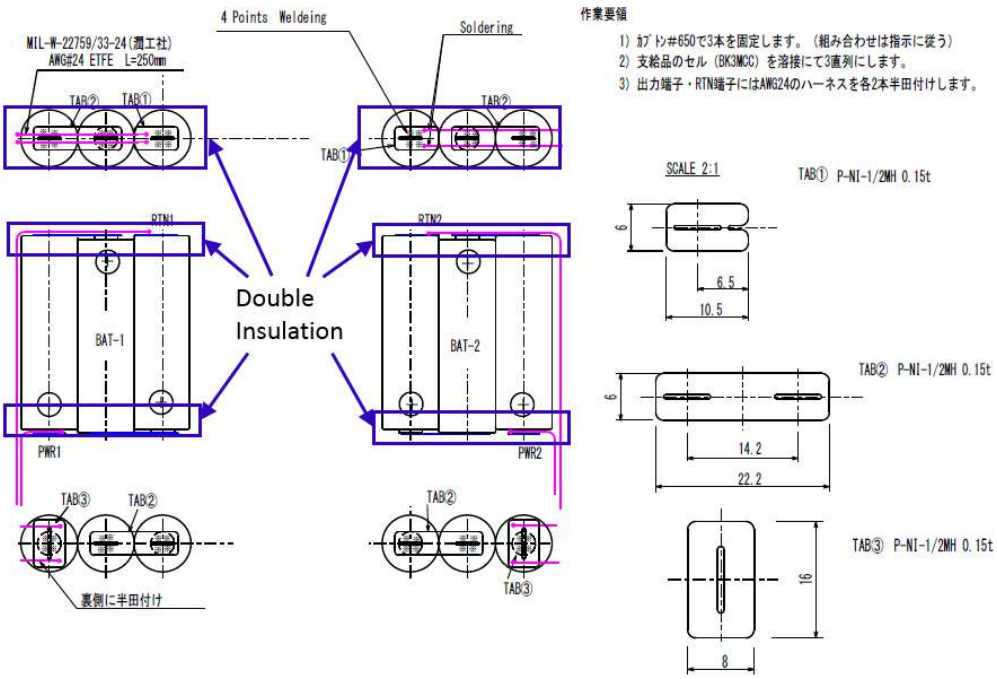
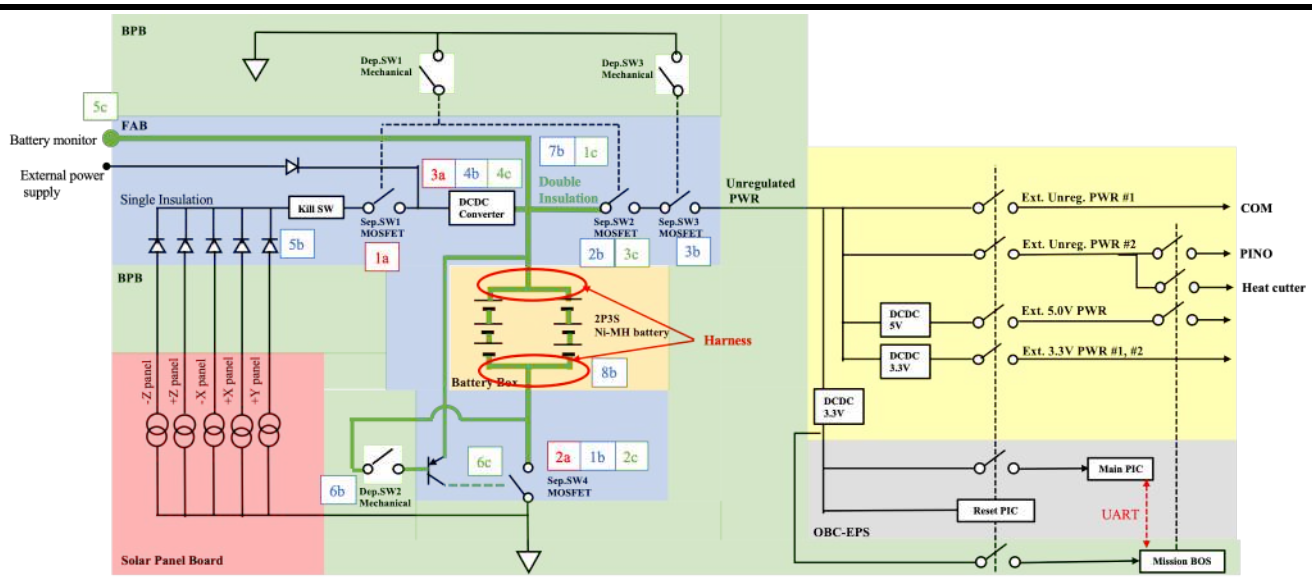



**Figure 2 Battery Connectivity**




| Hazard | | Hazard Control #1 | Hazard Control #2 | Hazard Control #3 |
|---|---|---|---|---|
| Over-charge | | SepSW1[1a] | SepSW4[2a] | DCDC converter[3a] |
| Over-discharge | Load side | SepSW4[1b] | SepSW2[2b] | SepSW3[3b] |
| | Solar cell side | | DCDC converter[4b] | Diode[5b] |
| | DepSW2 side | DepSW2 [6b] | Proper Insulation[7b] | Proper Charging [8b] |
| External short | Load side | Proper Insulation[1c] | SepSW4[2c] | SepSW2[3c] |
| | Solar cell side | | | DCDC converter[4c] |
| | External power supply side | | | DCDC converter[4c] |
| | Battery monitor side | | | Proper Insulation[5c] |
| | DepSW2 side | | | Proper Insulation[6c] |

Note: Proper insulation (double isolation is shown by green line in figure above, single isolation is black line)
All wires and components between the battery and the first power functions are assembled as double insulation. .
The DCDC converter (LTC3119) used for inhibit is a buck-boost converter and its internal FET configuration prevents reverse current.

**Figure 3 Inhibit schematics.**

### Overcharge protection

Sep SW1, Sep SW4 and DCDC converter are equipped in the solar cell side or the GND side of the battery as shown in Figure above. DCDC converter controls the input voltage from the solar cells to the battery pack to 4.2V.

### Over dis-charge protection

*(Load Side)*

Sep SW2, Sep SW3 and SepSW4 are equipped in the Load side of the battery as shown in Figure 3.

*(Solar cell Side)*

Sep SW4, DCDC converter and Diodes are equipped in the Solar cell side of the battery as shown in Figure 3.

*(DepSW2 Side)*

Dep SW2 is equipped in the HOT side of the battery as shown in Figure 3. Double Insulation is set between the battery and the DepSW2. Charge the battery to the final charging state (SOC > 80 %).

### External short protection

*(Load Side)*
Sep SW2 and Sep SW4 are equipped as shown in Figure 3. And double insulation is set between battery and Separation Switches.

*(Solar cell Side)*
Sep SW4 and DCDC converter are equipped as shown in Figure 3. And double insulation is set between battery and Separation Switch and DCDC converter.

*(External power supply Side)*
Sep SW4 and DCDC converter are equipped as shown in Figure 3. And double insulation is set between battery and Separation Switch and DCDC converter.

*(Battery monitor Side)*
Sep SW4 is equipped in the GND side of the battery as shown in Figure 3. Double insulation is set between battery and Battery monitor connector.

*(DepSW2 Side)*
Sep SW4 is equipped as shown in Figure 3. Double insulation is set between the Drain of SepSW4 and the battery and the Gate of Sep SW4.

**Summary of circuit protections and include trip/reset points (i.e. fuses, diodes, MOSFETs, resistors, source isolation, etc.):**

Covered cables are used from the battery to GND or the first inhibit. The covered cables are further covered with Kapton tape to provide double insulation. In addition, GND and HOT lines on a surface of a circuit board are separated by 1mm or more, and the surface is covered with Kapton tape for double insulation.

**Battery testing complete and report uploaded to this HR? Yes:** ☐ **No:** ☒
**If yes, provide a short summary test results including anomalies or failures.**

**Summary of safety testing performed or planned :**

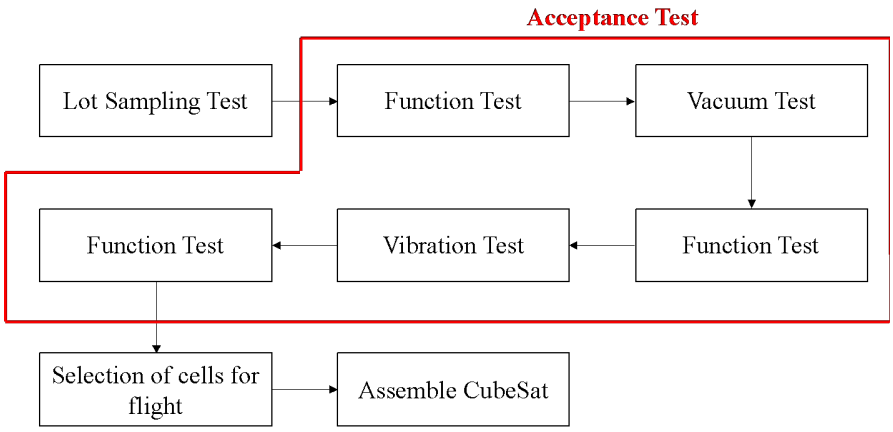



**Figure 4 Flowchart for Screening Test**

**1. Lot sampling Test**

(1) Thermal Test

Thermal test of battery cell is performed for confirmation of temperature tolerance. Test condition is summarized as below.

- Temperature: more than +60 degreeC
- Test Duration: Over 2 hours

Before and after the thermal test, several function tests below are performed to see that there is no change in characteristics.

- Visual inspection (scratches, misaligned seals, electrolyte leakage, etc.)
- Open Circuit Voltage (the change is less than 0.1%)
- Mass (the change is less than 0.1%)
- Capacity (the change is less than 5%)
- Charge/Discharge Characteristic
- Discharge Temperature

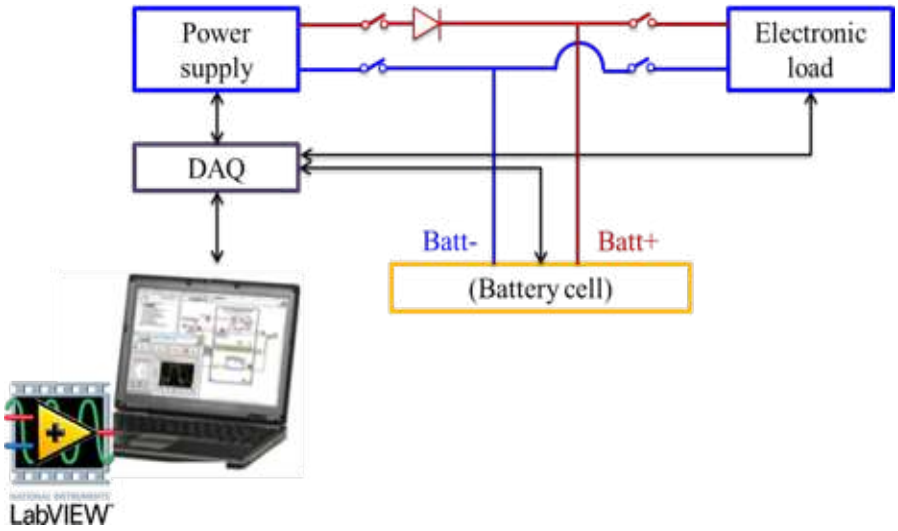



**Figure 5 Charge/Discharge Characteristics Test Configuration**

**2. Acceptance Test**

(1) Vacuum Test

Vacuum test of battery cell level is performed for screening. Test condition is summarized as below:

- Vacuum Level: less than 0.1 psia
- Test Duration: Over 6 hours

(2) Random Vibration Test

Random vibration test for flight cells is performed for screening purposes.

Test condition is summarized as below.

- Vibration Level: Minimum Screening Level (MSL)
- Tolerance: +/- 1.5dB for PSD
- Direction: 2 axes (Radial direction and Axial direction)
- Test Duration: Over 60 sec

Random Vibration Level for cells

| Freq. [Hz] | PSD [$G^2$/Hz] (MSL) |
|---|---|
| 20 | 0.01 |
| 80 | 0.04 |
| 350 | 0.04 |
| 2000 | 0.007 |
| Overall | 6.06 Grms |
| Duration | 1min/axis |

(3) Function Test

Before and after the environment tests (vibration test and vacuum test), several function tests below will be performed to see that there is no change in characteristics.

Note that the Charge/Discharge Characteristics test measures the range between maximum voltage and minimum voltage.

Test Load: 1.9 [A]

Test Contents;

- Visual inspection (scratches, misaligned seals, electrolyte leakage, etc.)
- Open Circuit Voltage (the change is less than 0.1%)
- Mass (the change is less than 0.1%)
- Capacity (the change is less than 5%)
- Charge/Discharge Characteristic
- Discharge Temperature

**3. Safety Function Test for system**

**(1) Function Test for system**

After assembling the satellite, before and after the environmental test, the following functional tests are performed to confirm that there are no problems with the assembled battery.

Test content

- Open Circuit Voltage (Everyday for a 5 days)
- Take measurements three times a day.
- No testing using the battery will be performed during this test.

**(2) Function Test for safety function**

Protective devices (DC/DC converter, Diode) are functionally tested for flight products before FM vibration test. Function of separation switches (Sep SW) will be confirmed after the FM vibration test. Functional tests are performed with the board assembled, but the solar panel is not assembled because the test is performed using the input line from the solar panel. To evaluate the electronic elements after the environmental test, the satellite must be disassembled, which is risky, so the evaluation is performed before the environmental test. The functionality of surface-mounted electronic elements is already verified in the QT level test for EM. Therefore, there are no problems with the electronic elements in the vibration test for FM. On the other hand, mechanical switches are inspected after the environmental test because they may be broken in the environmental test.

**・SepSW1 test**

SepSW1 is on the battery charging line from solar panels. Place the satellite in front of a solar simulator with all Dep SW pressed.

When the satellite is exposed to the light of the solar simulator, the Inhibit (SepSW1) prevents the battery from charging when DepSW1 is pressed. On the other hand, after DepSW1 is released, the voltage will be applied to the battery line. Therefore, the soundness of SepSW1 can be confirmed by checking the source voltage to the battery from the solar panels.

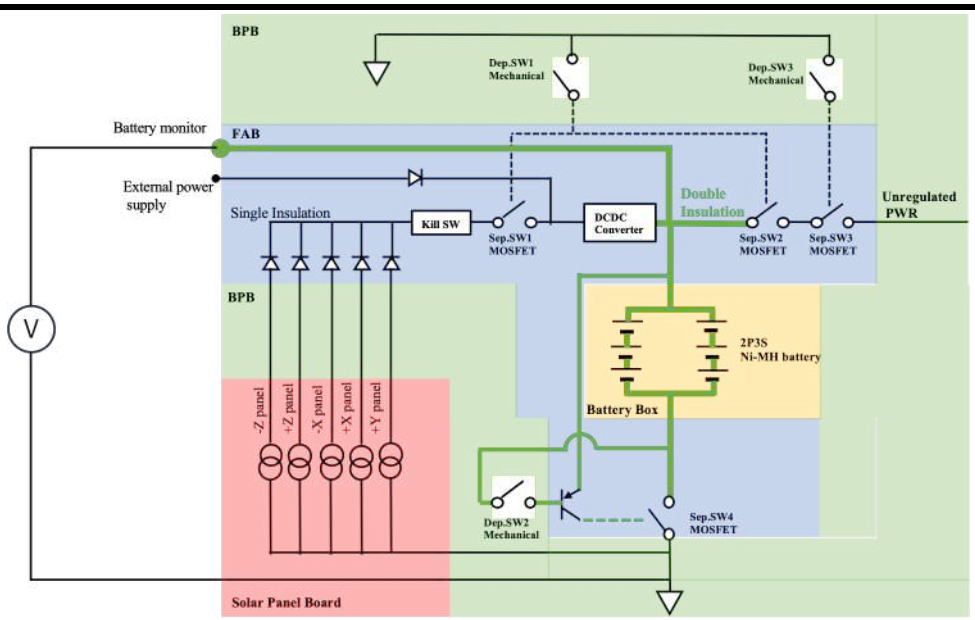



**Figure5 SepSW1 test configuration**

- **SepSW2,3,4 test**

Connect the debugger to the access port of the satellite so that serial data from the satellite can be output to an external PC.

When the satellite is in OFF-state and the debugger is connected between the satellite and the PC, there should be no output the serial data in the PC display. OFF-state of the satellite is when one of the Inhibits are pressed. On the other hand, the satellite is in ON-state when all inhibits are released, as such, the satellite should display information on the PC.

SepSW1 and SepSW2 can be operated by DepSW1, SepSW4 can be operated by DepSW2, and SepSW3 can be operated by DepSW3.

Check the serial data output when DepSW1, 2, and 3 are turned on respectively, and check the function of SepSW2, 3, and 4.

- **DCDC converter test (for overcharge)**

The output of the solar cell is regulated by one DCDC converter to charge the battery. Connect an external power supply to the Back Plane Board and simulate the input voltage from the solar cell. Then, measure the DCDC output voltage from the connector that connects to the battery. The battery overcharged voltage is 4.8 V or higher. When the DC/DC input voltage is supplied by 5.2 V, we inspect the DC/DC output voltage should be below 4.2V by measuring the voltage at the battery slot.

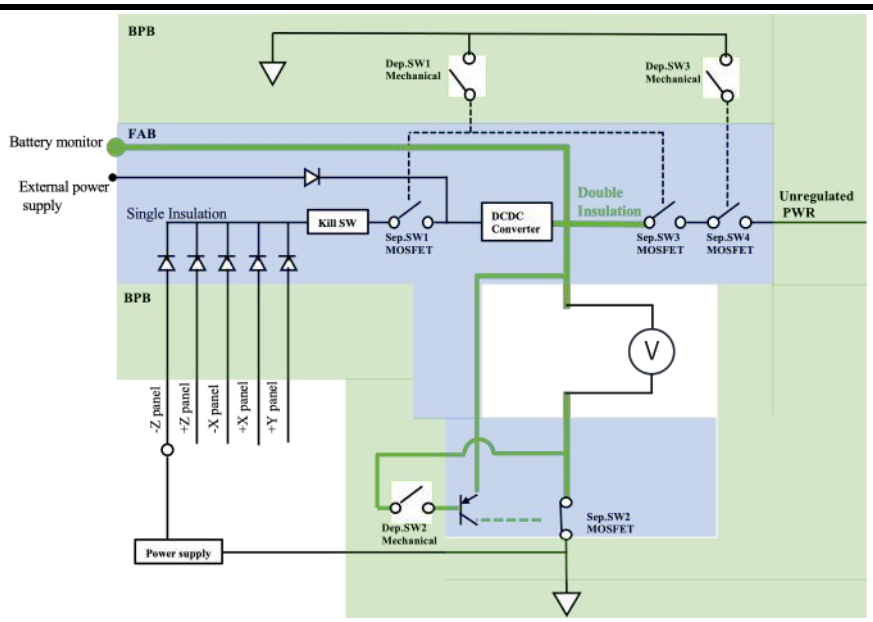



**Figure6 DCDC converter test configuration**

- **DCDC converter test (for over-discharge and external short)**

Ensure that the DCDC converter prevents reverse current flow. Connect an external power supply to the FAB board instead of the battery. With DepSW2 turned on, check the voltage on the input and output sides of the DCDC converter. Verify that the voltage from the external power supply is applied to the output side and that the voltage is not applied to the input side.

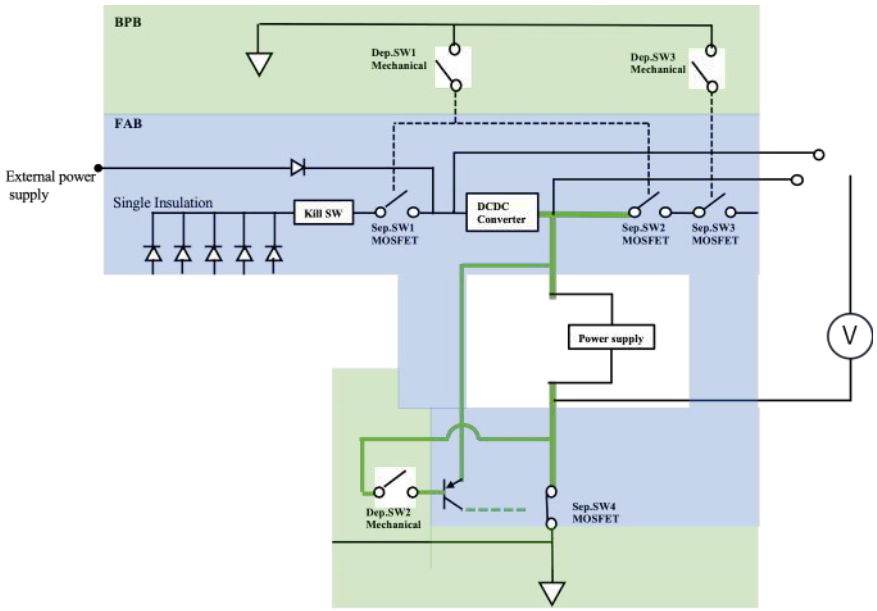



**Figure7 DCDC converter test configuration**

- **Diode test** (Solar cell side)

In order to check if current flows in the reverse direction of the diode, an electronic load is connected in place of the solar panels. The anode side of the diode under test is connected to the electronic load and a power supply is connected to the cathode side of the diode under test. When the electronic load is activated to draw current, the diode should prevent it from doing so.

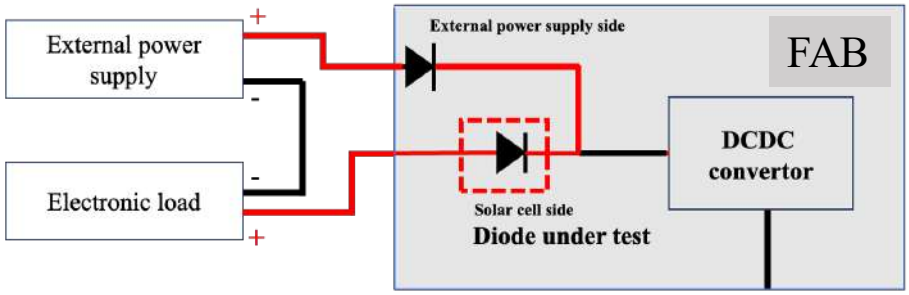



**Figure 8 Diode test configuration**

**Has the pre-flight processing, flight acceptance plan, on-orbit processes, operational constraints, and the on-orbit and post-flight battery disposal plan for the hardware been conducted in accordance with JWI 8705.3?**

Yes: ☐ No: ☒ If yes, provide accompanying documentation.

No on-orbit processing of the battery is planned before CubeSat deployment from ISS.

CubeSat including the battery will not return to the ground but will be burned up at atmospheric reentry.

(Document No.#21-011, Date of review: Sep.25.2026)

# G-3. Toxicity Analysis (HMST)

## Common Batteries HAZARDOUS MATERIALS SUMMARY TABLE Applicable sheet

**[BIRDS-5] ASCENT to ISS Increment ALL**

| Batteries Name | Chemicals or BiologicalMaterials<br>Commonly Flown Battery Chemistries | QTY | Applicable Parts<br>Batteris Including Parts Name (P/N,S/N) | QTY |
|---|---|---|---|---|
| BK-3MCC | Nickel Metal Hydride - NiOOH/metal alloy/KOH | 6 | PearlAfricaSat-1 | 1 |
| BK-3MCC | Nickel Metal Hydride - NiOOH/metal alloy/KOH | 6 | TAKA | 1 |
| BK-3MCC | Nickel Metal Hydride - NiOOH/metal alloy/KOH | 6 | ZIMSAT-1 | 1 |

## FINAL HAZARDOUS MATERIALS SUMMARY TABLE
## ASCENT to ISS Increment ALL

**Verification Status**
V-1: *Previously Flown*
V-2: *N/A*

*Expt /Hdwr/Item:* **Common Batteries**
*Expt/HW Acronym:* **NA**
*OpNom:*

Record # Common.Bat — PAYLOAD CHEMICAL or BIOLOGICAL INFORMATION

| Last Modified 8/22/19 Part # | Sub-System | Label | Chemicals or Biological Materials | Maximum Concentration | Maximum Volume or Amount | Tox Lvl | BioSafety Lvl | Flammability Haz Lvl |
|---|---|---|---|---|---|---|---|---|
| | Batteries | Batteries | Commonly Flown Battery Chemistries | NA | NA | 2 | NA | 0 |
| | | | **Alkaline** | | | | | |
| | | | Zn/MnO2/KOH | | | | | |
| | | | **Zinc Carbon** | | | | | |
| | | | Zn/carbon rod/NH4Cl (ZnCl2) | | | | | |
| | | | **Lead Acid** | | | | | |
| | | | Metallic Pb/PbO2/H2SO4 | | | | | |
| | | | **Lithium Ion** | | | | | |
| | | | LiCoO2, LiNMC, LiNCA | | | | | |
| | | | LiMO2, LiFePO4, or other | | | | | |
| | | | LiTi4O5 or lithium alloys of Si or Sn | | | | | |
| | | | LiPF6, LiBOB, LiBF4 or other | | | | | |
| | | | **Lithium Primary** | | | | | |
| | | | Li-MnO2; LiFeS2; LiCFx | | | | | |
| | | | **EXCLUDING:** Li-SOCl2, Li-SO2 | | | | | |
| | | | Li-SO2Cl2, and Li-BCX | | | | | |
| | | | **Nickel-Cadmium** | | | | | |
| | | | NiOOH/Cd(OH)2/KOH | | | | | |
| | | | **Nickel-Hydrogen** | | | | | |
| | | | H2/Ni(OH)2/KOH | | | | | |
| | | | **Nickel Metal Hydride** | | | | | |
| | | | NiOOH/metal alloy/KOH | | | | | |
| | | | **Silver-Zinc** | | | | | |
| | | | ZnO/Ag/KOH | | | | | |
| | | | **Silver Oxide** | | | | | |
| | | | Ag2O/Zn/KOH or NaOH | | | | | |
| | | | **Zinc Air** | | | | | |
| | | | Zn/Air/KOH | | | | | |

**HAZARD RESPONSE LEVEL (HRL) = 2**

**SPECIFIC HAZARD LEVELS**

Toxicity (THL) 2 | BioSafety (BSL) NA | Flammability 0

**Target Organ(s):** Eye, Skin

**Principal Adverse Effects:**

Electrolytes contained in batteries may cause severe eye irritation and possible permanent eye damage when they come into contact with the eyes.

SPECIAL NOTE: THIS RECORD DOES **NOT** COVER BATTERIES WITH THL > 2, WHICH ARE GENERALLY NOT ALLOWED IN THE HABITABLE VOLUME OF SPACECRAFT. Batteries containing sulfur dioxide and/or thionyl chloride may be THL 4 and **MUST** be assessed separately. Toxic vapors from batteries with THL 4 chemistry can be life threatening to crew (reference memo SF23-07-073).

**Contact:** [redacted]

**Comments:** Most batteries contain electrolytes that are rated THL 2. It is therefore prudent to assume that all batteries contain irritating or corrosive electrolytes. **NOTE:** As of May 21, 2019, specific battery information will now be contained within the battery/capacitor spreadsheet located in IHS record #48243.

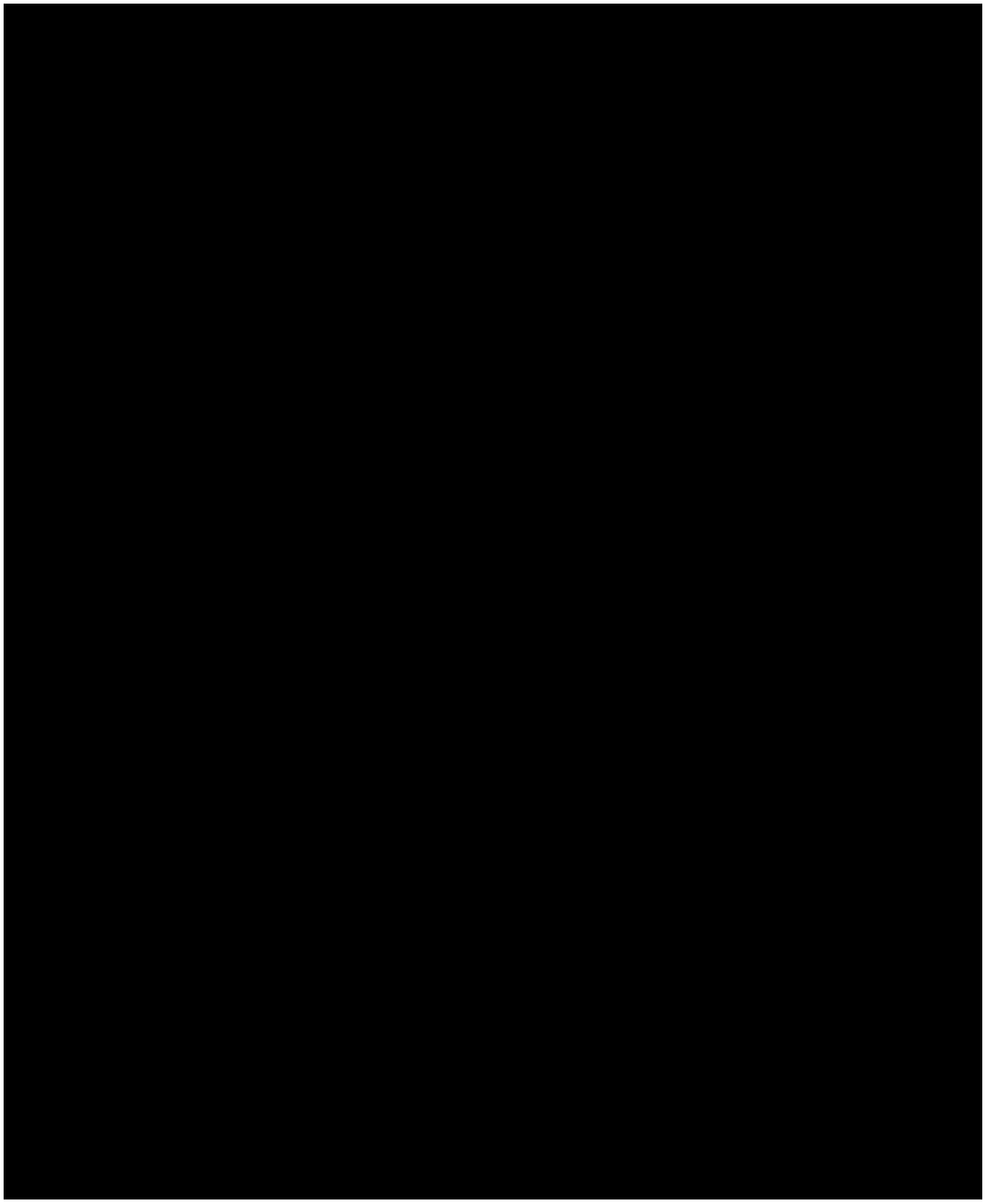
ECLS Environmental Rating = NA | ECLS Hardware Rating = NA

As of August 22, 2019 10:46:59 AM — Page 1 — **Record # Common.Bat**

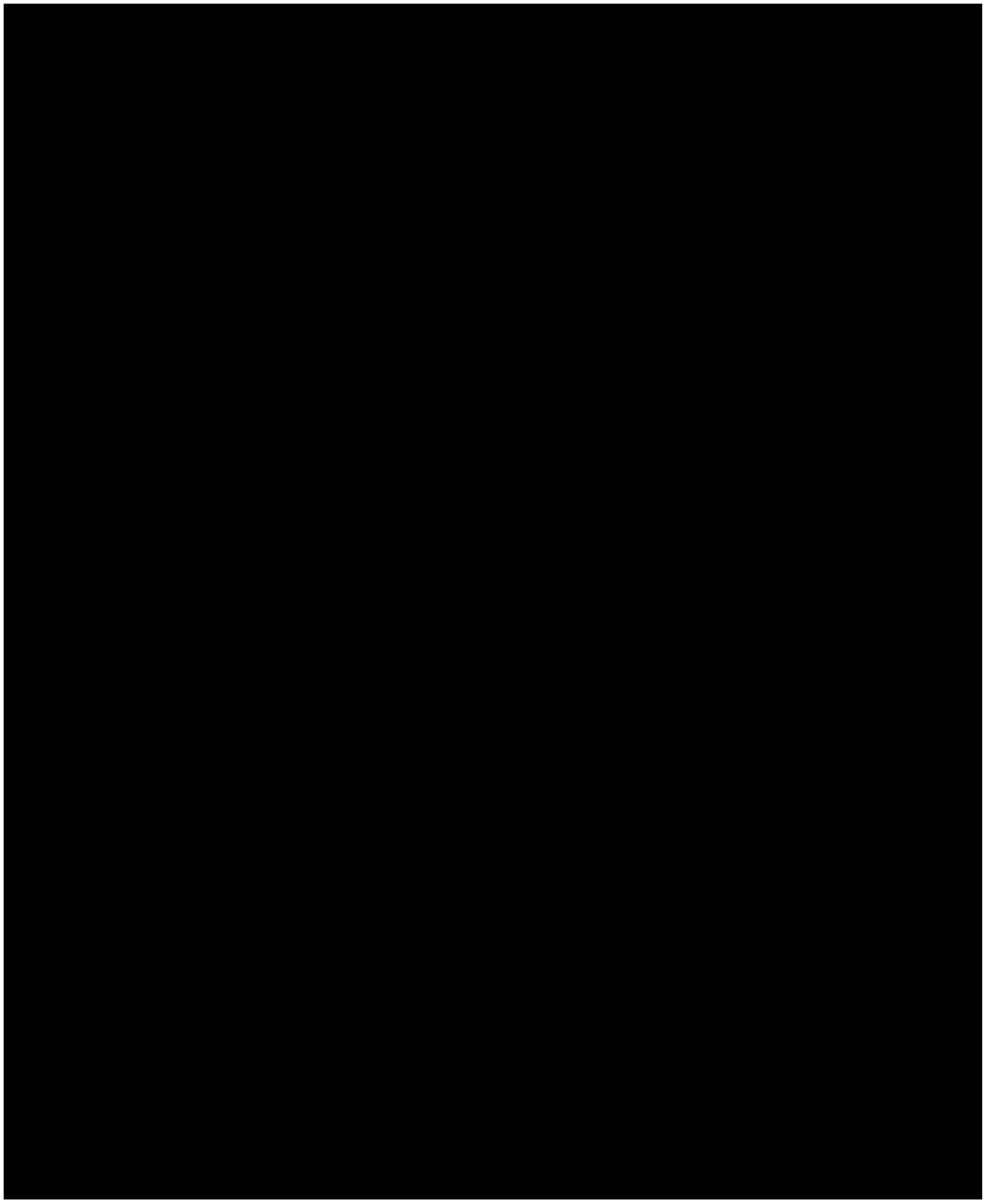
Currently assessed by:
Mission Toxicologist: JSC Toxicology

# G-4. ISS EME Tailoring/Interpretation Agreement

**ISS Electromagnetic Effects Panel**
**Tailoring/Interpretation Agreement**



**ISS Electromagnetic Effects Panel**
**Tailoring/Interpretation Agreement**
**Technical Concurrence**



# G-5. Hazard Analysis Verification

# for Space Radiation Analysis Group (SRAG)

| ISS System/Payload: | | Hazard Analysis Verification<br>Space Radiation Analysis Group (SRAG) |
|---|---|---|
| Safety Review Panel POC: | | Date: 09/19/2023 |
| System/Payload POC | Takashi Yamauchi | MM/DD/YY |

POC: [redacted]

Alternate POC: [redacted]

| RF Emitter Name | Frequency [MHz] | Max Power [W] | Attenuator (Gain loss) [dB] | Max Gain [dBi] | Emitter Power Density [W/m²] | Power Density Requirement [W/m²] | Minimum Safe Distance [m] | Hazard Severity | Payload Actions |
|---|---|---|---|---|---|---|---|---|---|
| BIRDS5 UHF COM | [redacted] | 0.9 | 0 | 0.5 | 0.89 | 2.92 | No Hazard | No Hazard | There is no Crew Hazard. Please submit this form to NIR SME and document on the THINCC form. |
| BIRDS5 VHF COM | [redacted] | 0.79 | 0 | 2 | 1.11 | 2.00 | No Hazard | No Hazard | There is no Crew Hazard. Please submit this form to NIR SME and document on the THINCC form. |

General Notes (for System/Payload input as applicable):

# Appendix H

## List of Safety-related failures and accidents

Not Applicable

# Appendix I

## Operational Control Matrix (OCM)

Not Applicable

# Appendix J

# Safety Verification Tracking Log (SVTL)

International Space Station
Safety Verification Tracking Log

| Mission/Element | | | J-SSOD | | | Date: | 2021/10/28 | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Completed by Project | | Confirmation by JAXA S&PA | |
| Log Number | Hazard Report Number | Safety Verification Number | Description (Identify Procedures by Number and Title) | Operation(s) Constrained | Independent Verification Required (Yes/No) | Scheduled Date | Completion Date | Method of Closure Comments/Verification Completion Notice (VCN) | Completion Date | Remarks As a Result of Independent Validation and Verification |
| 1 | BIRDS5-UNQ-01 | 1.1-1(1) | Inspection to verify that the satellites are installed per the approved packing requirement. | No | No | 31-5-22 | | | | |
| 2 | BIRDS5-UNQ-02 | 1.4-1(3) | Confirm the battery monitor terminals are covered with Kapton tape | No | No | 31-5-22 | | | | |
| 3 | BIRDS5-UNQ-02 | 1.4-2(2) | Confirm that the battery voltage is in a state of final charge (SOC > 80%) at the time of delivery. | No | No | 31-5-22 | | | | |